Mambo: Degree-Two Hashing for Zero-Knowledge Proof Systems

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Abstract

We introduce Mambo, a coupled-quadratic permutation family over prime fields: at state width $t = 2$, $\theta' = \varphi \cdot \theta + \psi^2 + c_i$ and $\psi' = \varphi \cdot \psi + (\theta')^2 + d_i$, with per-round constants derived from SHA-256. Its central security-relevant algebraic quantities are exact theorems rather than estimates. We prove that the algebraic degree after n rounds is exactly 4^n at an R1CS cost of $2n$ nonlinear constraints, that the CICO ideal admits an explicit Gröbner basis with closed-form Hilbert series $(1+z)^{2n-1}$ and regularity degree $d_{\mathrm{reg}} = 2n$, and that the maximum differential probability per round is exactly $1/p$. The $t = 2$ degree and regularity theorems are machine-checked in Lean 4; the formalisation verifies the mathematics, not the attack-cost model. As a corollary, Mambo attains the Algebraic Figure of Merit bound $\eta = 1.0$ of Boeuf and Perrin [1]. Because the Gröbner basis is explicit, algebraic attack cost reduces to a single quantity—the eliminant degree, $D_I = 2^{t(n-1)}$ at rate 1, and $D_{\mathrm{eff}} = 2^{t(n-1)-(t-2)}$ on the capacity-1 hash surface after a rate-specialization attack that we disclose against our own construction—and we size all parameters by the conservative requirement $D_{\mathrm{eff}} \geq 2^\lambda$ at the minimum round count, with recommended round counts adding a further 25% round margin against model risk. Assuming the CICO-hardness conjecture stated in the paper, the resulting recommended 128-bit 2-to-1 sponge hash costs 165 nonlinear constraints, against 240 for Poseidon at its own margined parameters. A self-cryptanalysis is included: the $(t-2)$-bit specialization attack above, a polynomial-time attack on additive PRNG extraction at $k = 1$, small-field distinguishers, and an explicit maturity caveat; the unmargined minimum-round cells are stated explicitly as falsification targets.

Artifact. The Lean 4 development, reference implementation, reproduction scripts, and Galois certificates are available at `https://github.com/mjbasso7/mambo-artifacts` (commit `0a4297f`) under the Business Source License 1.1 with a research, verification, and audit use grant.

Keywords: ZK-friendly permutation, algebraic degree growth, R1CS, ALFOM, coupled quadratic map, Gröbner basis, sponge indifferentiability, zero-knowledge proofs.

1 Introduction

1.1 The Algebraic Efficiency Gap

The design of arithmetization-oriented (AO) symmetric primitives has proceeded largely by empirical methods: candidate constructions are proposed, constraint counts are measured, and security is argued via a combination of heuristic Gröbner basis estimates and classical differential/linear analysis. This approach has produced designs with widely varying ratios of algebraic degree to arithmetic cost, yet no formal metric existed to compare these ratios—until the ALFOMs framework of Boeuf and Perrin [1] introduced the Algebraic Figure of Merit.

Ongoing cryptanalytic progress—for example improved resultant techniques [5] and Gröbner-basis analyses of coupled-variable designs [28]—highlights the value of constructions whose algebraic cost is matched tightly to the degree they produce. Motivated by these developments, we propose Mambo: a coupled-quadratic permutation family that attains the ALFOM upper bound of 1.0, and study its security in the standard attack catalogue.

This paper is best read as two coupled artifacts: an exact algebraic characterization of a coupled-quadratic chain—degree, Gröbner basis, Hilbert series, regularity, and differential profile as closed-form theorems rather than estimates—and an explicit cryptanalysis challenge built on that characterization, with named falsification targets: the minimum-round cells of Table 6, Conjecture 5.2, and the functional-indecomposability hypothesis of Section 5.3. It is not a claim that Mambo should replace deployed hashes today; the maturity caveat of Remark 5.3 governs that question. The efficiency results stand independently as measured facts.

1.2 The ALFOMs Framework and the R1CS Upper Bound

Boeuf and Perrin define the Algebraic Figure of Merit for a symmetric primitive as:

$$\eta = \frac{\log_2 D}{C},$$

where D is the algebraic degree of the round function after C nonlinear constraints in R1CS. Each nonlinear constraint computes one product $a \cdot b = c$; if $\deg(a) \le d_1$ and $\deg(b) \le d_2$, then $\deg(c) \le d_1 + d_2 \le 2\max(d_1, d_2)$. By induction, C constraints yield $D \le 2^C$, hence:

$$\eta \le 1.0\,.$$

This bound is tight: any design achieving $\eta = 1.0$ extracts the maximum possible algebraic degree from every nonlinear constraint. The ALFOMs paper introduced Clotho ($\beta = 2$) as a theoretical instance attaining the bound; the present work contributes an alternative family with concrete implementations and a security study.

1.3 Contributions

This paper makes four contributions:

(A) An exact, machine-checked algebraic characterization. The security-relevant algebraic quantities of the Mambo map are proven theorems, not estimates. We prove that the algebraic degree after n rounds is exactly $d_\psi(n) = 4^n$ at R1CS cost $C = 2n$ (Theorem 4.2; constructive, no leading-degree cancellation, valid over every prime field with $p > 2$ and every $\varphi \ge 2$); that the CICO ideal admits an explicit Gröbner basis with closed-form Hilbert series $(1+z)^{2n-1}$ and regularity degree $d_{\text{reg}} = 2n$ for all $n \ge 2$ (Theorem G.5, Appendix G); and that the maximum differential probability per round is exactly $1/p$ (Proposition 5.17). The $t = 2$ closed form and the degree theorem are machine-checked in Lean 4, as is the eliminant-degree closed form $D_I = 2^{t(n-1)}$ for the capacity-fixed hash attack surface (Section 5.5); the parameter-tier Macaulay floors are kernel-verified arithmetic at all nine minimum (analysis-equality) cells of Table 6, conditional on the measured d_{reg} at $t \in \{3, 4\}$; the development uses only the three standard axioms, and the production floors none. The formalisation verifies the mathematics, not the attack-cost model. The ALFOM value $\eta = 1.0$ [1] follows from the degree theorem as a corollary (Corollary 4.3).

(B) A security analysis with explicit conditionality. Mode-level security (sponge collision and indifferentiability, PRNG indistinguishability) is proven in the ideal-permutation model (Proposition 6.2, Lemma 7.4). What remains assumed is stated once, explicitly: a non-falsifiable design claim (Design Claim 5.1) and a single falsifiable CICO-hardness conjecture parameterized over (t, n, p) (Conjecture 5.2), with the supporting evidence collected in Section 5.3. Attack costs are priced under an explicit two-tier cost model (Section 5.2); all parameters are sized under the conservative tier, $D_{\text{eff}} = 2^{t(n-1)-\max(t-2,0)} \ge 2^\lambda$ at the minimum round count, which already prices in the capacity-1 specialization attack of Section 5.6; every recommended round count adds a 25% round margin on top of the minimum (Section 5.13).

(C) Efficient instantiations and implementations. At 128-bit security under the conservative cost model with the recommended 25% round margin, the 2-to-1 sponge hash costs 165 nonlinear constraints ($t = 3$, $n = 55$; the $t = 4$, $n = 43$ variant costs 172), against 240 for Poseidon at matched security—a margined-against-margined comparison; the permutation costs 164 NL ($t = 2$, $n = 82$) and the ZK-private PRNG 2 NL per output element. Cross-verified reference implementations are provided across multiple R1CS and STARK back-ends and five prime fields; permutation constraint counts match the theoretical $2n$ exactly, and the sponge-hash count matches modulo exactly one compiler-folded $0{\cdot}0$ squaring (259 measured versus 260 predicted at the $t = 2$ minimum cell $n = 65$; Section 8.4).

(D) Self-cryptanalysis. We attack our own construction and disclose the results: a rate-specialization attack on the capacity-1 hash surface at $t \geq 3$ that improves the best known CICO cost by exactly $t-2$ bits and is priced into every parameter set (Section 5.6); a polynomial-time attack on additive PRNG extraction at $k = 1$, stated under a precise attack model (Proposition 5.19) with its mitigation; the small-field distinguishers of Section 5.9; the self-exclusion of $t = 2$ on 31-bit fields (Section 5.13); and an explicit maturity caveat (Remark 5.3).

1.4 Comparison with Existing ZK-friendly Permutations

Table 1 reports the algebraic efficiency and nonlinear-constraint cost of the main ZK-friendly permutations at matched 128-bit security, alongside Mambo at its three recommended 128-bit operating points. Mambo parameters are sized under the conservative cost model of Section 5.2, $D_{\text{eff}} = 2^{t(n-1)-\max(t-2,0)} \geq 2^{128}$ at the minimum round count—which prices in the self-disclosed specialization attack of Section 5.6—plus the recommended 25% round margin of Section 5.13. Competitor figures are their published margined recommendations, so the headline ratio ($240/165 \approx 1.45\times$ versus Poseidon at $t = 3$) is a margined-versus-margined comparison. Normalizing both designs to a uniform +25%-over-best-attack policy gives $\approx 1.38\times$ (Mambo 165 versus a re-derived Poseidon2 228 at BN254, $t = 3$, where the best 128-bit attack equality sits at 180 NL per the ToSC 2025(2) reevaluation [9]); the two published margins differ by only ≈ 4 bits. Competitor constraint and degree figures are from the cited specifications; the η values are our own computations (see table footnotes).

Lookup-based designs. A parallel line of work obtains fast native hashing from table lookups rather than from low-degree algebra. Reinforced Concrete [10] introduced the approach for 256-bit primes; Tip5 [12] combines power maps with lookups over the 64-bit Goldilocks field; Monolith [11] extends it to constant-time lookups over 31- and 64-bit fields; and Skyscraper [16] applies it to arbitrary large primes inside a Feistel structure (discussed in Section 3.5). These designs are the native-speed leaders in proof systems equipped with lookup arguments, and Table 1 deliberately excludes them: their nonlinear layers have no meaningful R1CS representation, so neither the nonlinear-constraint counts nor the ALFOM metric — which is defined per R1CS constraint — transfers. Mambo does not compete in that regime; it targets the purely algebraic setting (R1CS and low-degree AIR), where no lookup argument is assumed. We note that the lookup line has accumulated a nontrivial cryptanalytic record, including collision attacks on round-reduced Tip5, Tip4, and Monolith [19] and statistical-plus-algebraic attacks on split-and-lookup constructions up to full-round Skyscraper [17, 18]; the revised Skyscraper-v2 [16] adds rounds at both ends of the permutation in response. Within the algebraic line itself, Rescue-Prime Optimized [13] is the deployed descendant of Rescue-Prime [36, 40] and inherits its cost profile, and Arion [14], built from generalized triangular dynamical systems [15], is the published framework closest in spirit to Mambo's width-t triangular coupling (Section 7.4); full-round ArionHash instances were broken by the FreeLunch attack discussed below.

Table 1: Algebraic figure of merit and constraint costs at matched 128-bit security across existing ZK-friendly permutations. Mambo is shown at its three recommended 128-bit operating points under the conservative cost model (Section 5.2): two wider-state rows (matched state width and rate against Poseidon $t = 3$) and one narrow-state rate-1 row. The Sec. column reports permutation-CICO / sponge-collision security in bits.

Design	NL/P	NL/H	NL/PRNG	Deg	η^e	Sec.	Cryptanalytic record
Mambo $t=3$, $n=55^a$	165	165	164*	2	1.0	128/127^c	this work^f
Mambo $t=4$, $n=43^a$	172	172	164*	2	1.0	128/127^c	this work^f
Mambo $t=2$, $n=82$	164	328	164*	2	1.0	128/127^c	this work^f
Clotho ($\beta=2$) [1]	—	—	—	2	1.0	—	theoretical only
Poseidon $t=3$ [2, 3]†	240	240	∼240	5	≤ 0.71	128/128	7 yrs public analysis
Anemoi [27]	120	∼120	∼120	3	≤ 0.58	128/128	full-round breaks [7, 28]
Griffin [4]	96	96	∼96	5	N/A^d	128/128	full-round breaks [7, 5]
Rescue-Prime [40]	264	264	—	3	≤ 0.39	128/128	6 yrs, no full break
MiMC [41]	324	324	∼324	3	≤ 0.79	128/128	$\mathbb{F}_{2^m}$ broken [20]; prime field unbroken

*Public-output PRNG cost at the recommended $k = 82$ ($2k$ NL per output element; the analysis-equality minimum is $k = 65$). The in-circuit private-witness PRNG costs 2 NL per output at $k = 1$; $k = 1$ with public output is broken (footnote f). Competitor PRNG figures are hash-based DRBG costs over public outputs; the modes are not like-for-like and the column is indicative only. †Poseidon and the faster-linear-layer variant Poseidon2 have identical NL counts and algebraic degree at matched (t, rate); both [2, 3] apply to this row. [a]The $t = 2$ row's algebraic quantities (degree, Hilbert series, d_{reg}) are proven theorems; the $t \in \{3, 4\}$ rows rest on the empirical regularity measurements of Remark 5.7 and the proven specialization floor of Section 5.6 (Proposition 5.16). All three Mambo rows are the recommended (margin-inclusive) L1 cells of Table 6; the corresponding analysis-minimum cells are $n = 65/44/34$. [c]Mambo collision figures are for capacity 1 over BN254 ($p \approx 2^{254}$, birthday bound 2^{127}); competitors are quoted at their published 128-bit hash configurations. [d]Griffin's round mixes the power maps x^d, $x^{1/d}$, and the multiplicative Horst layer; its maximal coordinate degree is field-sized ($\approx p$) while the attack-relevant dual representation is low-degree, so no single η value is meaningful. [e]All η entries are our own computations from the cited specifications via Definition 2.2; they do not appear in the design papers. MiMC: 324 NL = 162 cube rounds at 2 NL each, degree $\min(3^{162}, \sim p)$, hence $\eta \leq \log_2 3/2 \approx 0.79$. [f]Self-disclosed cryptanalytic record: the $(t-2)$-bit capacity-1 rate-specialization attack (Section 5.6, priced into every Mambo row) and the polynomial-time seed recovery from $k = 1$ additive PRNG extraction (Proposition 5.19), which is why the PRNG column quotes the $k = 82$ public-output cost.

Gröbner-basis attacks and FreeLunch systems. The modern algebraic attack methodology for AO permutations was set out by Bariant, Bouvier, Leurent, and Perrin [6], who reduced the security of sponge instances to CICO and introduced trivariate/bypass modelings of the resulting systems. Bariant et al. [7] subsequently observed that for many designs the monomial order can be chosen so that the natural CICO encoding *already is* a Gröbner basis — a *FreeLunch system* — making the basis computation free and leaving the change of ordering (FGLM-type) and eliminant steps as the entire attack cost; this broke full-round instances of Anemoi, Arion, and Griffin. Two follow-ups sharpen the resolution step: an improved resultant-based solver [5] and the CheapLunch extension [8], which carries the FreeLunch approach beyond single-output CICO via weighted orders and tightened ideal-degree bounds. Mambo's position with respect to this literature is stated explicitly rather than left for cryptanalysts to discover: the CICO system of $P_{\varphi,n}$ is a FreeLunch-type system by construction — Appendix G exhibits its degrevlex Gröbner basis in closed form — so we assign *zero* cost to the basis computation, and the security accounting in Section 5 rests entirely on the eliminant degree, which Theorem 4.2 and the closed-form Hilbert series make exact. The Macaulay-bound figures reported for the original generators serve only to rule out blind F_4/F_5 computation on the raw system [32, 31]; they are not part of the security claim.

1.5 Paper Organization

Section 2 establishes notation, the R1CS cost model, and the ALFOMs metric. Section 3 defines the Mambo map, its R1CS representation, and the round constant derivation. Section 4

contains the main algebraic result. Section 5 provides security analysis. Section 6 covers sponge indifferentiability. Section 7 describes the three instantiations. Section 8 covers implementations and benchmarks. Section 9 concludes.

2 Preliminaries

2.1 Notation

Throughout, p denotes an odd prime and $\mathbb{F}_p$ the prime field of order p. The state space is $\mathbb{F}_p^2$. For a polynomial $f \in \mathbb{F}_p[\theta_0, \psi_0]$, $\deg(f)$ denotes its total degree. Unless stated otherwise, all arithmetic is modular. We write $[n]$ for the set $\{1, 2, \ldots, n\}$ and λ for the target security level; the parameter tiers instantiate $\lambda \in \{128, 192, 256\}$, with $\lambda = 128$ the default when unstated. The multiplier φ is an integer with $2 \leq \varphi \leq p - 2$, identified with its image in $\mathbb{F}_p$.

2.2 Rank-1 Constraint Systems (R1CS)

Definition 2.1 (R1CS)**.** A Rank-1 Constraint System over $\mathbb{F}_p$ consists of matrices $A, B, C \in \mathbb{F}_p^{m \times n}$ and a satisfying assignment $\mathbf{w} \in \mathbb{F}_p^n$ such that $(A\mathbf{w}) \circ (B\mathbf{w}) = C\mathbf{w}$, where $\circ$ denotes the Hadamard (entrywise) product. A constraint is *nonlinear* if both $A\mathbf{w}$ and $B\mathbf{w}$ in the corresponding row depend on non-constant entries of $\mathbf{w}$.

The R1CS cost of a computation is the number of nonlinear constraints required to express it, denoted C. A squaring x^2 requires exactly one nonlinear constraint.

2.3 The ALFOMs Metric

Definition 2.2 (ALFOM [1])**.** For a permutation $\pi \colon \mathbb{F}_p^t \to \mathbb{F}_p^t$ expressed as an n-fold composition $\pi = f^n$ with R1CS cost C, the Algebraic Figure of Merit is

$$\eta = \frac{\log_2 D}{C},$$

where $D = \max_j \deg(\pi_j)$ is the maximum total degree of any output coordinate.

Proposition 2.3 (R1CS upper bound [1, Prop. 3.2])**.** *For any computation expressed in R1CS over $\mathbb{F}_p$ with C nonlinear constraints, $D \leq 2^C$. Consequently, $\eta \leq 1.0$.*

2.4 The Constrained-Input Constrained-Output (CICO) Problem

Definition 2.4 (CICO [27])**.** Let $P \colon \mathbb{F}_p^t \to \mathbb{F}_p^t$ be a permutation. The Constrained-Input Constrained-Output problem for P with input constraints $\mathcal{C}_{\text{in}}$ and output constraints $\mathcal{C}_{\text{out}}$ is: find $\mathbf{x} \in \mathbb{F}_p^t$ satisfying $\mathcal{C}_{\text{in}}$ such that $P(\mathbf{x})$ satisfies $\mathcal{C}_{\text{out}}$.

Definition 2.5 (CICO security, concrete)**.** Fix a cost model $\mathcal{M}$ assigning to each algorithm a cost in field operations. P is $(\lambda, \mathcal{M})$-CICO-secure if every algorithm solving the CICO problem for P with constant success probability has $\mathcal{M}$-cost at least 2^λ. Throughout this paper $\mathcal{M}$ counts field operations, with attacker memory priced separately where stated (Section 5.2). Since P is a fixed public permutation, asymptotic (probabilistic polynomial-time) language is not meaningful for it and is not used.

The logical relation between CICO and hash security runs in one proven direction: a hash adversary is already a CICO solver, so CICO hardness is a *necessary* condition.

Lemma 2.6 (CICO hardness is necessary)**.** *Let H be the Mambo sponge hash built from $P_{\varphi,n}$ with rate r and capacity c over $\mathbb{F}_p$. An adversary that finds a single-block preimage of a target digest τ at cost T yields a CICO solver for $P_{\varphi,n}$ (input constraint: the c capacity cells are zero; output constraint: the first rate cell equals τ) at cost $T + O(1)$.*

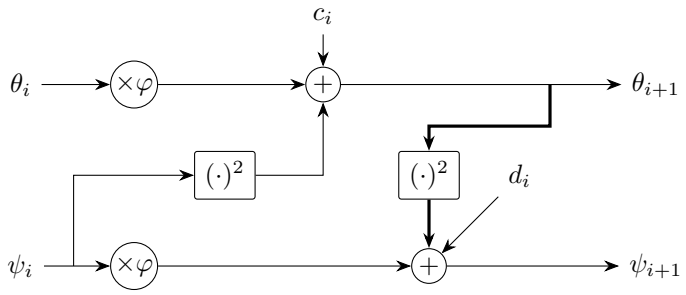


Figure 1: Dataflow of one Mambo round (Definition 3.1). Each round costs exactly two nonlinear constraints (the two squarings). The thick wire is the sequential coupling: the second squaring consumes the *updated* value θ_{i+1}, tapped after the first adder, not the input θ_i. The width-t generalization (Section 7.4) chains t such squarings, each feeding the next update in the same way.

Proof. For a single-block message $\mathbf{m} \in \mathbb{F}_p^r$, the sponge computes $H(\mathbf{m})$ as the first rate cell of $P_{\varphi,n}(\mathbf{m}, 0^c)$. A preimage $\mathbf{m}$ of τ is therefore, verbatim, a solution of the stated CICO instance. □

CICO hardness is not known to be *sufficient* for hash security; sufficiency of the mode bounds is obtained in the ideal-permutation model (Section 6), calibrated by the design claim of Section 5.1. The paper's parameter sizing enforces the necessary condition quantitatively under the explicit cost model of Section 5.2.

3 The Mambo Map

3.1 Definition

Definition 3.1 (Mambo map)**.** Let p be an odd prime and let φ be an integer with $2 \le \varphi \le p-2$, identified with its image in $\mathbb{F}_p$ (in particular $\varphi \not\equiv 0, \pm 1 \bmod p$; cf. Remark 5.20). For round i, let $(c_i, d_i) \in \mathbb{F}_p^2$ be the round constants (Section 3.4). The Mambo map $M_{\varphi,i} \colon \mathbb{F}_p^2 \to \mathbb{F}_p^2$ at round i is defined by $M_{\varphi,i}(\theta, \psi) = (\theta', \psi')$, where

$$\theta' = \varphi \cdot \theta + \psi^2 + c_i \,, \tag{1}$$

$$\psi' = \varphi \cdot \psi + (\theta')^2 + d_i \,. \tag{2}$$

The sequential coupling is the defining structural feature: ψ' depends on the *updated* value θ', not on the original θ. The round constants (c_i, d_i) are distinct for each round, eliminating round symmetry.

Remark 3.2 (Invertibility)**.** For $\varphi \neq 0$, each round map $M_{\varphi,i}$ is a bijection on $\mathbb{F}_p^2$. Given (θ', ψ') and the round constants (c_i, d_i), one recovers $\psi = \varphi^{-1}(\psi' - (\theta')^2 - d_i)$ and $\theta = \varphi^{-1}(\theta' - \psi^2 - c_i)$. The Jacobian determinant is $\varphi^2 \neq 0$.

Definition 3.3 (n-fold Mambo permutation)**.** $P_{\varphi,n} = M_{\varphi,n-1} \circ \cdots \circ M_{\varphi,1} \circ M_{\varphi,0}$. We write $(\theta_n, \psi_n) = P_{\varphi,n}(\theta_0, \psi_0)$. Default parameters: $n = 82$, $\varphi = 7$ (the recommended 128-bit cell at $t = 2$; the analysis-minimum cell is $n = 65$; see Table 6 for all tiers).

State width and the t-branch family. The definition above is the standard state-width-2 instance ($t = 2$) of a triangular-coupling family indexed by the state width $t \ge 2$. The generalization $M_{\varphi,i} \colon \mathbb{F}_p^t \to \mathbb{F}_p^t$, in which t sequential squarings each feed into the next, is specified in Section 7.4. Every member of the family preserves ALFOM $\eta = 1.0$ (Proposition 7.6). State width is an instantiation parameter: selected jointly with the base field $\mathbb{F}_p$ and the number of rounds n to meet a target security level (Section 5.13, Table 8).

3.2 R1CS Representation

Each application of $M_{\varphi,i}$ requires exactly two nonlinear constraints:

1. $\psi_i \cdot \psi_i = u_i$ (computes $u_i = \psi_i^2$), then $\theta_{i+1} = \varphi \cdot \theta_i + u_i + c_i$ is linear.
2. $\theta_{i+1} \cdot \theta_{i+1} = v_i$ (computes $v_i = \theta_{i+1}^2$), then $\psi_{i+1} = \varphi \cdot \psi_i + v_i + d_i$ is linear.

(The auxiliary witness symbols u_i, v_i are chosen to avoid collision with the state width t.) The round constants c_i and d_i enter only the linear constraints and therefore do not contribute any additional nonlinear constraints.

Proposition 3.4 (R1CS cost)**.** *The n-fold Mambo permutation $P_{\varphi,n}$ has R1CS cost $C = 2n$ nonlinear constraints.*

MPC online cost. In the standard collaborative-SNARK / secret-shared-witness setting, where each non-linear gate over a secret-shared field element costs one online multiplication round and linear operations are local, the per-permutation online MPC cost of $P_{\varphi,n}$ is exactly $2n$ multiplication rounds (164 online multiplications at the recommended $n = 82$).

3.3 Design Rationale

Three design choices are critical:

Coupled quadratics. A single iterated squaring $\theta_{i+1} = \theta_i^2 + c$ achieves degree 2^n at cost n, giving $\eta = 1.0$. However, this univariate map admits efficient algebraic attacks. Coupling two variables forces each squaring to amplify the degree of both components.

Sequential dependency. In a parallel coupling (using the old θ), the degree growth is $d(n) = 2^n$, yielding $\eta = 0.5$. The sequential coupling—ψ' depending on θ'—doubles the effective degree per step from 2^n to 4^n at no additional constraint cost, pushing η from 0.5 to 1.0.

Field independence. The map uses only field multiplication, field addition, the constant φ, and per-round constants (c_i, d_i). No S-boxes, no MDS matrices. The same definition works on any prime field with $p \geq 2^{31}$.

3.4 Round Constants

The Mambo permutation uses distinct round constants $(c_i, d_i) \in \mathbb{F}_p^2$ for each round $i \in \{0, \ldots, n-1\}$. These constants serve three purposes:

1. **Fixed-point elimination.** Without round constants, $M_\varphi(0,0) = (0,0)$: the origin is a fixed point. Round constants ensure that the all-zero state is mapped to a nonzero output at every round.
2. **Invariant subspace destruction.** Without constants, certain diagonal inputs (α, α) map to diagonal outputs (β, β), preserving a one-dimensional subspace. Distinct per-round constants break this symmetry.
3. **Slide attack prevention.** Round constants destroy the shift invariance that otherwise permits slide attacks (Section 5.8).

Derivation. The $2n$ constants are derived deterministically from SHA-256 in counter mode with the seed string `"Mambo_round_constants"`. Specifically, for $j = 0, 1, \ldots, 2n-1$:

$$h_j = \text{SHA-256}\big(\texttt{"Mambo_round_constants"} \,\|\, \text{BE}_{32}(j)\big)\,,$$

where $\text{BE}_{32}(j)$ is the 32-bit big-endian encoding of j. Each h_j is interpreted as a 256-bit unsigned integer and reduced modulo p; if the result is zero it is replaced with 1. The round constants are then $c_i = h_i \bmod p$ for $i \in \{0, \ldots, n-1\}$ and $d_i = h_{n+i} \bmod p$ for $i \in \{0, \ldots, n-1\}$.

Impact on algebraic degree. The round constants are degree-0 terms in $\mathbb{F}_p[\theta_0, \psi_0]$ and therefore do not affect the leading monomial at any step. The degree recurrence (Lemma 4.1) and the ALFOM of 1.0 (Corollary 4.3) are preserved exactly.

Impact on R1CS cost. In the R1CS representation, round constants enter only the linear constraints (additions to θ_{i+1} and ψ_{i+1}). Adding a constant is a free linear combination in R1CS: the nonlinear constraint count remains $C = 2n$.

Domain separation across state widths. Wider-state variants ($t \geq 3$) use a distinct seed string so that round constants for the narrow ($t = 2$) primitive cannot be re-used as constants for a wider primitive (or vice versa). The $t = 2$ seed is `"Mambo_round_constants"` as above; the $t = 4$ seed is `"Mambo_t4_round_constants"`. Higher widths follow the same `Mambo_t`$\langle t \rangle$`_round_constants` pattern. This enforces cross-variant domain separation at the primitive level so that a commitment produced under one state width cannot collide with a commitment produced under a different width even at identical inputs. Reference implementations verify consistency against these seeds at build time.

3.5 Relation to Feistel Networks and Existing Quadratic Structures

The Feistel reading. One Mambo round (Definition 3.1) can be read as two consecutive unbalanced Feistel steps with round function $f(x) = x^2$ and branch multiplier φ: the first step updates θ in place as $\theta \mapsto \varphi\theta + f(\psi) + c_i$, the second updates ψ in place as $\psi \mapsto \varphi\psi + f(\theta') + d_i$. Up to the φ-scaling of the recycled branch and the absence of a branch swap (the update is written in place; the swap is absorbed into the alternation of the two steps), this is the oldest skeleton in symmetric cryptography, instantiated with the lowest-degree nonlinearity a prime field offers. We state this plainly because the skeleton is also current: the algebraic rounds of Skyscraper [16] are exactly such steps, $(x_L, x_R) \mapsto (x_R + x_L^2/\sigma + \gamma_i,\ x_L)$ with σ a Montgomery constant, and Feistel–MiMC [41, 6] is the same skeleton with $f(x) = x^3$. Mambo's contribution is not the skeleton; it is the exact algebraic characterization that this particular instantiation of the skeleton admits, which we make precise below.

Composition versus parallel application. The Feistel reading makes the role of the sequential coupling exact. Because ψ' consumes the *updated* value θ' within the same round, the $2n$ squarings of $P_{\varphi,n}$ form a single composition chain: every squaring acts on a state component carrying the maximal degree produced so far, and the degree doubles at every nonlinear constraint, giving total degree 4^n at cost $2n$ (Theorem 4.2). The structural alternative — applying the two quadratics in parallel to the *old* state, $\theta' = \varphi\theta + \psi^2 + c_i$, $\psi' = \varphi\psi + \theta^2 + d_i$ — is a two-branch SPN rather than a Feistel network; its degree growth is 2^n at the same cost, i.e. $\eta = 0.5$. Squaring is the unique round function for which this composition extracts the full R1CS bound: a power map x^α costs $\lceil \log_2 \alpha \rceil$ constraints but multiplies the degree by α, with equality of the two only at $\alpha = 2$. The φ-scaling of the recycled branch keeps each step invertible without a swap (Remark 3.2), breaks the branch-interchange and slide symmetries of an unscaled Feistel chain, and is the hypothesis ($\varphi \geq 2$) under which the no-cancellation argument of Theorem 4.2 holds for every prime $p > 2$.

Delineation from existing quadratic and Feistel-type AO designs. MiMC [41] iterates the univariate cube: each round costs two constraints and multiplies the degree by 3, so n rounds give degree 3^n at cost $2n$, i.e. $\eta = \log_2 3/2 \approx 0.79$; the same accounting applies to its Feistel variant. The open Flystel [27] (Anemoi's butterfly) deliberately splits the two directions: the verification relation is quadratic while one evaluation direction involves a high-degree power map $x^{1/\alpha}$, so the security argument rests on the degree asymmetry between CCZ-equivalent

representations rather than on proven degree growth of the iterated map — and the published algebraic attacks on Anemoi target precisely the low-degree representation [28, 7]. Horst [4], Griffin's nonlinear mode, replaces Feistel's additive combination $L + f(R)$ by the multiplicative $L \cdot G(R)$: degrees of distinct branches *add*, which is efficient but does not double the maximal degree per constraint, and the resulting systems proved vulnerable to dedicated solvers [7, 5]. Skyscraper [16] is the closest published relative at the level of the round step, but the design intent is orthogonal: of its ten Feistel rounds only six are squarings (used for cheap native nonlinearity), the remaining four are lookup-based "Bars" rounds carrying the statistical security, and no exact algebraic claims are made for the composition. Its cryptanalytic record — a practical rebound distinguisher on the full permutation [17] and full-round attacks combining cheap polynomial solving across the quadratic rounds with statistical properties of the lookup rounds [18], answered by the revised Skyscraper-v2 [16] — illustrates that a *short* quadratic Feistel chain is algebraically cheap to traverse. Mambo draws the opposite conclusion from the same skeleton: it uses *only* the quadratic Feistel step, iterates it $2n$ times with n sized so that the eliminant degree meets the security target, and prices the entire attack surface by the proven growth of the algebraic invariants rather than by interleaved non-algebraic components.

Prior exact-degree results, and what Theorem 4.2 adds. The degree growth of iterated low-degree maps has been studied most intensively for MiMC. Eichlseder et al. [20] proved a nontrivial *upper* bound on the growth of the $\mathbb{F}_2$-algebraic degree of MiMC-like iterates over binary fields, yielding a higher-order distinguisher on full MiMC; Bouvier, Canteaut, and Perrin [21] subsequently determined the *exact* algebraic degree of iterated x^3 for large round ranges, exhibiting plateaus where the degree does not grow at all for consecutive rounds; and Cid et al. [22] showed that the linear layer can degrade the growth from exponential to linear in SP-networks. Three features distinguish those results from Theorem 4.2. First, they concern the $\mathbb{F}_2$-algebraic degree over binary extension fields — the invariant relevant to higher-order differential attacks there — whereas over a prime field the relevant invariant is the total degree of the iterated polynomials, which is the one Mambo characterizes. Second, they are obtained a posteriori, by intricate exponent tracing applied to a fixed published design, and are exact only for specific maps and round ranges; the plateau phenomenon they uncover shows that maximal growth cannot be assumed in general. Third, none of them yields the ideal-theoretic data that algebraic attack costs are actually priced in. For Mambo the situation is inverted by construction: the sequential quadratic coupling forces the degree recurrence to meet its upper branch with equality at every step, for every $p > 2$ and every $\varphi \geq 2$, and the same triangular structure makes the CICO ideal explicit enough that its Gröbner basis, Hilbert series, regularity degree $d_{\text{reg}} = 2n$, and eliminant degree are closed-form theorems (Section 5.5, Appendix G) rather than empirical estimates. The Feistel lineage is thus a feature of the design: the skeleton is what makes the construction fully analyzable, and the analysis — including the resulting attack costs — is carried out in this paper rather than left to future cryptanalysis.

4 Degree Growth Analysis

This section contains the main algebraic result.

4.1 Degree Recurrence

Let $d_\theta(n)$ and $d_\psi(n)$ denote the total degree of θ_n and ψ_n as polynomials in $\mathbb{F}_p[\theta_0, \psi_0]$.

Lemma 4.1 (Degree recurrence)**.**

$$d_\theta(n) = \max\big(d_\theta(n-1),\ 2\,d_\psi(n-1)\big)\,, \tag{3}$$

$$d_\psi(n) = 2\,d_\theta(n)\,. \tag{4}$$

Proof. From (1): $\theta_n = \varphi \cdot \theta_{n-1} + \psi_{n-1}^2 + c_{n-1}$. The term $\varphi \cdot \theta_{n-1}$ has degree $d_\theta(n-1)$, ψ_{n-1}^2 has degree $2\,d_\psi(n-1)$, and the round constant c_{n-1} has degree 0. Their sum has degree $\max(d_\theta(n-1), 2\,d_\psi(n-1))$, provided the leading-degree terms do not cancel (see Theorem 4.2).

From (2): $\psi_n = \varphi \cdot \psi_{n-1} + \theta_n^2 + d_{n-1}$. Since $2\,d_\theta(n) > d_\psi(n-1)$ for $n \geq 1$, the squared term dominates and the degree-0 constant d_{n-1} is absorbed: $d_\psi(n) = 2\,d_\theta(n)$. □

4.2 Main Result: Exact Degree Growth

Theorem 4.2 (Degree growth)**.** *For all $n \geq 1$,*

$$d_\psi(n) = 4^n\,, \qquad d_\theta(n) = \tfrac{4^n}{2} = 2 \cdot 4^{n-1}\,.$$

Moreover, the leading-degree homogeneous components of θ_n and ψ_n are nonzero; no cancellation occurs at the top degree.

Proof. We proceed by strong induction on n.

Base case ($n = 1$). Starting from (θ_0, ψ_0), both of degree 1:

$$\begin{aligned}\theta_1 &= \varphi \cdot \theta_0 + \psi_0^2 + c_0\,,\\ \psi_1 &= \varphi \cdot \psi_0 + (\varphi \cdot \theta_0 + \psi_0^2 + c_0)^2 + d_0\,.\end{aligned}$$

Expanding, the leading term of ψ_1 is ψ_0^4; the constants c_0, d_0 contribute only degree-0 and lower-degree cross terms. We have $d_\theta(1) = 2$ and $d_\psi(1) = 4 = 4^1$.

Inductive step. Assume for all $k \leq n-1$: $d_\psi(k) = 4^k$ and $d_\theta(k) = 4^k/2$. By Lemma 4.1:

$$\begin{aligned}d_\theta(n) &= \max\Big(\tfrac{4^{n-1}}{2},\ 2 \cdot 4^{n-1}\Big) = 2 \cdot 4^{n-1} = \tfrac{4^n}{2}\,,\\ d_\psi(n) &= 2\,d_\theta(n) = 4^n\,.\end{aligned}$$

Non-cancellation argument. Consider $\theta_n = \varphi \cdot \theta_{n-1} + \psi_{n-1}^2 + c_{n-1}$. The degree of $\varphi \cdot \theta_{n-1}$ is $4^{n-1}/2$, ψ_{n-1}^2 has degree $2 \cdot 4^{n-1}$, and c_{n-1} has degree 0. Since $2 \cdot 4^{n-1} > 4^{n-1}/2$, the leading terms come *exclusively* from ψ_{n-1}^2—the square of the leading homogeneous component of ψ_{n-1}, which is nonzero by induction (the square of a nonzero polynomial over a field of characteristic $p > 2$ is nonzero). The constant c_{n-1} cannot affect the leading monomial.

For $\psi_n = \varphi \cdot \psi_{n-1} + \theta_n^2 + d_{n-1}$: the degree of $\varphi \cdot \psi_{n-1}$ is 4^{n-1}, while θ_n^2 has degree 4^n and d_{n-1} has degree 0. Since $4^n > 4^{n-1}$, the leading terms come exclusively from θ_n^2.

At no step do leading-degree terms from different sources combine, so no cancellation can occur. □

The non-cancellation argument is stronger than typical for iterated polynomial maps: the strict degree separation between the linear term and the quadratic term ensures single-source dominance at every step—a structural consequence of the sequential coupling.

4.3 Corollary: ALFOM = 1.0

Corollary 4.3 (ALFOM = 1.0)**.** *The Mambo permutation $P_{\varphi,n}$ has*

$$\eta = \frac{\log_2(d_\psi(n))}{C} = \frac{\log_2(4^n)}{2n} = \frac{2n}{2n} = 1.0\,.$$

This holds for all $n \geq 1$, independently of the field $\mathbb{F}_p$ and the parameter φ.

The ALFOM of 1.0 is an unconditional algebraic fact: it does not depend on any security assumption or heuristic, and round constants, being degree-0 elements of $\mathbb{F}_p[\theta_0, \psi_0]$, cannot alter the leading monomial or the degree count at any step. The metric should also be deflated correctly: η measures algebraic degree per nonlinear constraint, not attack cost, and the trivial

iterated squaring chain $x \mapsto x^2 + c$ attains $\eta = 1.0$ while being cryptographically worthless (univariate and invertible by repeated square-root extraction; Section 3.3). Hence $\eta = 1.0$ is necessary for constraint-optimal degree growth but not sufficient for security, and the security content of this paper lives in the exact characterization and the analysis of Section 5, not in the metric.

4.4 Comparison with Clotho

Clotho ($\beta = 2$), introduced in [1], also achieves $\eta = 1.0$. The two constructions reach the bound by different mechanisms: Clotho uses a power map $x \mapsto x^\beta$ with a carefully chosen linear layer; Mambo uses two coupled quadratic equations with sequential dependency. Whether all $\eta = 1.0$ R1CS constructions share a common algebraic characterization is an open question.

4.5 Non-Telescoping Property

Throughout this subsection fix an odd prime p, a parameter $\varphi \in \mathbb{F}_p$, and round constants $c_0, d_0, \ldots, c_{n-1}, d_{n-1} \in \mathbb{F}_p$, and write $\Theta_i, \Psi_i \in \mathbb{F}_p[\theta_0, \psi_0]$ for the round-i iterates ($\Theta_0 = \theta_0$, $\Psi_0 = \psi_0$). The sponge-CICO specialisation sets $\psi_0 = 0$ and fixes the rate output $\theta_n = \tau$; we write $F_i(a) := \Theta_i(a, 0)$, $G_i(a) := \Psi_i(a, 0)$, and $f_n(a) := F_n(a) - \tau$.

For integer weights $w_x, w_y \geq 1$ let $\deg_w$ denote the weighted degree on $\mathbb{F}_p[x, y]$ with $\deg_w x = w_x$, $\deg_w y = w_y$, and call the sum of the terms of maximal weighted degree the *top w-component*. Top w-components are multiplicative over a field. Two lemmas isolate the structure that prevents telescoping.

Lemma 4.4 (Single-monomial dominance under substitution)**.** *Let K be a field and let $P \in K[x, y]$ have, for weights (w_x, w_y), top w-component equal to a single monomial λy^J with $\lambda \neq 0$ and $J \geq 1$. Let $F, G \in K[a]$ satisfy $\deg F \leq d\, w_x$ and $\deg G = d\, w_y$ exactly, for some integer $d \geq 1$. Then*

$$\deg_a P(F, G) \;=\; d\, J\, w_y, \qquad \mathrm{lc}\big(P(F, G)\big) \;=\; \lambda\, \mathrm{lc}(G)^J.$$

The same statement holds with the roles of x and y exchanged.

Proof. A monomial $x^i y^j$ of P contributes degree at most $i \deg F + j \deg G \leq d\,(i w_x + j w_y)$ to $P(F, G)$. Every monomial of P other than y^J has weighted degree at most $J w_y - 1$ (a monomial of weighted degree $J w_y$ would belong to the top w-component), hence contributes degree at most $d(J w_y - 1) < d J w_y$. The monomial λy^J contributes $\lambda\, \mathrm{lc}(G)^J a^{d J w_y}$ plus lower-order terms, and $\lambda\, \mathrm{lc}(G)^J \neq 0$ over a field. □

Lemma 4.5 (Weighted structure of partial compositions)**.** *Fix $r \geq 1$ and constants $c'_0, d'_0, \ldots, c'_{r-1}, d'_{r-1} \in \mathbb{F}_p$, and let $\Theta'_k, \Psi'_k \in \mathbb{F}_p[x, y]$ be the k-round iterates from a free state (x, y) under these constants. For each of the weightings $(w_x, w_y) \in \{(1, 1), (1, 2)\}$ and every $1 \leq k \leq r$, the top w-components of Θ'_k and Ψ'_k are the single monomials*

$$\mathrm{top}_w\, \Theta'_k \;=\; y^{2 \cdot 4^{k-1}}, \qquad \mathrm{top}_w\, \Psi'_k \;=\; y^{4^k},$$

each with coefficient 1. The analogous statement holds for a chain entered at the ψ-update (first map $(u, v) \mapsto \varphi v + u^2 + d'$, then full rounds) with weights $(w_u, w_v) = (2, 1)$: the dominant monomials are the corresponding pure powers of u, with coefficient 1. All statements hold in every characteristic, for every φ, and for arbitrary constants.

Proof. Induction on k; we write the case $(w_x, w_y) = (1, 2)$, the other two being the same computation. For $k = 1$: $\Theta'_1 = \varphi x + y^2 + c'_0$ has summands of w-degree $1, 4, 0$, so $\mathrm{top}_w\, \Theta'_1 = y^2$; $\Psi'_1 = \varphi y + (\Theta'_1)^2 + d'_0$, and $\mathrm{top}_w (\Theta'_1)^2 = (y^2)^2 = y^4$ has w-degree $8 > 2$, so $\mathrm{top}_w\, \Psi'_1 = y^4$. For the step: in $\Theta'_{k+1} = \varphi \Theta'_k + (\Psi'_k)^2 + c'_k$ the square has top w-component $(y^{4^k})^2 = y^{2 \cdot 4^k}$ of

w-degree 4^{k+1}, strictly exceeding $\deg_w \Theta'_k = 4^k$; in $\Psi'_{k+1} = \varphi\Psi'_k + (\Theta'_{k+1})^2 + d'_k$ the square has top w-component $y^{4^{k+1}}$ of w-degree $2 \cdot 4^{k+1} > 2 \cdot 4^k$. At every step the top w-component comes from a single source and is the square of a single monomial with coefficient 1. □

Proposition 4.6 (Non-telescoping)**.** *Let $S = \mathbb{F}_p[\theta_0, \psi_0, \theta_1, \psi_1, \ldots, \theta_n, \psi_n]$ and let $J_n \subset S$ be the ideal generated by the $2n$ round relations $\theta_i - \varphi\theta_{i-1} - \psi_{i-1}^2 - c_{i-1}$ and $\psi_i - \varphi\psi_{i-1} - \theta_i^2 - d_{i-1}$, $1 \le i \le n$.*

(i) ***Free-input elimination floor.*** *$\big(J_n + (\psi_n - \tau)\big) \cap \mathbb{F}_p[\theta_0, \psi_0] = \big(\Psi_n(\theta_0, \psi_0) - \tau\big)$ is principal, generated by a polynomial of total degree exactly 4^n whose leading form is the single monomial $\psi_0^{4^n}$. Hence every nonzero polynomial in (θ_0, ψ_0) alone lying in the ideal of the round system has total degree at least 4^n: the intermediate states cannot be algebraically eliminated below degree 4^n. This holds over every field, for every φ, and for every choice of round constants.*

(ii) ***CICO elimination floor.*** *Assume $\varphi \neq 0$ in $\mathbb{F}_p$. Then $\big(J_n + (\psi_0,\, \theta_n - \tau)\big) \cap \mathbb{F}_p[\theta_0] = (f_n)$ with $\deg f_n = 4^{n-1} = D_I$ and $\mathrm{lc}(f_n) = \varphi^{4^{n-1}} \neq 0$. Hence every nonzero univariate consequence of the sponge-CICO system in the rate input—the output of any elimination procedure whatsoever, including iterated resultants and Gröbner bases, since both produce elements of the ideal—has degree at least D_I.*

(iii) ***Variety floor.*** *For every fixed round-constant tuple (with $\varphi \neq 0$), f_n is squarefree for all but at most $D_I - 1$ targets $\tau \in \mathbb{F}_p$. For such τ the CICO system has exactly D_I distinct solutions over $\overline{\mathbb{F}_p}$, and consequently* any *polynomial reformulation of the system—in particular any meet-in-the-middle organisation, one- or two-sided, at any cut point—admits no nonzero $g \in \overline{\mathbb{F}_p}[\theta_0]$ of degree $< D_I$ whose root set contains all solutions.*

(iv) ***Split exactness.*** *For every cut of the $2n$-equation chain—after the ψ-update of round m (middle state (θ_m, ψ_m), $1 \le m \le n-1$) or after the θ-update of round m (middle state (θ_m, ψ_{m-1}), $2 \le m \le n-1$)—the meeting polynomial obtained by substituting the forward half-iterates into the backward half-constraint equals f_n identically: its degree is exactly 4^{n-1} with leading coefficient $\varphi^{4^{n-1}}$, independent of the cut. No choice of meeting point lowers the degree at which the two halves meet.*

Proof. (i) Order the variables $\theta_0 \prec \psi_0 \prec \theta_1 \prec \psi_1 \prec \cdots$; each round relation rewrites one variable as a polynomial in strictly earlier ones. Let $\epsilon : S \to \mathbb{F}_p[\theta_0, \psi_0]$ be the evaluation homomorphism fixing θ_0, ψ_0 and sending $\theta_i \mapsto \Theta_i$, $\psi_i \mapsto \Psi_i$. By induction along the order, $\theta_i \equiv \Theta_i$ and $\psi_i \equiv \Psi_i \pmod{J_n}$, hence $g \equiv \epsilon(g) \pmod{J_n}$ for every $g \in S$ (substituting congruent elements into a polynomial preserves congruence). Write $I = J_n + (\psi_n - \tau)$. Since ϵ kills every round relation and $\epsilon(\psi_n - \tau) = \Psi_n - \tau$, and ϵ is a ring homomorphism into $\mathbb{F}_p[\theta_0, \psi_0]$, we get $\epsilon(I) \subseteq (\Psi_n - \tau)$. If $g \in I \cap \mathbb{F}_p[\theta_0, \psi_0]$ then $g = \epsilon(g) \in (\Psi_n - \tau)$; conversely $\Psi_n - \tau = (\psi_n - \tau) - (\psi_n - \Psi_n) \in I$. The degree 4^n and the monic leading form $\psi_0^{4^n}$ are Theorem 4.2 together with the $(1,1)$-weighted case of Lemma 4.5 at $k = n$. Degrees add over a domain, so every nonzero multiple of $\Psi_n - \tau$ in $\mathbb{F}_p[\theta_0, \psi_0]$ has total degree at least 4^n.

(ii) Identical, with the evaluation $\epsilon^\circ : S \to \mathbb{F}_p[\theta_0]$ sending $\psi_0 \mapsto 0$, $\theta_i \mapsto F_i$, $\psi_i \mapsto G_i$; the elimination ideal is (f_n). Degree and leading coefficient follow from the specialised recursion $F_1 = \varphi\theta_0 + c_0$, $G_1 = F_1^2 + d_0$, and, for $i \ge 1$, $F_{i+1} = \varphi F_i + G_i^2 + c_i$, $G_{i+1} = \varphi G_i + F_{i+1}^2 + d_i$: since $\deg G_i^2 = 4^i > 4^{i-1} = \deg F_i$ and $\deg F_{i+1}^2 = 2 \cdot 4^i > 2 \cdot 4^{i-1} = \deg G_i$, the top term comes from a single source at every step, giving by induction

$$\deg F_i = 4^{i-1}, \quad \mathrm{lc}\, F_i = \varphi^{4^{i-1}}, \quad \deg G_i = 2 \cdot 4^{i-1}, \quad \mathrm{lc}\, G_i = \varphi^{2 \cdot 4^{i-1}},$$

all nonzero because $\varphi \neq 0$. (If $p \mid \varphi$ the system degenerates—F_1 is constant—so the hypothesis is sharp.)

(iii) Write $f_n = h - \tau$ with $h := F_n$. Since p is odd and $D_I = 2^{2(n-1)}$, we have $p \nmid D_I$; by the discriminant formula for $h(x) - c$ already used in the proof of Theorem 5.12 (Gelfand–Kapranov–Zelevinsky [39], Ch. 12), $\mathrm{disc}(f_n)$ is a polynomial in τ of degree exactly $D_I - 1$ with

leading coefficient $\pm D_I^{D_I}\varphi^{D_I(D_I-1)} \neq 0$, so it vanishes for at most $D_I - 1$ values of τ. For all other τ, f_n has D_I distinct roots in $\overline{\mathbb{F}_p}$, each a CICO solution, and any g vanishing on all of them is a multiple of f_n up to a scalar, hence $\deg g \geq D_I$. This argument references only the solution set, not the algebraic form in which an attacker carries the system; it therefore applies to every sound reorganisation of the equations, including meet-in-the-middle systems built from forward iterates, from inverse-round iterates entering at the output side, or both.

(iv) Cut after the ψ-update of round m, middle state $(x, y) = (\theta_m, \psi_m)$. The backward half-constraint is $P_{n-m}(x, y) := \Theta'_{n-m}(x, y) - \tau$, where Θ'_{n-m} is the $(n-m)$-round iterate of Lemma 4.5 under the constants of rounds $m+1, \ldots, n$; the additive constant $-\tau$ does not change the top w-component, which is $y^{2\cdot 4^{n-m-1}}$ with coefficient 1 for the weights $(1, 2)$. The forward half-iterates satisfy $\deg F_m = 4^{m-1} = d \cdot 1$ and $\deg G_m = 2 \cdot 4^{m-1} = d \cdot 2$ exactly, with $d = 4^{m-1}$ and $\mathrm{lc}(G_m) = \varphi^{2\cdot 4^{m-1}} \neq 0$, by the recursion in (ii). Lemma 4.4 gives

$$\deg_a P_{n-m}(F_m, G_m) = 4^{m-1} \cdot 2 \cdot 4^{n-m-1} \cdot 2 = 4^{n-1}, \qquad \mathrm{lc} = \left(\varphi^{2\cdot 4^{m-1}}\right)^{2\cdot 4^{n-m-1}} = \varphi^{4^{n-1}},$$

and $P_{n-m}(F_m, G_m) = f_n$ identically by associativity of composition. For a cut after the θ-update of round m ($m \geq 2$), the middle state is $(u, v) = (\theta_m, \psi_{m-1})$ with $\deg F_m = 4^{m-1} = d' \cdot 2$ and $\deg G_{m-1} = 4^{m-1}/2 = d' \cdot 1$, $d' = 4^{m-1}/2$; the backward chain is entered at the ψ-update, and by the last part of Lemma 4.5 its top w-component for the weights $(2, 1)$ is the single monomial $u^{4^{n-m}}$ with coefficient 1. Lemma 4.4, with the roles of the variables exchanged, gives degree $d' \cdot 4^{n-m} \cdot 2 = 4^{n-1}$ and leading coefficient $\left(\varphi^{4^{m-1}}\right)^{4^{n-m}} = \varphi^{4^{n-1}}$, and the meeting polynomial is again f_n identically. □

Corollary 4.7 (Middle-state eliminant)**.** *Assume $\varphi \neq 0$, fix a cut after round m, and let $A := \mathbb{F}_p[a]/(f_n)$ and $\beta := F_m(a) \bmod f_n \in A$, the image of the middle coordinate θ_m. The univariate consequences of the CICO system in θ_m form the principal ideal generated by the minimal polynomial μ_β of β, and the canonical eliminant computable from the half-systems,*

$$E_m(x) \;=\; \det(x \cdot \mathrm{Id} - M_\beta) \;=\; \mathrm{lc}(f_n)^{-4^{m-1}}\, \mathrm{Res}_a(f_n(a),\, x - F_m(a)),$$

with M_β the multiplication-by-β matrix on A, is monic of degree exactly D_I. If moreover f_n is squarefree and F_m separates the roots of f_n in $\overline{\mathbb{F}_p}$, then $\mu_\beta = E_m$ has degree D_I: every nonzero univariate consequence in the middle coordinate also has degree at least D_I, so meeting in the middle does not reduce the effective univariate degree on the middle coordinate either. The separation condition is Zariski-open in $(c_0, \ldots, d_{n-1}, \tau)$: the coefficients of E_m are polynomials in τ of degree at most 4^{m-1}, so once E_m is squarefree at a single target τ^*, separation can fail for at most $(2D_I - 1)\, 4^{m-1}$ targets τ. For the production constants of Section 3.4 such certificates were computed for all splits $1 \leq m \leq n-1$ at $n \in \{2, \ldots, 6\}$ over both production fields (Section 4.6).

Proof. By the evaluation argument of Proposition 4.6(ii), a polynomial $q(\theta_m)$ lies in the CICO ideal iff $q(F_m) \in (f_n)$, i.e. iff $q(\beta) = 0$ in A, i.e. iff $\mu_\beta \mid q$. The algebra A is free of rank D_I with basis $1, a, \ldots, a^{D_I-1}$, so E_m is monic of degree D_I; the resultant expression is the standard product formula $\mathrm{Res}_a(f_n, x - F_m) = \mathrm{lc}(f_n)^{4^{m-1}} \prod_i (x - F_m(a_i))$ over the roots a_i of f_n with multiplicity. If f_n is squarefree, $A \otimes \overline{\mathbb{F}_p} \cong \overline{\mathbb{F}_p}^{\,D_I}$ with $\beta \mapsto (F_m(a_i))_i$, so $\mu_\beta = \prod_{v \in \{F_m(a_i)\}} (x - v)$ and $\deg \mu_\beta = D_I$ iff the values $F_m(a_i)$ are pairwise distinct, iff E_m is squarefree. For the τ-degree bound: in $\mathrm{Res}_a(F_n(a) - \tau,\, x - F_m(a))$ the Sylvester matrix contains τ only in the 4^{m-1} rows carrying the coefficients of f_n, each linearly, so $\deg_\tau E_m \leq 4^{m-1}$; hence $\mathrm{disc}_x(E_m) \in \mathbb{F}_p[\tau]$ has degree at most $(2D_I - 1)\, 4^{m-1}$, and a single τ^* with $\mathrm{disc}_x(E_m)(\tau^*) \neq 0$ shows it is not identically zero. □

Remark 4.8 (Scope)**.** Proposition 4.6 is a statement about algebraic degrees, not running time: it proves that no elimination strategy—in particular no meet-in-the-middle organisation of the

rounds—produces a univariate constraint of degree below D_I, so the cost model of Section 5.2 prices every such strategy at the same $\widetilde{O}(\min(D_I, p))$ univariate figure. It does not (and cannot) rule out faster algorithms for the root-finding step itself; that is the content of Conjecture 5.2. The separation hypothesis of Corollary 4.7—certified at the production constants for all splits at $n \in \{2, \ldots, 6\}$ over both production fields—gates only the middle-coordinate meet-in-the-middle variant; the unconditional rate-input floors (i)–(iv) do not depend on it.

4.6 Symbolic Verification

We verified the degree formulas symbolically using SageMath for $n = 1, \ldots, 8$ by computing the explicit polynomials over $\mathbb{Q}(\varphi)$. The non-telescoping battery of Proposition 4.6 and Corollary 4.7—per-round degrees and leading coefficients, monic free-input leading forms, split exactness at every cut, and the full-degree middle-state minimal-polynomial certificates—is reproduced by `analysis/nontelescoping_verify.sage` at the production constants for $n = 2, \ldots, 6$ over both production fields.

Table 2: Symbolic verification of degree growth.

n	$d_\theta(n)$	$d_\psi(n)$	$\log_2 d_\psi$	Status
1	2	4	2	Verified
2	8	16	4	Verified
3	32	64	6	Verified
4	128	256	8	Verified
5	512	1 024	10	Verified
6	2 048	4 096	12	Verified
7	8 192	16 384	14	Verified
8	32 768	65 536	16	Verified

5 Security Analysis

5.1 Security Framework and Assumptions

The security of Mambo is organised in three layers, following the standard architecture for fixed unkeyed permutations:

1. **Mode bounds, proven in the ideal-permutation model.** Replacing $P_{\varphi,n}$ by a uniformly random permutation, the sponge collision and indifferentiability bounds (Proposition 6.2, Corollary 6.3, via BDPV08 [42]) and the PRNG indistinguishability bound (Lemma 7.4) hold unconditionally with the stated birthday terms.
2. **One explicitly labeled design claim** (Design Claim 5.1) asserting that the concrete permutation admits no structural distinguisher below the target cost. This calibrates when the ideal-permutation modeling of layer 1 is sound; it is the single assumption of the paper.
3. **One falsifiable CICO-hardness conjecture** (Conjecture 5.2), parameterized over (t, n, p) so that it covers every parameter set of Table 6, minimum and recommended rows alike. It is the sharpest concrete consequence of the design claim on the algebraic attack surface, and the statement we invite the community to attack. The remainder of this section collects the evidence for it.

Design Claim 5.1 (No structural distinguisher)**.** *For every parameter set $(t, n, \mathbb{F}_p, \varphi)$ of Table 6 (minimum or recommended row) at security level λ, no algorithm given the full circuit description distinguishes the Mambo permutation $P_{\varphi,n}$ on $\mathbb{F}_p^t$ from a uniformly random permutation at cost below 2^λ, beyond the generic and algebraic attacks accounted for in this section.*

Design Claim 5.1 is a design claim in the sense customary for fixed unkeyed permutations (Keccak, Poseidon): read as a formal indistinguishability statement it would be vacuous, since a fixed public permutation is trivially distinguishable from random by evaluating it at one point and comparing with one oracle query. It is an engineering assertion, falsifiable only by exhibiting a concrete distinguisher. Its sharpest falsifiable consequence on the algebraic attack surface is the following conjecture, which any such distinguisher of the CICO type would violate.

Conjecture 5.2 (CICO hardness)**.** *Let $t \geq 2$, $n \geq 2$, let p be an odd prime, and let $P_{\varphi,n}$ be the Mambo-t permutation on $\mathbb{F}_p^t$ with φ and the round constants of Section 3.4. Consider the sponge CICO problem: given a uniformly random target $\tau \in \mathbb{F}_p$, find $\mathbf{x} \in \mathbb{F}_p^t$ whose $t-1$ capacity coordinates are zero and such that the first coordinate of $P_{\varphi,n}(\mathbf{x})$ equals τ. Every algorithm solving this problem with constant success probability requires $\widetilde{\Omega}(\min(2^{t(n-1)}, p))$ field operations.*

The quantity $2^{t(n-1)}$ is the eliminant degree D_I of the sponge CICO system (Section 5.4); the $\min(\cdot, p)$ accounts for exhaustive search over the single free rate coordinate, which costs $O(p{\cdot}tn)$ field operations. A matching $\widetilde{O}(\min(D_I, p))$ attack exists (Section 5.2), so Conjecture 5.2 asserts exactly that the known generic algorithms are optimal for this system. Any violation of Conjecture 5.2 falsifies Design Claim 5.1; the converse does not hold (a PRP-mode distinguisher need not solve CICO), which is why the design claim, not the conjecture, is the assumption from which the conditional mode bounds are calibrated.

Remark 5.3. This conjecture occupies the same epistemic position the ideal-permutation assumption held for Keccak in the years before its public cryptanalytic record (15+ years to date), for Poseidon (7 years of public cryptanalysis to date) and for Anemoi during its pre-cryptanalysis window prior to the resultant attack of [28]. The maturity gap is the primary risk factor: arithmetisation-oriented designs in this family routinely accumulate attack surface only after multi-year community scrutiny, and Mambo has had months. The proven degree growth 4^n and the proven degree floor for every elimination strategy (Proposition 4.6) constrain the algebraic attack surface, but they do not substitute for the years of independent review the listed comparators have undergone.

5.2 Attack-Cost Model and Parameter Sizing

All parameter recommendations in this paper are sized under an explicit two-tier cost model for the algebraic attack surface, stated here once and referenced throughout. The classical algebraic attack pipeline has three stages: (i) a Gröbner basis computation in degree-reverse-lexicographic order, (ii) a change of monomial order or eliminant extraction (FGLM), and (iii) univariate root-finding. For Mambo, stage (i) contributes no security: the degrevlex Gröbner basis of the CICO ideal is explicit and free— Appendix G exhibits it in closed form. The load-bearing quantity is therefore the degree of the zero-dimensional ideal, equivalently the degree of the univariate eliminant,

$$D_I = 2^{t(n-1)}$$

(Section 5.4 at $t = 2$; Section 7.4 for general t).

Conservative tier (used for all recommendations). At sponge rate $r = 1$ and capacity $c = t - 1$, the CICO system is a single univariate polynomial f_n of degree D_I in the free rate input. Given the circuit, the dense coefficient vector of f_n is computable by direct iteration with FFT polynomial arithmetic in $\widetilde{O}(D_I)$ field operations (round r squares dense polynomials of degree $\sim 2^{t(r-1)}$, and the geometric sum over rounds is dominated by its last term), and an $\mathbb{F}_p$-root is then found in $\widetilde{O}(D_I \log p)$ time by computing $x^p \bmod f_n$, taking a gcd with $x^p - x$, and running Cantor–Zassenhaus—at $\Theta(D_I \log p)$ bits of memory. Exhaustive search over the rate coordinate costs $O(p \cdot tn)$ field operations. The binding known attack on the rate-1 CICO

surface is therefore $\widetilde{O}(\min(D_I, p))$ time; it uses no Gröbner basis computation and no FGLM step. On the capacity-1 surface of the deployed hash at $t \geq 3$, the attacker additionally holds $t-2$ free rate inputs, and the best measured specialization of those inputs (the self-disclosed attack of Section 5.6) lowers the eliminant degree to

$$D_{\text{eff}} \;=\; 2^{\,t(n-1)-\max(t-2,\,0)}.$$

The conservative model prices CICO at $\widetilde{O}(\min(D_{\text{eff}}, p))$ and sizes parameters by

$$D_{\text{eff}} \;\geq\; 2^{\lambda}, \qquad \text{i.e.} \qquad n \;\geq\; \left\lceil \tfrac{\lambda+\max(t-2,\,0)}{t} \right\rceil + 1,$$

together with the field-saturation condition $t \log_2 p \geq \lambda$ (the closure count D_I saturates at p^t over $\mathbb{F}_p$; Section 5.4). At $t = 2$ there are no free rate inputs and $D_{\text{eff}} = D_I = 4^{n-1}$. The smallest n meeting this requirement is the *minimum* round count $n_{\min}$ of Table 6, sized at exact analysis equality with zero margin; the *recommended* round count adds a 25% round margin, $n_{\text{rec}} = \lceil 1.25\, n_{\min} \rceil$ (Section 5.13). The minimum row is retained deliberately: it is the falsification target of the cryptanalysis challenge—any attack beating $\widetilde{O}(\min(D_{\text{eff}}, p))$ at a minimum cell falsifies Conjecture 5.2 directly, without needing to penetrate the margin. Every parameter set in this paper satisfies the requirement at its minimum row, and all integrator-facing recommendations use n_{rec}. Because the Gröbner basis is free, the relevant external solvers are the dedicated FreeLunch resolution algorithm [7] and the resultant and CheapLunch refinements of the change-of-ordering step [5, 8]; all of them process the full eliminant and therefore cost at least the $\widetilde{O}(\min(D_{\text{eff}}, p))$ figure that the conservative tier already prices, while the model-dependent tier below is precisely the accounting that such change-of-ordering improvements erode.

Model-dependent tier (not used for recommendations). The historical sizing convention for AO primitives prices the algebraic attack at the dense-FGLM figure $O(n \cdot D_I^2)$ and requires only $D_I \geq 2^{\lambda/2}$. For Mambo this convention is not conservative: the univariate path above costs $\widetilde{O}(D_I)$ time, so a $D_I = 2^{\lambda/2}$ parameter set resists it only if the attacker's $\Theta(D_I \log p)$ bits of writable memory are priced at cost (an area–time metric; at $\lambda = 128$, $D_I = 2^{64}$, and $\log_2 p = 254$ this is on the order of 2^{72} bits). Parameter sets sized under this convention are reported separately in Table 7, labeled with the cost model that certifies them; they are not recommendations.

5.3 Evidence for the Conjecture

This subsection collects, in one place, the evidence supporting Design Claim 5.1 and Conjecture 5.2. Each item states its exact scope; none is a proof of the conjecture, and the maturity caveat of Remark 5.3 applies to all of them.

(i) Eliminant-degree battery. The load-bearing quantity of the cost model is the eliminant degree, and the claim that no algebraic dependency lowers it below its formal value has been tested exhaustively at computationally accessible sizes. Fixing $\theta_0 = \alpha$ and $\psi_k = \tau$ yields a univariate polynomial of formal degree 4^k, and direct Gröbner-basis computation confirms $\dim_{\mathbb{F}_p}(\mathbb{F}_p[\theta_0, \psi_0]/(\theta_0 - \alpha, \psi_k - \tau)) = 4^k$ for every tested (p, k, α, φ) across the prime set $\{257, 1009, 4099, 2^{31}-1\}$, $k \in \{1, \ldots, 6\}$, $\alpha \in \{0, 1, 31, 1000\}$, and $\varphi \in \{2, 3, 7, p-1\}$ (34 of 34 experiments at $t = 2$). The wider-state slice ideal $(x_1 - \alpha_1, \ldots, x_{t-1} - \alpha_{t-1}, x_t^{(k)} - \tau)$ likewise confirms $D_I = 2^{tk}$ at every tested cell:

state	k range	primes	confirming runs
$t=2$	1–6	$\{257, 1009, 4099, 2^{31}-1\}$	34 / 34
$t=3$	1–4	$\{257, 1009, 4099\}$	12 / 12
$t=3$	$k=3$, $\varphi \in \{2, 3, 7, p-1\}$	$\{4099\}$	6 / 6
$t=4$	1–4	$\{257, 1009, 4099\}$ ($k=4$ only $p=257$)	10 / 10
Total wider-state + extended-k evidence			62 / 62

No counterexample was observed in any configuration; reproducer scripts are in the supplementary materials. The sponge-convention analogue ($D_I = 2^{t(n-1)}$, square-free eliminant) was verified through $n = 5$ at $t \in \{3, 4\}$ (Section 7.4), and the capacity-1 specialization optimum D_{eff} was measured exactly at every cell of $t \in \{3, 4, 5\}$, $n \in \{2, \ldots, 5\}$ (Section 5.6).

(ii) Closed-form regularity, verified beyond its proof range. The $t = 2$ Hilbert series and $d_{\text{reg}} = 2n$ are theorems (Theorem G.5), machine-checked in Lean 4 and verified bit-exact over the rationals at $n \in \{2, \ldots, 10\}$ and at $p \in \{101, 257, 1009, 7919\}$ (Section 5.5); the wider-width regularity measurements are bit-exact at $t = 3$, $n \leq 7$ and $t = 4$, $n \leq 6$ (Remark 5.7).

(iii) Squarefreeness of the eliminant. Generic squarefreeness is a theorem (Theorem 5.12); the production constants were checked to lie outside the bad locus at all tested cells ($n \in \{3, \ldots, 6\}$, and by full elimination at $p = 257$). A square-free eliminant has no repeated-root shortcut and forces the full Cantor–Zassenhaus cost.

(iv) Monodromy of the sponge eliminant. A root-finding shortcut below the $\widetilde{O}(\min(D_{\text{eff}}, p))$ figure of the cost model would need the eliminant $f_n(x) = \theta_n(x) - \tau$ to carry special structure: a functional decomposition $f_n = g \circ h$ (splitting the solve into two lower-degree stages), an imprimitive or solvable Galois group (a subfield or radical tower), or sparsity. We certify the opposite at every computationally accessible size. For the *production* round constants over BN254, the Galois group of $\theta_n(x) - \tau$ over $\mathbb{F}_p(\tau)$ is the full symmetric group S_{D_I} for $n \in \{3, 4, 5, 6\}$ ($D_I = 16$ up to 1024), by deterministic Dedekind certificates: a squarefree specialization whose factorization pattern contains a unique prime part q with $D_I/2 < q \leq D_I - 3$ forces, via an elementary primitivity lemma and Jordan's theorem [25], a subgroup containing A_{D_I}, and an odd-pattern witness pins the full S_{D_I}. The $n \in \{3, 4, 5\}$ cells were independently re-certified by a second method (a Morse-type certificate: the critical values of θ_n are pairwise distinct, so inertia is generated by transpositions). For every $n \geq 3$ and *any* round constants, the same conclusion reduces to a single falsifiable hypothesis—functional indecomposability of θ_n—via tame total ramification at infinity (a regular D_I-cycle), Schur's theorem on primitive groups of composite degree with a regular cyclic subgroup [24, 25], the classification of primitive groups containing such a subgroup [23], and an elementary arithmetic lemma excluding the projective-linear cases at degree 2^{2n-2}; the correspondence between functional decompositions and block systems of the monodromy is classical [26]. The indecomposability hypothesis is itself proven unconditionally for every decomposition with outer degree at most 4 or inner degree 2 (the latter by a symmetry obstruction that telescopes down the coupled rounds), and verified exactly for all remaining splits up to $n = 6$ with the production constants. We state the scope plainly: full symmetric monodromy at the production round counts ($n_{\min} = 65$, $n_{\text{rec}} = 82$) is *not* claimed unconditionally—it holds exactly if θ_n is indecomposable, a stated, falsifiable hypothesis. What the certified range already rules out is precisely the composition, subfield, and sparsity structure that known root-finding shortcuts exploit (every coefficient of θ_n is moreover nonzero at the tested sizes); the certificates (specializations, cycle types, and verification scripts) are part of the supplementary materials.

(v) Differential and statistical behaviour. The per-round maximum differential probability is exactly $1/p$ (Proposition 5.17), with multi-round decay measured through $n = 4$ and the Bézout ceiling (6) as the extrapolating bound (Section 5.7); DDT and BCT enumerations on three small primes show no anomalous structure beyond the disclosed small-field artifacts (Section 5.9). Statistical testing of the permutation output passes PractRand to 1 TB and TestU01 BigCrush (160/160) on two 64-bit fields:

Test suite	Field	Result	Notes
PractRand	Goldilocks	1 TB PASS	357 tests, 0 anomalies
BigCrush	Goldilocks	160/160	All p-values in $[0.001, 0.999]$
BigCrush	Mersenne-61	160/160	All p-values in $[0.001, 0.999]$
Multi-seed	2 fields	15/15 pass	Zero failures
Avalanche	4 primes	0.49–0.50	Expected: 0.50

Tests were run on Goldilocks and Mersenne-61, not BN254 (impractical due to field size).

(vi) Round-reduced solve experiments. We ran both stages of the attack pipeline end-to-end on round-reduced sponge-CICO instances (rate-1 convention: capacity cells zeroed at the input, one rate cell fixed at the output, single free rate input), strictly sequentially on a consumer M-series laptop with an 8-minute cap per cell. Fields: KoalaBear ($p = 2^{31} - 2^{24} + 1$, primary), the BN254 scalar field (spot cells), and a 20-bit prime $p_{\mathrm{m}} = 1048573$ on which the Gröbner solver supports full root extraction. Two attacks: `uni` builds the dense univariate eliminant f_n by iterated composition and root-finds it ($\mathbb{F}_p$ dense root-finding, not the $\widetilde{O}(D_I)$ optimum of Section 5.2); `gb` runs F4 on the iterated multivariate quadratic system, computes the colength, and extracts the variety via FGLM where the solver supports the field. Every recovered root was forward-verified through the permutation.

Attack	Field	t	n	D_I	Time (s)	Outcome
`uni`	KoalaBear	2	5	2^8	0.018	root verified
`uni`	KoalaBear	2	6	2^{10}	0.29	root verified
`uni`	KoalaBear	2	7	2^{12}	7.1	root verified
`uni`	KoalaBear	2	8	2^{14}	88.3	root verified
`uni`	KoalaBear	2	9	2^{16}	> 480	DNF (time cap)
`uni`	KoalaBear	3	4	2^9	0.052	root verified
`uni`	KoalaBear	3	5	2^{12}	2.1	root verified
`uni`	BN254	2	4	2^6	0.45	root verified
`uni`	BN254	2	6	2^{10}	20.7	root verified
`gb` (F4 + colength)	KoalaBear	2	2–10	$\leq 2^{18}$	≤ 0.07	colength $= D_I$; no root
`gb` (F4 + colength)	KoalaBear	3	2–6	$\leq 2^{15}$	≤ 0.35	colength $= D_I$; no root
`gb` (F4 + FGLM)	p_{m}	2	6	2^{10}	0.36	roots verified
`gb` (F4 + FGLM)	p_{m}	2	7	2^{12}	20.5	roots verified
`uni`	p_{m}	2	6	2^{10}	0.12	root verified
`uni`	p_{m}	2	7	2^{12}	2.2	root verified

The largest instance broken end-to-end on the laptop is $t = 2$, $n = 8$: $D_I = 2^{14}$, 88 s; $n = 9$ ($D_I = 2^{16}$) did not finish within the 480 s cap. At every cell the measured colength equals $2^{t(n-1)}$ exactly ($n = 2$–10 at $t = 2$, $n = 2$–6 at $t = 3$), confirming the eliminant-degree model bit-exactly across the broken range. The Gröbner route never beats the direct eliminant: F4 on the iterated system is nearly free (the basis is explicit; Appendix G) but recovers no root, and once the FGLM extraction is counted the full Gröbner solve is $\sim 9\times$ slower than the eliminant at equal $D_I = 2^{12}$ (20.5 s vs 2.2 s, same prime; $\sim 3\times$ at 2^{10})—a third independent confirmation, after the closed-form basis and the cost-model analysis of Section 5.2, that the univariate eliminant

is the binding attack. Measured scaling: over $D_I \in [2^8, 2^{14}]$ ($t = 2$, KoalaBear) the uni solve time fits $D_I^{2.0}$ (per-step log-log slopes 2.03/2.31/1.81), and $t = 3$ over $D_I \in [2^9, 2^{12}]$ gives slope 1.80: cost tracks D_I, not (t, n) separately. The measured exponent 2.0 reflects the dense root-finder implementation used and is an upper bound consistent with—and above—the conservative $\widetilde{O}(D_I)$ model; no figure is extrapolated past the largest measured cell. The degree data are field-shape independent (colength and eliminant degree track D_I identically on 31-bit and 254-bit fields), while absolute time at matched D_I is 70–130× worse at 254-bit, a poly($\log p$) factor on top of the D_I model. The minimum conservative 128-bit cells ($n = 65/44/34$ at $t = 2/3/4$, the falsification targets of the cost model) sit far beyond this frontier—57 rounds at $t = 2$ and 39 at $t = 3$ (no $t = 4$ cell was broken at any n)—and in eliminant degree the gap is 2^{14} broken against $\geq 2^{128}$ required, with every added round multiplying D_{eff} by 2^t; the recommended cells ($n = 82/55/43$) add a further 25% round margin on top. No shortcut below the D_I barrier was found. Raw per-cell data (build/solve/total wall-clock and peak memory) are in the supplementary materials.

None of (i)–(vi) substitutes for the multi-year community scrutiny discussed in Remark 5.3; the items are offered as the falsification surface we could build ourselves.

5.4 Interpolation and CICO Resistance

An interpolation attack on a function of algebraic degree D requires at least $D+1$ input-output pairs. Already at the minimum cell $n = 65$: $D = d_\psi(65) = 4^{65} \approx 2^{130}$, exceeding the target security level; the recommended $n = 82$ gives $4^{82} \approx 2^{164}$.

For the CICO problem, an attacker must solve a constrained system of degree 4^n in two unknowns. In sponge mode this is exactly univariate polynomial root-finding at degree $D_I = 4^{n-1}$, which the conservative cost model (Section 5.2) prices at $\widetilde{O}(\min(D_I, p))$; the 128-bit requirement $D_I \geq 2^{128}$ holds at $n \geq 65$.

Graeffe-accelerated root recovery. Zhao, Sanso, Vitto, and Ding [29] accelerate univariate root recovery (a Graeffe transform inside the NTT pipeline) by constant powers-of-two factors in wall time and memory. The Mambo sponge CICO is exactly such a univariate problem, so their methodology applies directly; it improves the constants in front of the $\widetilde{O}(\min(D_I, p))$ attack figure but not its exponent, which is what the conservative sizing rule $D_{\text{eff}} \geq 2^\lambda$ (Section 5.2) already prices.

Degree of the ideal (D_I). In the ALFOMs framework [1], the cost of algebraic attacks is decomposed into two steps: (1) the Gröbner basis computation, and (2) the FGLM basis-conversion step, whose cost is polynomial in D_I (the degree of the zero-dimensional ideal). Following standard convention [1, 5, 28], D_I counts solutions over the algebraic closure $\overline{\mathbb{F}_p}$, without including the field equations $x^p - x = 0$.

For Mambo in sponge mode with rate $r = 1$ and capacity $c = 1$, fixing the capacity ($\psi_0 = 0$) and the rate output ($\theta_n = \tau$ for some target $\tau \in \mathbb{F}_p$) reduces the system to a single univariate polynomial in θ_0 of formal degree 4^{n-1} (see the sponge-mode degree paragraph below). Since the Mambo round equations introduce no algebraic dependencies that reduce this degree, the ideal degree equals the formal degree:

$$D_I = 4^{n-1}.$$

At the minimum cell $n = 65$: $D_I = 4^{64} = 2^{128}$, matching the target security level exactly; at the recommended $n = 82$: $D_I = 4^{81} = 2^{162}$, a 34-bit margin.

Field-saturation bound. The expression $D_I = 4^{n-1}$ counts common zeros over the algebraic closure $\overline{\mathbb{F}_p}$. When restricting to solutions in $\mathbb{F}_p^t$, this quantity is additionally bounded by the total number of points $|\mathbb{F}_p^t| = p^t$, giving the effective bound

$$D_I^{(\mathbb{F}_p)} \;=\; \min\!\left(4^{n-1},\; p^t\right)$$

at state width $t = 2$, and analogously $\min(2^{t(n-1)}, p^t)$ for general t. For the recommended parameters $(\varphi, n, t) = (7, 82, 2)$ over BN254 ($p \approx 2^{254}$), $p^2 \approx 2^{508} \gg 4^{81} = 2^{162}$ and the saturation bound is vacuous: the formal ideal degree $D_I = 2^{162}$ holds, and the conservative requirement $D_I \geq 2^\lambda$ of Section 5.2 is met with the design margin (it is met exactly at the minimum cell $n = 65$). For small-prime instances (e.g. M31, BabyBear with $p \approx 2^{31}$), the saturation bound binds once $n - 1 > \log_2 p$, in which case the effective security derives from p^t rather than from n. Table 8 in Section 5.13 makes the field dependence explicit by tabulating which $(t, \mathbb{F}_p)$ combinations remain above the conservative threshold $D_I \geq 2^\lambda$ for each tier.

The Gröbner stage of the algebraic attack contributes no security margin: the degrevlex basis is explicit and free (Appendix G). The single load-bearing quantity is D_I, priced at $\widetilde{O}(\min(D_I, p))$ under the conservative cost model of Section 5.2. The per-round growth factor of D_I is 4, comparable to other AO constructions (Anemoi: 5, Rescue: α^b).

Sponge-mode degree and the minimum cell $n = 65$. In sponge mode with rate $r = 1$ and capacity $c = 1$, the capacity is fixed ($\psi_0 = 0$), leaving a single free variable θ_0. The algebraic degree of θ_n as a univariate polynomial in θ_0 satisfies $\deg(\theta_n) = 4^{n-1}$ (one factor of 4 is lost at the first round because $\psi_0 = 0$ contributes degree 0). A higher-order differential [35] of order $4^{n-1} + 1$ therefore vanishes, giving a distinguisher. At $n = 64$ this yields only 2^{126} security; at $n = 65$ we obtain $\deg(\theta_{65}) = 4^{64} = 2^{128}$, matching the target exactly. This equality under the conservative sizing rule $D_I \geq 2^\lambda$ of Section 5.2 defines the minimum cell $n_{\min} = 65$ at $t = 2$; the recommended round count is $n_{\text{rec}} = 82$ (Section 5.13).

5.5 Gröbner Basis Complexity

The most direct algebraic attack writes the system of $2n$ quadratic equations in $2(n + 1)$ unknowns. In the preimage scenario, substituting the known output reduces this to $2n = 130$ equations in $2n = 130$ unknowns—a *determined* system.

Proposition 5.4 (Macaulay bound)**.** *The degree of regularity of the Mambo preimage system, measured via the second Hilbert series of the determined ideal over the rationals, equals $d_{\text{reg}} = 2n$ for every $n \geq 2$ (Theorem G.5 in Appendix G; empirically verified bit-exact at $n \in \{2, 3, \ldots, 10\}$ in Table 3). At the minimum cell $n = 65$ we have $d_{\text{reg}} = 130$, and the Macaulay bound is:*

$$\binom{d_{\text{reg}} + m}{m}^{\omega} = \binom{130 + 130}{130}^{2.81} > 2^{300},$$

where $m = 130$ is the number of unknowns and $\omega < 2.81$ is the linear algebra exponent.

This bound rules out a blind F4/F5 computation on the original generators only. It is not the binding attack cost: the degrevlex Gröbner basis of the CICO ideal is explicit (Appendix G), so an attacker never needs to run F4/F5, and the cost model of Section 5.2 prices the algebraic attack by the eliminant degree D_I instead.

Composition versus regularity degree. The algebraic degree of the round map composition in sponge mode with the capacity fixed to zero, $\deg(\theta_n) = 4^{n-1}$ (cf. Section 5.4, sponge-mode degree paragraph; this is distinct from the full bivariate degrees $d_\theta(n) = 2 \cdot 4^{n-1}$ and $d_\psi(n) = 4^n$ of Theorem 4.2), is distinct from the regularity degree of the CICO ideal, $d_{\text{reg}} = 2n$ (Table 3).

Table 3: Empirical regularity degree of the Mambo preimage system.

n	Vars	Eqs	d_{reg} (obs.)	Wall-clock
2	4	4	4	0.05 s
3	6	6	6	0.06 s
4	8	8	8	0.07 s
5	10	10	10	0.08 s
6	12	12	12	0.08 s
7	14	14	14	0.10 s
8	16	16	16	0.17 s
9	18	18	18	0.35 s
10	20	20	20	0.58 s
65	130	130	130 (thm.)	$\gg 2^{300}$

Measured on a single-host Linux cloud VM using Singular 4.3.2. The ideal of the Mambo CICO preimage system is built over the rationals in degree-reverse-lexicographic order; the degree of regularity is read off the second Hilbert series `hilb(G,2)` of the standard basis G as the index of the first non-positive coefficient. The observed relation $d_{\mathrm{reg}} = 2n$ holds bit-exactly over all nine tested round counts. This is one below the value $2n+1$ that the strict semi-regularity assumption would predict for a determined system of $2n$ quadratic equations in $2n$ unknowns, which indicates that the Mambo CICO ideal admits one extra non-generic syzygy beyond the semi-regular case. Since the equations contain no characteristic-specific identities, the Hilbert series is independent of the ground field, so the empirical values apply equally at the BN254 scalar field used in the hash instantiation. The $n = 65$ row uses $d_{\mathrm{reg}} = 130$ from the proven closed form $d_{\mathrm{reg}} = 2n$ (Theorem G.5); it is not an extrapolation.

The first measures how a polynomial grows under iteration; the second measures the degree at which Buchberger-style S-polynomial reduction terminates for the preimage ideal. The two quantities play different roles in the analysis and are not interchangeable.

Remark 5.5 (Relationship to ALFOM and FGLM)**.** The ALFOMs framework [1] decomposes algebraic attack cost into a Gröbner basis step and an FGLM step. In constructions such as Griffin [4], the GB step can be surprisingly inexpensive (cf. the FreeLunch attack [7] and its resultant refinement [5]), leaving the ideal degree as the bottleneck. Mambo is, by its own Appendix G, an extreme instance of the same phenomenon: the degrevlex Gröbner basis is explicit and free, so the GB step contributes no security and the single load-bearing quantity is the ideal degree $D_I = 2^{t(n-1)}$, priced at $\widetilde{O}(\min(D_I, p))$ under the conservative model of Section 5.2. The Macaulay bound of Proposition 5.4 retains only the role of ruling out blind F4/F5 computation on the original generators. The per-round D_I growth factor of 4 is comparable to other AO constructions and shows that the ALFOM of 1.0 captures genuine growth of the load-bearing degree, not merely formal degree inflation.

Corollary 5.6 (Closed-form Hilbert series)**.** *For every $n \geq 2$, the second Hilbert series numerator of the Mambo CICO determined ideal (as in Proposition 5.4) equals*

$$\mathrm{hilb}(G, 2) \;=\; (1+z)^{2n-1}.$$

Hence $\dim_{\mathbb{Q}}(R/I) = 2^{2n-1}$, which is exactly half of the Bezout bound 2^{2n} for a determined system of $2n$ quadratic equations in $2n$ unknowns.

A structural proof of Corollary 5.6, proceeding via an explicit Gröbner basis with pairwise coprime leading monomials and Buchberger's first criterion, is given in Appendix G (Theorem G.5). The proof simultaneously establishes $d_{\mathrm{reg}} = 2n$ for all $n \geq 2$, lifting Proposition 5.4 from an empirical observation through $n = 10$ to a uniform statement. The degree-growth theorem (Theorem 4.2) and the $t = 2$ closed-form regularity and half-density dimension $\dim_{\mathbb{Q}}(R/I) = 2^{2n-1}$ of Theorem G.5 are additionally machine-checked in Lean 4. The Macaulay floor arithmetic

is kernel-verified at all nine minimum (analysis-equality) cells of Table 6, and additionally at the nine cells of the superseded model-dependent grid of Table 7; both grids are conditional on the measured d_{reg} at $t \in \{3, 4\}$ (Remark 5.7); the development is `sorry`-free and uses only the three standard axioms, with the floor computations verified as axiom-free kernel proofs. The formalisation verifies the mathematics (closed forms, dimensions, and bound arithmetic), not the attack-cost model of Section 5.2. The division of labour in the formalisation is precise, and worth stating to preempt a natural objection: the eliminant *degree* $D_I = 2^{t(n-1)}$—the quantity on which the conservative attack cost depends—is machine-checked for the capacity-fixed attack-surface ideal I_{sp} itself, as a kernel-verified `finrank` computation of the quotient-algebra dimension; what remains empirical for I_{sp} is its *regularity*, on which the eliminant attack does not load-bear; and the proven regularity theorems ($d_{\mathrm{reg}} = 2n$, the Hilbert series) concern the rate-fixed ideal I_{gb} and are labeled as such throughout (Definition 5.9, Remark 5.11). Scripts reproducing the structural identities are included in the supplementary materials, and characteristic independence was additionally empirically verified at $p \in \{101, 257, 1009, 7919\}$ for $n \in \{2, \ldots, 10\}$; coefficient patterns matched the closed form bit-for-bit.

Half-density Bezout gap (heuristic observation). The closed form $(1 + z)^{2n-1}$ is one factor of $(1 + z)$ shorter than the semi-regular baseline $(1 + z)^{2n}$: Mambo's CICO ideal has half the generic solution-space density of a random $2n$-quadratic system in $2n$ unknowns. This is a *solution-count* observation and not a cryptographic attack-cost reduction. Gröbner attack cost scales primarily with d_{reg}, which Proposition 5.4 already bounds as $2n$. FGLM cost scales with the colon-length; the one-bit density gap enters FGLM via D_I but is not a direct one-bit slack on attack cost.

A formal reduction from Mambo's CICO hardness to the hardness of random MQ in $2n$ variables remains open work. We note the half-density structure here because it is tight, explicit, and testable.

Remark 5.7 (Wider state widths $t \in \{3, 4\}$: empirical regularity)**.** Proposition 5.4 and Theorem G.5 are stated for the narrow-state $t = 2$ construction. The wider-state variants instantiated in Table 6 ($t \in \{3, 4\}$, defined in Section 7.4) admit an analogous *empirical* regularity bound under the paper-exact CICO convention (capacity-in cells fixed to zero, single rate-out cell fixed to the target). Measuring the second Hilbert series of the determined ideal in degree-reverse-lex order via Singular yields

$$d_{\mathrm{reg}}(t, n) \;=\; \begin{cases} 2n, & t = 2 \text{ (Theorem G.5, all } n \geq 2), \\ 3n - 1, & t = 3 \text{ (empirical, } n \in \{2, \ldots, 7\} \text{ bit-exact}), \\ 4n - 3, & t = 4 \text{ (Theorem 5.8, all } n \geq 4;\ n \in \{2, 3\} \text{ by direct computation}). \end{cases}$$

At the minimum (analysis-equality) 128-bit cells of Table 6, the Macaulay bound on F4/F5 cost is therefore

(t, n)	$m = tn$	d_{reg}	$\binom{m+d_{\mathrm{reg}}}{d_{\mathrm{reg}}}^{\omega}$	Margin vs 128-bit
$(2, 65)$	130	130	$> 2^{700}$	$\gg 128$
$(3, 44)$	132	131	$> 2^{700}$	$\gg 128$
$(4, 34)$	136	133	$> 2^{700}$	$\gg 128$

with $\omega < 2.81$ as in Proposition 5.4 (at the recommended cells $n = 82/55/43$ the bound is larger still). The $t = 3$ measurement extends, but does not generalize, the structural argument of Theorem G.5: the closed-form Boolean-quotient bijection used there (Lemma G.1) is specific to the $t = 2$ generator pattern. The $t = 4$ case is itself now closed in closed form (Theorem 5.8 below), by a different route—an explicit leading-monomial classification rather than the $t = 2$ Boolean-quotient bijection. A unified formula $d_{\mathrm{reg}}(t, n) = tn - \binom{t-1}{2}$ fits all three widths $t \in$

$\{2, 3, 4\}$ at production n but is not asserted as a theorem beyond the stated bit-exact ranges; small-n edge cases at wider widths give larger d_{reg} than the asymptotic formula, and extending the closed form to $t \geq 5$ remains open. The Macaulay figures above rule out blind F4/F5 computation on the original generators at all recommended widths; they are not the binding attack cost, which under the cost model of Section 5.2 is the eliminant figure $\widetilde{O}(\min(D_I, p))$.

Theorem 5.8 (Closed-form Hilbert series at $t = 4$)**.** *For the rate-fixed (regularity) CICO ideal I at width $t = 4$ and every $n \geq 4$, the Hilbert series is*

$$\mathrm{HS}_{R/I}(z) = (1+z)^{4n-9}\, Q_4(z), \qquad Q_4(z) = z^5 + 11z^4 + 25z^3 + 19z^2 + 7z + 1,$$

so that, since $Q_4(1) = 64 = 2^6$, $\dim_{\mathbb{F}_p}(R/I) = 2^{4n-3}$ and $d_{\text{reg}} = 4n - 3$. The cases $n \in \{2, 3\}$ hold by direct computation. The statement is established for the production round-constant schedule, with constant-independence of the underlying leading-monomial set certified by random specialization rather than by a deterministic symbolic Gröbner computation (see below).

Proof sketch. By Lemma G.4 (Appendix G) the degree-reverse-lex initial ideal of I equals an explicit finite monomial certificate C for every $n \geq 4$. Counting the standard monomials of C by inclusion–exclusion gives the displayed series and $\dim_{\mathbb{F}_p}(R/C) = (1+z)^{4n-9} Q_4(z)\big|_{z=1} = 2^{4n-9} \cdot 2^6 = 2^{4n-3}$; the wider-width colength relation $\dim_{\mathbb{F}_p}(R/I) = 2^{tn-(t-1)} = 2^{4n-3}$ established following Proposition 5.10 then forces $\mathrm{in}_<(I) = C$, whence the Hilbert series of R/I coincides with that of R/C and $d_{\text{reg}} = \deg \mathrm{HS} + 1 = (4n-4)+1 = 4n-3$. The classification of Lemma G.4 is proven by window-local membership together with upper-semicontinuity of the initial ideal; by upper-semicontinuity the parameter points at which the initial ideal degenerates away from the generic certificate C form a proper Zariski-closed subset $B \subsetneq \mathbb{A}^m$, so by the Schwartz–Zippel lemma a uniformly sampled point lands in B with probability at most $\deg B/p$ per prime; the 80 sampled points across two primes therefore certify $\mathrm{in}_<(I) = C$ at the production constants with failure probability bounded by $(\deg B/p)$ in each of two coprime moduli ($p \in \{2^{31} - 1, 7919\}$), i.e. below 2^{-40} for the certificate's bounded degree. The fully deterministic symbolic-constant Gröbner computation that would discharge B unconditionally is beyond current compute; the result is stated for the production constant schedule with this explicitly bounded genericity certificate. □

Definition 5.9 (Boundary conventions)**.** Two CICO boundary conventions recur in this paper and must not be conflated. Write the wider-state cells as $(x_1, \ldots, x_t)$ (Section 7.4); at $t = 2$, $(x_1, x_2) = (\theta, \psi)$, with θ the rate cell and ψ the capacity cell.

(i) *Capacity-fixed (sponge) convention* I_{sp}: the capacity cells are zeroed at the input and one rate cell is fixed at the output—at $t = 2$, $\psi_0 = 0$ and $\theta_n = \tau$, with the rate input θ_0 free. The free input enters round 1 *linearly* ($\theta_1 = \varphi\theta_0 + c_0$). This is the attack surface of the sponge hash; its eliminant has degree $D_I = 2^{t(n-1)}$ ($= 4^{n-1}$ at $t = 2$; Section 5.4).

(ii) *Rate-fixed (regularity) convention* I_{gb}: the input cells $x_1, \ldots, x_{t-1}$ are zeroed, the free input is x_t—the cell that enters round 1 through a squaring—and one rate cell is fixed at the output. At $t = 2$: $\theta_0 = 0$ and $\theta_n = \tau$, with ψ_0 free ($\theta_1 = \psi_0^2 + c_0$). This is the ideal of Appendix G (and, at $t \geq 3$, of Remark 5.7), for which the closed forms $\mathrm{HS} = (1+z)^{2n-1}$, $\dim(R/I) = 2^{2n-1}$, and $d_{\text{reg}} = 2n$ are proven at $t = 2$.

Proposition 5.10 (Factor-2 colength relation)**.** *At $t = 2$, for every $n \geq 2$,*

$$\dim_{\mathbb{F}_p}(R/I_{\text{gb}}) \;=\; 2^{2n-1} \;=\; 2 \cdot 4^{n-1} \;=\; 2\, D_I(I_{\text{sp}}).$$

Proof. The left-hand equality is Theorem G.5. The right-hand equality is the sponge-mode eliminant degree $D_I = 4^{n-1}$ of Section 5.4, which follows from Theorem 4.2 after the first-round factor-4 loss at $\psi_0 = 0$. □

The factor of 2 has a structural origin: the two ideals share the round map from round 2 onward and differ exactly in round 1, where the free input of I_{gb} enters through a squaring (degree 2) while the free input of I_{sp} enters linearly (degree 1). The degree recurrence of Lemma 4.1 is multiplicative in the initial degrees, so the single round-1 factor of 2 propagates unchanged through all subsequent rounds to the eliminant degree. This identifies the half-density gap of Theorem G.5 as precisely the round-1 squaring. At wider widths the same relation $\dim(R/I_{\mathrm{gb}}) = 2 \cdot D_I = 2^{tn-(t-1)}$ was verified bit-exact at $t \in \{3, \ldots, 7\}$ (with the elimination polynomial of the free input squarefree of full colength degree at the tested cells, so that the colength equals the elimination degree), and the round-1 degree-doubling propagation was confirmed by the integer degree recurrence at $t \in \{3, \ldots, 7, 16\}$, $n \leq 5$.

Remark 5.11 (Scope of the proven regularity)**.** Theorem G.5 and the wider-state measurements of Remark 5.7 concern the rate-fixed ideal I_{gb}. For the capacity-fixed sponge ideal I_{sp}—the convention that matches the hash attack surface—the eliminant-degree closed form $D_I = 2^{t(n-1)}$ is machine-checked in Lean 4 (a kernel-verified `finrank` computation of the quotient-algebra dimension; supplementary material) and is additionally confirmed empirically across the 34/34 ($t = 2$) and 62/62 ($t \leq 4$) Gröbner dimension runs (Section 5.3); squarefreeness of the eliminant is verified at $n \in \{3, \ldots, 6\}$ (Theorem 5.12 gives genericity; the production constants were checked to lie outside the bad locus at the tested cells). What remains empirical for I_{sp} is its regularity, on which the eliminant attack does not load-bear: we do not claim transfer of the proven $d_{\mathrm{reg}} = 2n$ statement to the capacity-fixed ideal.

Separately, the wider-state CICO analysis of Section 7.4 fixes capacity $t-1$ (rate 1), whereas the instantiated 2-to-1 hash at $t \geq 3$ uses capacity 1 and rate $t-1$. The collision-relevant CICO for the latter leaves $t-2$ rate input variables free—an underdetermined system in which an attacker may specialize the free variables to optimize the resulting eliminant. The attacker's best such specialization is now proven (Proposition 5.16, Section 5.6): it costs the construction exactly $t-2$ bits, giving $D_{\mathrm{eff}} = 2^{t(n-1)-(t-2)}$, and all parameter sizing uses this figure rather than the rate-1 eliminant degree.

The first-syzygy and joint-signal structural observations attached to the closed form—the Koszul syzygy computation and the common origin of the half-density Hilbert gap and the exact per-round MDP $= 1/p$—are collected in Appendix F (Remarks F.1 and F.2).

Theorem 5.12 (Generic squarefreeness of the elimination polynomial)**.** *For every $n \geq 2$ and every odd prime p, let $f_n(\theta_0) \in \mathbb{F}_p(c_0, d_0, \ldots, c_{n-1}, d_{n-1})[\theta_0]$ denote the elimination polynomial of the Mambo CICO ideal at n rounds, with round constants treated as algebraic parameters. Then $\mathrm{disc}(f_n) \neq 0$ as an element of $\mathbb{F}_p[c_0, d_0, \ldots, c_{n-1}, d_{n-1}]$.*

Equivalently, the set $\mathcal{B}_n := \{\, (c_0, d_0, \ldots, c_{n-1}, d_{n-1}) \in \mathbb{F}_p^{2n} : \mathrm{disc}(f_n) = 0 \,\}$ of round-constant tuples for which f_n is non-squarefree is a proper algebraic subset of $\mathbb{F}_p^{2n}$, hence of codimension at least one.

Proof. By induction on n.

Base case. For $n \in \{2, 3, 4\}$, direct computation in Singular in the parameter ring yields $\gcd(f_n, f_n') = 1$, hence $\mathrm{disc}(f_n) \neq 0$.

Inductive step. Assume the theorem holds at level n. The level-$(n+1)$ system adds the equations $\theta_{n+1} - \varphi\,\theta_n - \psi_n^2 - c_n = 0$ and $\psi_{n+1} - \varphi\,\psi_n - \theta_{n+1}^2 - d_n = 0$. Substituting $\theta_{n+1} = \tau$ and using the level-n chain expansions $\theta_n(\theta_0)$ and $\psi_n(\theta_0)$, the elimination polynomial at level $n+1$ takes the form

$$f_{n+1}(\theta_0) = h_n(\theta_0) - c_n, \qquad h_n(\theta_0) := \tau - \varphi\,\theta_n(\theta_0) - \psi_n(\theta_0)^2, \tag{5}$$

where h_n has degree $\deg_{\theta_0}(\psi_n^2) = 2 \cdot 4^{n-1} = 4^n$ in θ_0 (Theorem 4.2). The variable d_n does not appear in f_{n+1}.

The discriminant of a polynomial $h(x)-c$ of degree D over a field of characteristic coprime to D is a polynomial in c of degree exactly $D-1$, with leading coefficient $(-1)^{D(D-1)/2}\,D^D\,(\mathrm{lc}_x\,h)^{D-1}$ (Gelfand–Kapranov–Zelevinsky [39], Ch. 12). Applied to (5) with $D = 4^n$ and $\mathrm{char}(\mathbb{F}_p) = p$ odd: the leading coefficient of $\mathrm{disc}(f_{n+1})$ as a polynomial in c_n equals $\pm(4^n)^{4^n}(\mathrm{lc}_{\theta_0}\,h_n)^{4^n-1} = \pm(4^n)^{4^n}(\mathrm{lc}_{\theta_0}\,\psi_n)^{2(4^n-1)}$, which is nonzero by Theorem 4.2 (no leading-coefficient cancellation). Hence $\mathrm{disc}(f_{n+1})$ is a non-trivial polynomial in c_n over the base ring $\mathbb{F}_p[c_0,\ldots,d_{n-1},\tau,\varphi]$, and consequently a non-trivial polynomial in $(c_0,\ldots,d_{n-1},c_n) \in \mathbb{F}_p^{2n+1}$. □

Remark 5.13 (Implication for meet-in-the-middle FGLM)**.** Squarefreeness of $f_n(\theta_0)$ is equivalent to the multiplication-by-θ_0 matrix M_{θ_0} in R/I having minimal polynomial of full degree D_I. By Theorem 5.12, this holds for all round-constant tuples outside the proper algebraic subset $\mathcal{B}_n \subset \mathbb{F}_p^{2n}$. The SHA-256-derived constants of Section 3.4 avoid $\mathcal{B}_n$ for the configurations we tested empirically (squarefreeness verified at $n \in \{3,4,5,6\}$ and at $p \in \{257\}$ by full elimination).

Full minimal-polynomial degree removes the first structural condition that the meet-in-the-middle FGLM time-memory tradeoff [38] exploits: a low-degree algebraic relation among the powers of M_{θ_0}. Empirically, the matrix powers $M_{\theta_0}^k$ densify monotonically (no row stays sparse past $k \sim \sqrt{D_I}$), the second enabling condition. We therefore found no structure enabling a sublinear-memory MITM variant in the tested range ($n \le 6$, $p = 257$); this is an evidence-scoped statement about the tested configurations, not a proof that no such tradeoff exists. The same full-degree property holds for the multiplication matrix of every middle coordinate F_m (Corollary 4.7), verified at $n \le 6$ for all splits over both production fields, so the MITM-relevant matrices on the middle coordinates carry no shorter minimal polynomial than M_{θ_0}. We note moreover that FGLM cost figures are not an attack-cost floor for the sponge CICO surface: the univariate path of Section 5.2 bypasses the FGLM step entirely at $\widetilde{O}(D_I)$ time.

5.6 The Capacity-1 Rate-Specialization Attack ($t \ge 3$, Self-Disclosed)

The deployed 2-to-1 hash at $t \ge 3$ runs the sponge at capacity $c = 1$ and rate $t-1$ (Section 7.2), whereas the eliminant degree $D_I = 2^{t(n-1)}$ of Conjecture 5.2 is computed for the rate-1 (capacity $t-1$) convention. At $c = 1$ the CICO system—capacity coordinate zeroed at the input, one output coordinate fixed to τ—is *one* equation in $t-1$ free rate inputs, an underdetermined system whose solution set is generically a $(t-2)$-dimensional variety. The attacker does not need the variety: one point suffices. The cheapest known strategy fixes $t-2$ of the free rate inputs to constants and root-finds the remaining univariate, choosing which input to leave free so as to minimize its degree. This is our own best attack on the deployed surface, adjacent to the multi-output CICO line of CheapLunch [8], and we disclose it together with its measured cost.

Freeing the rate input $x_j(0)$ (all other rate inputs set to constants, capacity $x_t(0) = 0$) yields $\deg_{x_j(0)} x_1(n) = 2^{\,t(n-1)-(j-1)}$: the degrees form a ladder, minimized at $j = t-1$, the rate input adjacent to the capacity. The attack therefore solves a univariate of degree

$$D_{\mathrm{eff}}(t,n) \;=\; 2^{\,t(n-1)-(t-2)} \;=\; \frac{D_I}{2^{\,t-2}},$$

a flat $(t-2)$-bit improvement over the rate-1 figure. This degree ladder, and the optimality of the latest-axis specialization, are now established unconditionally.

Lemma 5.14 (Per-variable degree vector)**.** *Fix width $t \ge 2$ and free a single rate input $x_j(0) = z$ ($1 \le j \le t-1$), all other rate inputs constant and $x_t(0) = 0$. Writing $d_i^{(r)} = \deg_z x_i(r)$, one round of the Mambo-t map gives $d_i^{(r+1)} = \max\!\big(\max_{1\le k\le i} 2^{\,i-k} d_k^{(r)},\, 2^{\,i} d_t^{(r)}\big)$ as an upper bound, with equality once the dominant term is established (Lemma 5.15). For $n \ge 2$, $\deg_{x_j(0)} x_1(n) = 2^{\,t(n-1)-(j-1)}$.*

Proof. The initial vector has $d_i^{(0)} = \mathbf{1}[i = j]$. Zero degrees are fixed by the recursion, so z propagates only forward: $d_i^{(1)} = 2^{\,i-j}$ for $i \geq j$ and 0 otherwise, whence $d_t^{(1)} = 2^{\,t-j}$. At round 2 the feedback term $2^{\,i} d_t^{(1)} = 2^{\,i+t-j}$ dominates the in-round terms $2^{\,i-j}$ by the factor 2^t, so $d_i^{(2)} = 2^{\,i+t-j}$ for all i. The profile $d_k = 2^{\,k+a}$ is fixed up to $a \mapsto a + t$ by one further round, giving $d_i^{(r)} = 2^{\,i+(r-1)t-j}$ for $r \geq 2$; set $i = 1$, $r = n$. □

Lemma 5.15 (No leading-coefficient cancellation)**.** *Along the dominant cascade,* $\mathrm{lc}_z\, x_i(r) = \varphi^{\deg_z x_i(r)}$*. In particular* $\mathrm{lc}_z\, x_1(n) = \varphi^{\,2^{\,t(n-1)-(j-1)}} \neq 0$*, independently of the round constants and the specialized rate inputs, so the degree of Lemma 5.14 is attained for every parameter choice.*

Proof. At each round the unique top-degree contribution is a squaring branch ($2 \deg_z x'_{i-1} > \deg_z x_i$ by the strict separation $t > 0$), and $\mathrm{lc}(p^2) = (\mathrm{lc}\, p)^2$ over the domain $\mathbb{F}_p$ (p odd); the linear summand φx_i and the degree-0 constant c_i lie strictly below the top degree and cannot cancel it. The variable z enters monic through $\varphi x_j(0)$, giving $\mathrm{lc} = \varphi$ at degree 1, and squaring preserves $\mathrm{lc} = \varphi^{\deg}$ at every step. The separation is independent of all constants, so no specialization lowers the degree. □

Proposition 5.16 (Specialization floor)**.** *For $t \geq 3$, $n \geq 2$, every 1-parameter specialization of the capacity-1 rate inputs yields an eliminant of degree at least* $D_{\mathrm{eff}} = 2^{\,t(n-1)-(t-2)}$*, attained by freeing $x_{t-1}(0)$. Hence D_{eff} is an unconditional lower bound on the per-permutation specialization-attack eliminant degree.*

Proof. A 1-parameter curve $x_k(0) = \gamma_k(s)$ has initial vector $\delta_k = \deg_s \gamma_k$; by Lemma 5.14 the minimal output degree over $\sum_k \delta_k \geq 1$ is the single axis $\delta_{t-1} = 1$, all other $\delta = 0$, giving $2^{\,t(n-1)-(t-2)}$; any earlier, multiple, or nonlinear dependence strictly raises it and, by Lemma 5.15, the raised leading term (again a pure square) does not cancel. □

The genericity condition is vacuous for the degree floor: by Lemma 5.15 the leading coefficient of the freed-axis eliminant is the fixed nonzero scalar $\varphi^{D_{\mathrm{eff}}}$, a degree-0 function of the specialized values and round constants, so its zero set is empty and no choice of specialization or constants lowers the degree below D_{eff}. The only residual constant-dependent property is whether the freed-axis eliminant is squarefree—which governs the root-finding structure, not the degree—and that is exactly the proper bad locus $\mathcal{B}_n$ of Theorem 5.12 (codimension ≥ 1); the self-disclosed cost is $\widetilde{O}(D_{\mathrm{eff}})$ regardless.

Empirical confirmation. Proposition 5.16 is corroborated bit-exactly: the closed form $D_{\mathrm{eff}} = 2^{t(n-1)-(t-2)}$ was measured at every cell of $t \in \{3, 4, 5\}$, $n \in \{2, \ldots, 5\}$ over $\mathbb{F}_{65521}$ with $\varphi = 7$, for two independent constant seeds, and the specialized univariate was squarefree in every cell (so Cantor–Zassenhaus root-finding genuinely runs in $\widetilde{O}(D_{\mathrm{eff}})$ with no repeated-factor savings). Adversarial attempts to beat the single-axis optimum failed: 300 random constant tuples per cell plus structured tuples (all-zero, all-one, ramp, all-φ) at $n \in \{3, 4\}$ produced degree exactly D_{eff} in every trial with the leading coefficient equal to $\varphi^{D_{\mathrm{eff}}}$ throughout (no cancellation ever observed, matching Lemma 5.15); specializing a coordinate to an affine function of an auxiliary variable only raises the degree; and a generic linear direction recovers the full $2^{t(n-1)}$. Reproducer scripts are included in the supplementary materials.

Consequences. The attack does not collapse wider state to the $t = 2$ security level (for $n \geq 3$, D_{eff} strictly exceeds $2^{2(n-1)}$), but it is a genuine weakening of the naive wide-state accounting, and we treat it as the binding figure: the falsifiable hardness statement for the $c = 1$ surface is Conjecture 5.2 with the exponent $t(n-1)$ replaced by $t(n-1) - (t-2)$, and every parameter set is sized by the rule $D_{\mathrm{eff}} = 2^{\,t(n-1)-\max(t-2,0)} \geq 2^{\lambda}$ of Section 5.2. At $t = 2$ there are no free rate inputs and nothing changes. On the 128-bit minimum row the rule costs one extra round

at $t = 4$ ($n_{\min} = 34$ instead of 33) and leaves $t = 3$ at $n_{\min} = 44$ unchanged; the recommended row adds the 25% round margin on top (Section 5.13).

5.7 Differential and Linear Cryptanalysis

Single-step differential uniformity. The map $x \mapsto x^2$ over $\mathbb{F}_p$ (p odd) has differential uniformity exactly 2: for any nonzero input difference δ_{in}, the equation $(x + \delta_{\text{in}})^2 - x^2 = \delta_{\text{out}}$ has at most 2 solutions.

Exact single-round MDP.

Proposition 5.17 (Per-round MDP $= 1/p$, exact)**.** *For every odd prime p and every round map $M_{\varphi,i}$ with $\varphi \neq 0$, every entry of the difference distribution table on $\mathbb{F}_p^2$ equals 0, 1, or p, and the value p is attained. Hence the maximum differential probability per round is exactly* MDP $= 1/p$*, half the generic upper bound $2/p$ for a degree-2 permutation on $\mathbb{F}_p^2$.*

Proof. Fix an input difference $(\Delta\theta, \Delta\psi) \neq (0, 0)$ and an output difference $(\Delta\theta', \Delta\psi')$. Differencing the two update equations of Definition 3.1 gives

$$\Delta\theta' = \varphi\Delta\theta + 2\psi\Delta\psi + \Delta\psi^2, \qquad \Delta\psi' = \varphi\Delta\psi + 2\theta'\Delta\theta' + (\Delta\theta')^2,$$

with $\theta' = \varphi\theta + \psi^2 + c_i$.

Case $\Delta\psi \neq 0$. The first equation is affine in ψ with nonzero slope $2\Delta\psi$, so each value of $\Delta\theta'$ is realized by exactly one ψ. For that ψ, $\theta \mapsto \theta'$ is a bijection of $\mathbb{F}_p$. If $\Delta\theta' \neq 0$, the second equation is affine in θ' with nonzero slope $2\Delta\theta'$ and determines θ' (hence θ) uniquely: the entry is 1. If $\Delta\theta' = 0$, the second equation collapses to $\Delta\psi' = \varphi\Delta\psi$ for every θ: the single entry $(0, \varphi\Delta\psi)$ receives all p preimages and the other $(0, \cdot)$ entries are 0.

Case $\Delta\psi = 0$, $\Delta\theta \neq 0$. Then $\Delta\theta' = \varphi\Delta\theta \neq 0$ identically, and the second equation is affine in θ' with nonzero slope $2\varphi\Delta\theta$: each $\Delta\psi'$ determines θ' uniquely, and each of the p choices of ψ yields exactly one θ. Every entry in the column $\Delta\theta' = \varphi\Delta\theta$ equals p and all others are 0.

In both cases the maximum entry is p, so MDP $= p/p^2 = 1/p$ exactly. The numerical argmax at $p = 31$, $\varphi = 7$ with the published round constants matches the predicted witnesses with 31 preimages. □

The structural origin of this factor-2 gap below the generic degree-2 ceiling, and its relation to the half-density Hilbert series of Theorem G.5, are discussed in Appendix F (Remark F.2). Under the standard trail (Markov) heuristic, a fixed n-round differential trail then has probability at most p^{-n}—already at the minimum cell $n = 65$ over BN254, below 2^{-16000}—while the rigorous multi-round bound that does not assume round independence is the Bézout ceiling (6) of Section 5.9.

Empirical evidence from small fields. After $n = 2$ steps, the max probability drops to approximately $10/p^2$—only a small constant above random. For deployment fields ($p > 2^{254}$), even $n = 1$ gives $1/p < 2^{-254}$.

Boomerang and rectangle distinguishers. We computed the full Difference Distribution Table (DDT) and Boomerang Connectivity Table (BCT) [34] for the Mambo round function over $\mathbb{F}_{31}$, $\mathbb{F}_{67}$, and $\mathbb{F}_{101}$.

The enumerated DDTs confirm Proposition 5.17 (maximum entry exactly p) across all tested primes. Multi-round trails were computed (not extrapolated) through $n = 3$: at $n = 2$ the best trail drops to $\sim 8/p^2$ (approximately $8\times$ random), confirming rapid decay. The plateau behavior at $n \geq 2$ and its extrapolation to large fields are analyzed in Section 5.9(c).

Table 4: Maximum differential probability on small fields.

p	n	Max diff. prob.	Random perm. expected	Ratio
31	1	$1/31$	$\sim 1/p^2$	$\sim p$
31	2	$7/31^2$	$\sim 1/p^2$	7
31	3	$7/31^2$	$\sim 1/p^2$	7
67	1	$1/67$	$\sim 1/p^2$	$\sim p$
67	2	$9/67^2$	$\sim 1/p^2$	9
101	1	$1/101$	$\sim 1/p^2$	$\sim p$
101	2	$8/101^2$	$\sim 1/p^2$	8
101	3	$8/101^2$	$\sim 1/p^2$	8

For the BCT with various splits ($E_0 = 1$–2 rounds, $E_1 = 1$–2 rounds), the maximum entry is 4–6× random. For BN254 ($p \approx 2^{254}$): a 1-round boomerang split gives probability $\sim 1/p^2 \approx 2^{-508}$, which is at the random threshold. Already with the minimum $n = 65$ and any nontrivial split, boomerang and rectangle attacks are infeasible.

Linear cryptanalysis. The linear approximation probability of $x \mapsto x^2$ is $O(1/\sqrt{p})$ by the Weil bound, leading to an n-step bias of $O(p^{-n})$; nonlinear approximations [30] do not improve this for a single quadratic layer.

5.8 Slide Attack Analysis

Slide attacks [43] exploit shift invariance: when every round applies the same function, an attacker can seek "slid pairs" $(\mathbf{x}, \mathbf{x}')$ with $\mathbf{x}' = M(\mathbf{x})$ and reduce the permutation to a single-round relation.

The introduction of distinct per-round constants (c_i, d_i) (Section 3.4) eliminates this attack vector entirely. Because $M_{\varphi,i} \neq M_{\varphi,j}$ for $i \neq j$, the permutation $P_{\varphi,n} = M_{\varphi,n-1} \circ \cdots \circ M_{\varphi,0}$ has no shift invariance, and the slide condition cannot be satisfied.

Even in an earlier version of Mambo without round constants, the birthday cost of finding a slid pair was $O(p)$ and fixed points of M_φ yielded only trivial slid pairs; the round constants are a defense-in-depth measure that eliminates the attack class unconditionally rather than relying on the large field size.

5.9 Small-Field Cryptanalysis Results (Full Disclosure)

(a) Integral distinguisher at $p = 101$. At $p = 101$ with low n, integral-style sums over structured input sets show non-random behavior. This does not scale to deployment fields where the field is astronomically larger than the degree.

(b) Impossible differentials at $p = 101$. Certain input/output difference pairs are impossible at $p = 101$—a small-field artifact where the algebraic structure is visible when p^2 is small.

(c) Differential uniformity plateau. For $n \geq 2$ rounds, the maximum differential probability is bounded by

$$\text{MDP}(n \text{ rounds}) \leq \frac{(2^n - 1)^2}{p^2}, \tag{6}$$

obtained by viewing the difference system as two equations of degree $2^n - 1$ in two unknowns and applying Bézout's theorem in $\mathbb{F}_p^2$. The bound assumes the difference equations form a regular sequence, a condition we verified empirically for $n \leq 4$ via full-DDT enumeration (no defect locus was observed where the intersection multiplicity exceeded $(2^n - 1)^2$).

The Bézout ceiling counts intersection points in the projective closure over the algebraic closure, including points at infinity, and is therefore conservative when restricted to affine $\mathbb{F}_p^2$-rational solutions. For Mambo specifically, the conservatism is structural: the same bilinear-quadratic asymmetry that produces the half-density Hilbert observation (Remark F.2, signal 1) makes the leading-degree forms of the iterated difference equations factor through one variable per equation. The projective intersection at infinity is therefore degenerate, and the affine $\mathbb{F}_p^2$-rational subset is far smaller than the $(2^n-1)^2$ projective Bezout count. The empirical max-DDT prefactor ≤ 10 at $n \leq 4$ reflects this gap.

Empirically, the prefactor of $1/p^2$ is much smaller than the Bézout ceiling: full-DDT enumeration at $p \in \{31, 61\}$ for $n \in \{2, 3, 4\}$ yields max-DDT-entry ≤ 10 in every measured case, whereas the Bézout ceiling $(2^n-1)^2$ grows as $9, 49, 225$ for $n = 2, 3, 4$. Consistent with the small-field figures reported in the differential table of Section 5.7. At $p = 65521 \approx 2^{16}$, a sampled enumeration with 10^6 random inputs at the structural-witness input difference $\Delta_{\text{in}} = (1, 0)$ produces a modal output-difference count of 32, in agreement with the Gumbel maximum $\mu + \sqrt{2\sigma^2 \ln p} \approx 33$ predicted under a $1/p$ multinomial for that fixed input difference. This confirms that the DDT row at the structural witness behaves as a $1/p$ multinomial: the $(a, 0)$ family achieves $1/p$ exactly (Proposition 5.17) with the remainder of the row consistent with random. It does not, by itself, certify the global multi-round MDP prefactor across all $(\Delta_{\text{in}}, \Delta_{\text{out}})$ pairs at $p = 65521$; that prefactor inherits the conservative Bézout ceiling (6) via the leading-form degeneracy noted above.

For deployment-scale primes ($p \approx 2^{254}$) the plateau prefactor is not empirically resolvable by random sampling, since detecting any collision in the differential histogram would require $\Theta(p)$ samples per input difference. The security argument therefore relies on (6) as the upper bound that extrapolates to large fields, not on the empirical prefactor. Whether the empirical prefactor remains bounded by a small constant for n at the deployment-scale $n = 82$ is open and is not load-bearing: the proven Bézout ceiling $(2^{82} - 1)^2/p^2 < 2^{-344}$ at $p \approx 2^{254}$ is already far below cryptographic relevance. The single-round identity $\text{MDP}_{n=1} = 1/p$ established in Proposition 5.17 (and exhibited by its two structural witness families) is the only multi-round-independent component of the bound.

These anomalies are expected for any algebraic construction and are analogous to small-field distinguishers known for Poseidon and Rescue-Prime.

5.10 The $k = 1$ Additive Attack (Self-Disclosed)

Definition 5.18 (Additive extraction)**.** Given a Mambo state (θ, ψ), the additive extraction outputs $\theta + \psi \pmod p$.

Proposition 5.19 ($k = 1$ seed recovery)**.** *Consider the PRNG of Definition 7.2 at $k = 1$ with public outputs. The attack model is the standard one for public-output generators: the adversary observes consecutive outputs and knows the circuit. Suppose the adversary holds, besides the first output equation $\theta_1 + \psi_1 = \sigma_1$ (degree 4 in the secret seed (θ_0, ψ_0)), any one additional polynomial constraint $g(\theta_0, \psi_0) = 0$ of degree d that is algebraically independent of it. Then the adversary recovers the seed in $O(\text{poly}(d, \log p))$ field operations: the bivariate resultant eliminates one variable, leaving a univariate of degree at most $4d$ whose $\mathbb{F}_p$-roots are found by Cantor–Zassenhaus and checked forward through the step map.*

The hypothesis is not hypothetical—the attack model itself supplies the second constraint. The *second consecutive output* gives $\theta_2 + \psi_2 = \sigma_2$, of degree 16 in the seed, so two observed outputs determine the seed up to at most $4 \cdot 16 = 64$ candidates (Bézout) at cost $O(\text{poly}(\log p))$; in our experiments at $p \in \{101, 257, 1009\}$ with the production constant schedule, the two-output system was zero-dimensional in every trial and the number of $\mathbb{F}_p$-rational candidate seeds was at most 4. Alternatively, partial seed knowledge (any affine relation $\ell(\theta_0, \psi_0) = b$, e.g. a public counter in one coordinate) is a $d = 1$ instance, reducing recovery to a single quartic. The $k = 1$

public-output configuration is therefore broken outright, with no auxiliary assumption, and is listed in the cryptanalytic-record column of Table 1 as a self-disclosed attack.

Mitigation. Three mitigations:

1. Hash and permutation modes are unaffected (full state output or sponge mode).
2. In ZK, the output is a private witness—no information leaks.
3. For public-output PRNG the same iterated-degree analysis as for CICO applies: outputs at step distance k have degree 4^k in the seed, and the analysis-equality minimum for λ-bit security is $k \geq 33$ for $\Omega(2^{64})$ and $k \geq 65$ for $\Omega(2^{128})$. The recommended public-output configuration applies the 25% round margin of Section 5.13: $k = 82$ at the 128-bit level, costing 164 NL per output.

5.11 Linearization and Other Algebraic Attacks

Kipnis–Shamir linearization. Each step introduces 2 nonlinear monomials. After n steps, auxiliary variables yield a system with $4n + 2$ unknowns and $2n$ equations, so the system is underdetermined by $2n + 2$. Determining polynomial relationships reduces to the Gröbner basis problem of Section 5.5.

XL and FXL attacks. Require working at degree $d_{\text{reg}} = 130$ for the determined system already at the minimum $n = 65$ (164 at the recommended $n = 82$). The resulting matrix dimensions are infeasible.

SAT/SMT solvers. Require bit-decomposition producing $O(n \cdot \log^2 p)$ boolean variables—exceeding 10^6 variables with highly nonlinear structure at $n = 82$, $\log_2 p = 254$.

Statistical test results (PractRand, BigCrush, avalanche) are part of the consolidated evidence of Section 5.3.

5.12 Sponge-Level Coverage: Multi-Block Inputs, Fixed Points, and Invariant Subspaces

The CICO analysis above prices the single-permutation, single-output instance. This subsection records what is and is not known one level up, at the sponge.

Multi-block and two-output CICO. Collision-style attacks constrain *two* outputs, or chain constraints across absorbed blocks; both give the attacker more freedom than the one-output instance of Conjecture 5.2. What is known exactly: the two boundary systems that bracket every such instance have known quotient dimensions with no degree collapse. The one-output system has eliminant degree D_{eff} (measured exactly at every cell of $t \in \{3, 4, 5\}$, $n \leq 5$; Section 5.6), and the fully determined system—every remaining input and output cell fixed—has colength exactly $2D_I = 2^{tn-(t-1)}$ (Proposition 5.10; proven at $t = 2$, verified bit-exact at $t \in \{3, \ldots, 7\}$). A two-output instance sits between these brackets: the attacker gains constraint-selection freedom, but every quotient dimension that bounds the resulting ideal is known exactly rather than estimated. The relevant external attack family is the multi-output line of CheapLunch [8], which extends FreeLunch resolution past single-output CICO via weighted orders and tightened ideal-degree bounds; its resolution cost is driven by the same ideal degrees that the brackets above pin down. The cross-block analogue of the within-block specialization of Section 5.6—multi-block absorption in which the attacker chooses the intermediate chaining values, each absorbed block adding a free rate variable mid-computation—we have measured at small cells ($t, n \in \{2, 3\}$, two seeds, plus a BN254 spot-check): the two-block effective degree equals the single-block $D_{\text{eff}} = 2^{t(n-2)+2}$ exactly, with *no* gain beyond the per-block $(t-2)$ bits. Structurally, the chaining value occupies the capacity cell, the deepest position in the squaring cascade, so

treating it as free *raises* the eliminant degree (e.g. 2^7 versus $D_{\text{eff}} = 2^5$ at $t = 3$, $n = 3$); the blocks compose in series and an attacker cannot amortize one root-find against the other, so the two-block cost never falls below the single-block floor and the parameter margin is unaffected. A general-(t, n) proof—the degree-tracking argument of Proposition 5.16 applied to a nonzero incoming capacity—remains open and is listed in Section 9; the ideal-permutation sponge bound (Section 6) covers the general case meanwhile.

Fixed points and cycle structure. For a uniformly random permutation of $\mathbb{F}_p^t$ the fixed-point count is asymptotically Poisson with mean 1, and the expected number of m-cycles is $1/m$. For one Mambo round the fixed-point condition is the determined quadratic pair $\theta = \varphi\theta+\psi^2+c_i$, $\psi = \varphi\psi + \theta^2 + d_i$, with Bézout count 4 (at width t: t quadrics, Bézout 2^t), so each round map has at most 4 fixed points. A point fixed by *every* round map would force $\psi^2 = (1 - \varphi)\theta - c_i$ for all i simultaneously, hence $c_0 = c_1 = \cdots = c_{n-1}$; the SHA-256-derived constants are pairwise distinct, so no such point exists—this is the fixed-point-structure issue of the constant-free design that motivated the per-round constants. For the full permutation $P_{\varphi,n}$ the fixed-point system $(\theta_n-\theta_0, \psi_n-\psi_0)$ is a determined pair of degrees $(2{\cdot}4^{n-1}, 4^n)$, and we know of no structure forcing $\mathbb{F}_p$-rational solutions beyond the random-permutation expectation. We state the limits honestly: cycle structure at production size is not enumerable, the statistical batteries of Section 5.3 do not probe global cycle statistics, and what the algebra supplies is the per-round Bézout ceiling, the impossibility of common round fixed points, and the absence of structural fixed subspaces (next paragraph).

Invariant subspaces. At $t = 2$ the proper nontrivial affine subspaces of $\mathbb{F}_p^2$ are points and lines, so the following is a complete classification at width 2. Points: covered above (no common fixed point of the round maps). Lines: no affine line is mapped to an affine line by any round map $M_{\varphi,i}$—in particular none is invariant, and no line-to-line subspace trail of the Anemoi/Griffin type exists. The argument is an unconditional leading-coefficient comparison: parameterize a line as $x \mapsto (x, ax + b)$ (or the vertical line $\theta = b$); its image under $M_{\varphi,i}$ has θ-component of degree 2 in the parameter when $a \neq 0$ (degree 1 when $a = 0$) and ψ-component of degree 4 (resp. 2), with leading coefficients a^4 (resp. φ^2) that cannot vanish; an affine image would force the higher-degree component to be an affine function of the lower-degree one, which the nonvanishing leading coefficient excludes. For the vertical line $\theta = b$ the image components have degrees 2 and 4 in the parameter with unit leading coefficients, with the same conclusion. This rules out, for every round map and hence for every window of consecutive rounds, the entire class of one-dimensional affine invariant structures over $\mathbb{F}_p$ and over every extension field. What remains heuristic: at $t \geq 3$ the same squaring-cascade degree argument rules out the coordinate-aligned families (any subspace with a free coordinate feeding the squaring chain), but we have not exhaustively classified invariant affine subspaces of intermediate dimension at $t \geq 3$, and a sponge-relevant invariant set need not be affine at all. The non-affine and wider-width cases are part of the cryptanalysis challenge of Section 9.

5.13 Parameter Selection and Safety Margins

The recommended parameters at $t = 2$ are $\varphi = 7$ and $n_{\text{rec}} = 82$ over BN254 ($n_{\text{min}} = 65$).

Parameter tiers: minimum and recommended rows. We propose three tiers aligned to NIST security levels. Each tier cell is stated as two round counts. The *minimum* count n_{min} is the smallest n satisfying the conservative cost model (Section 5.2), $D_{\text{eff}} = 2^{t(n-1)-\max(t-2,0)} \geq 2^\lambda$, i.e. $n \geq \lceil(\lambda + \max(t - 2, 0))/t\rceil + 1$—exact analysis equality, zero margin. The *recommended* count is $n_{\text{rec}} = \lceil 1.25\, n_{\text{min}}\rceil$. All integrator-facing figures in this paper use n_{rec}; the minimum row is kept visible because it anchors the analysis and is the falsification target of the cryptanalysis

Table 5: Minimum rounds for 128-bit security at state width $t = 2$ under the conservative cost model (Section 5.2). The round-count requirement is field-independent: $n_{\min} = \lceil \lambda/t \rceil + 1$, from $D_I = 2^{t(n-1)} \geq 2^\lambda$ (at $t = 2$ the specialization margin $\max(t-2, 0)$ of Section 5.6 vanishes), giving $n_{\min} = 65$ at $t = 2$, $\lambda = 128$; the recommended round count adds the 25% margin, $n_{\text{rec}} = 82$. The field enters only through the saturation condition $t \log_2 p \geq \lambda$.

Field	$\log_2 p$	$n_{\min}$ (t=2)	Saturation ($2 \log_2 p \geq 128$)
BN254 ($\mathbb{F}_r$)	254	65	met
Goldilocks	64	65	met exactly
Mersenne-61	61	—	**fails; use $t \geq 3$**
BabyBear	31	—	**fails; no recommended width**

For Mersenne-61, width $t \geq 3$ restores admissibility ($3 \cdot 61 = 183 \geq 128$). For BabyBear and comparable 31-bit primes, $t \log_2 p \geq 128$ first holds at $t = 5$, but the widths $t \geq 5$ are research configurations without security analysis (see the field-applicability discussion below): no recommended 128-bit configuration currently exists over 31-bit fields.

challenge (Sections 5.2 and 9). The state width t generalizes Mambo from the (θ, ψ) pair ($t = 2$) to triangular coupling on t elements (Section 7.4); each width preserves ALFOM = 1.

The size of the margin is a deliberate choice, and we justify it honestly rather than presenting it as convention. The degree analysis behind D_{eff} is exact and certified—closed forms, machine-checked at $t = 2$, measured bit-exact elsewhere—so the margin guards against *model risk*, not degree uncertainty: future improvements in eliminant processing (change-of-ordering refinements, root-finding constants, memory-pricing erosion of the kind already exhibited by the model-dependent tier) could cheapen the attack pipeline without touching the degree. Because each added round multiplies D_{eff} by 2^t, the +25% round margin buys at least 32 bits of additional attack cost at L1 ($+34/+33/+36$ bits at $t = 2/3/4$), i.e. any future improvement would have to recover more than 2^{32} in cost before the recommended cells degrade to the (already conservative) analysis equality.

The $t = 3$ and $t = 4$ cells rest on Conjecture 5.2 alone, exactly as the $t = 2$ cells do. The specialization floor $D_{\text{eff}} = 2^{t(n-1)-(t-2)}$ used to size these cells is now a proven unconditional lower bound (Proposition 5.16), and the non-telescoping degree floor invoked at every width is a full theorem for every elimination strategy (Proposition 4.6); neither is any longer a separate hypothesis. The single remaining assumption at $t \geq 3$ is therefore Conjecture 5.2, identical to the $t = 2$ case, where there is no rate-specialization freedom and the binding degree $D_{\text{eff}} = D_I = 4^{n-1}$ is likewise a proven closed form.

The recommended 128-bit cells achieve the 2-to-1 sponge hash at 165 nonlinear constraints ($t = 3$, $n = 55$; the $t = 4$, $n = 43$ variant costs 172; Table 1); the narrow-state $t = 2$ variant costs 164 nonlinear constraints per permutation (Section 7.5).

For reference, Table 7 lists the parameter sets obtained under the dense-FGLM cost model. They are certified only under that model's memory-priced accounting (Section 5.2) and are not recommendations.

Field applicability. The field-saturation bound $D_I^{(\mathbb{F}_p)} \leq p^t$ (Section 5.4) restricts the set of base fields for which a given (t, n) parameter choice remains valid. Writing λ for the target bit-security level, the conservative requirement $D_I \geq 2^\lambda$ (Section 5.2) combined with $D_I^{(\mathbb{F}_p)} \leq p^t$ gives the field-admissibility condition

$$p^t \;\geq\; 2^\lambda \quad\Longleftrightarrow\quad t \log_2 p \;\geq\; \lambda\,.$$

Below this threshold the formal degree $2^{t(n-1)}$ is short-circuited by field saturation and the tier is not achievable at width t over that field. For fields with $\log_2 p \approx 31$, no tier is reachable at the widths $t \in \{2, 3, 4\}$ under the conservative model ($4 \cdot 31 = 124 < 128$): clearing the saturation

Table 6: Mambo parameter tiers (9 cells across 3 state widths and 3 security levels), sized under the conservative cost model: $D_{\text{eff}} = 2^{t(n-1)-\max(t-2,0)} \geq 2^{\lambda}$ (Sections 5.2 and 5.6). Each cell carries a minimum row (analysis equality; the falsification target) and a recommended row ($n_{\text{rec}} = \lceil 1.25\, n_{\min} \rceil$); deployments and all headline figures use the recommended row. Tier labels denote the permutation-CICO level λ; collision resistance of the capacity-1 sponge over BN254 is 127 bits at L1 (birthday bound over $p \approx 2^{254}$; Corollary 6.3), and this qualifier applies to every L1 hash figure in the paper.

Tier (NIST level)	Row	Target	$t = 2$	$t = 3$	$t = 4$	Min D_{eff} (bits)
L1	minimum	128-bit	n=65	n=44	n=34	≥ 128
	recommended	128-bit	n=82	n=55	n=43	≥ 160
L3	minimum	192-bit	n=97	n=66	n=50	≥ 192
	recommended	192-bit	n=122	n=83	n=63	≥ 240
L5	minimum	256-bit	n=129	n=87	n=66	≥ 256
	recommended	256-bit	n=162	n=109	n=83	≥ 320

Each minimum entry is the smallest n with $t(n-1) - \max(t-2, 0) \geq \lambda$; each recommended entry is $\lceil 1.25\, n_{\min} \rceil$. The achieved D_{eff} exponents on the minimum row are: L1, $128, 128, 130$ for $t = 2, 3, 4$ respectively; L3, $192, 194, 194$; L5, $256, 257, 258$. On the recommended row: L1, $162, 161, 166$; L3, $242, 245, 246$; L5, $322, 323, 326$. Every column rests on Conjecture 5.2 alone: the $t \in \{3, 4\}$ specialization floor D_{eff} is the proven Proposition 5.16, so no column carries a hypothesis beyond the CICO conjecture (discussed above the table).

Table 7: Parameter sets under the dense-FGLM cost model ($n \cdot D_I^2$, requiring $D_I \geq 2^{\lambda/2}$; see the cost-model discussion in Section 5.2). Against the $\widetilde{O}(D_I)$-time univariate attack these sets retain λ-bit security only if the attacker's $\Theta(D_I \log p)$-bit memory is priced (area–time metric); at $t \geq 3$ the capacity-1 specialization of Section 5.6 erodes them by a further $t - 2$ bits. Not recommendations.

Level	Target	$t = 2$	$t = 3$	$t = 4$	Min D_I (bits)
L1	128-bit	n=33	n=23	n=18	≥ 64
L3	192-bit	n=49	n=33	n=25	≥ 96
L5	256-bit	n=65	n=44	n=33	≥ 128

bound there requires wider triangular coupling, with $t = 5$ the smallest width passing the bound at $\lambda = 128$. We state the consequence plainly: *no recommended 128-bit configuration currently exists on 31-bit fields.* Saturation admissibility is necessary, not sufficient—every width $t \geq 5$ is a *research configuration, insecure-until-analyzed*, because its small tier-rule round counts fall in the $n \leq t$ regime, outside the analyzed range of both the regularity measurements ($t \leq 4$; Remark 5.7) and the specialization measurements ($t \leq 5$, $n \leq 5$; Section 5.6). Closing the $n \leq t$ analysis gap is the named open problem whose resolution would change this (Section 9); until then the $t \in \{16, 32\}$ Circle-STARK measurements of Appendix E are per-round-count references, not endorsements. Table 8 tabulates admissibility for the base fields of principal interest.

Table 8: Field admissibility of the recommended parameter tiers under the conservative saturation condition $t \log_2 p \geq \lambda$. A check (✓) means the saturation bound does not bind; a dash (—) means the field saturates below target and the tier is not achievable at that width. BN254 admits all nine entries; 31-bit fields fail at every analyzed width, and the widths $t \geq 5$ that would pass the saturation bound are research configurations—insecure-until-analyzed (see text).

	L1 ($\lambda = 128$)			L3 ($\lambda = 192$)			L5 ($\lambda = 256$)		
Field (approx. $\log_2 p$)	t=2	t=3	t=4	t=2	t=3	t=4	t=2	t=3	t=4
BN254 (≈ 254)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Goldilocks (≈ 64)	✓*	✓	✓	—	✓*	✓	—	—	✓*
Mersenne-61 (≈ 61)	—	✓	✓	—	—	✓	—	—	—
M31 (≈ 31)	—	—	—	—	—	—	—	—	—
BabyBear (≈ 31)	—	—	—	—	—	—	—	—	—

* Goldilocks meets the threshold exactly at these entries ($t \log_2 p = \lambda$, e.g. $p^2 = 2^{128}$ at (t=2, λ=128)). We recommend the next wider width at those tiers over Goldilocks to retain margin. KoalaBear ($\log_2 p \approx 31$), used by the SP1 prover, shares the admissibility of BabyBear and M31 under the same $t \log_2 p \geq \lambda$ bound and is omitted from the table only to avoid duplication. The saturation bound alone would first pass on 31-bit fields at $t = 5$ ($\lambda = 128$), $t = 7$ (192), and $t = 9$ (256), but passing saturation is not a recommendation: every width $t \geq 5$ is a research configuration in the unanalyzed $n \leq t$ regime (see text), and the $t \in \{16, 32\}$ M31 measurements of Appendix E are per-round-count references only.

Remark 5.20 (Choice of $\varphi = 7$)**.** The multiplier φ must satisfy $\varphi \neq 0$ (non-degeneracy) and $\varphi \neq \pm 1$ (these yield weaker mixing and additional fixed points in our experiments). Beyond these exclusions, φ should have high multiplicative order in $\mathbb{F}_p^*$ to prevent short periodic structures. For the BN254 scalar field, $\varphi = 7$ is the smallest prime satisfying: (i) non-quadratic-residue ($7^{(p-1)/2} \neq 1$), (ii) non-cubic-residue ($7^{(p-1)/3} \neq 1$), and (iii) multiplicative order equal to $p - 1$ on the BN254 scalar field (verified directly by computing $7^{(p-1)/q} \bmod p$ for every prime divisor q of $p - 1$ in the BN254 factorisation; the test rules out small subgroup orders that would otherwise allow short periodic structures). These properties prevent algebraic shortcuts via square-root or cube-root extraction. The small magnitude (3 bits) also minimizes circuit cost, since multiplication by a small constant can be decomposed efficiently.

The same selection procedure applies field-generically: exclude 0 and ± 1, require a non-quadratic (and, when $3 \mid p-1$, non-cubic) residue of maximal multiplicative order, and take the smallest such prime for circuit cost. On Goldilocks ($p = 2^{64} - 2^{32} + 1$, $p - 1 = 2^{32} \cdot 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$), $\varphi = 7$ satisfies all criteria: it is a non-quadratic residue ($7^{(p-1)/2} \neq 1$), a non-cubic residue ($7^{(p-1)/3} \neq 1$), and has multiplicative order exactly $p - 1$ (verified by direct exponentiation against every prime divisor of $p - 1$). The published Goldilocks test vector (Appendix B) is therefore pinned at the conforming $\varphi = 7$, consistent with the BN254 vector. An earlier draft used a non-conforming Goldilocks multiplier (a cubic residue of multiplicative order $(p - 1)/3$); it has been retired in favour of $\varphi = 7$.

6 Sponge Indifferentiability and Field-Size Requirements

6.1 The Sponge Indifferentiability Theorem

Theorem 6.1 (BDPV08 [42])**.** *Let P be a random permutation on a state of size N, and let* Sponge[P] *be the sponge with rate r and capacity c. Then* Sponge[P] *is indifferentiable from a random oracle up to*

$$\mathrm{Adv}^{\mathrm{indiff}}(q) \leq \frac{q(q+1)}{2N_c},$$

where N_c is the size of the capacity space.

6.2 Conditions for Applying the Theorem

1. P is a permutation (bijection). **Proven** (Remark 3.2).
2. P is efficiently computable. **Proven**: $O(n)$ field operations.
3. P is efficiently invertible. **Proven**: $O(n)$ field operations.
4. P behaves as an ideal random permutation. **Modeled** (Design Claim 5.1; falsifiable form Conjecture 5.2).
5. The capacity must be large enough. **Field-dependent**.

6.3 Security Levels by Field

Proposition 6.2 (Sponge security, conditional)**.** *If $P_{\varphi,n}$ is modeled as a random permutation, the Mambo sponge hash with $r = 1$, $c = 1$ achieves:*

$$\mathrm{Adv}^{\mathrm{indiff}}(q) \leq \frac{q(q+1)}{2p}\,.$$

Corollary 6.3 (Mambo collision resistance, ideal-permutation model)**.** *Let H_{Mambo} denote the Mambo sponge hash with rate r and capacity c over $\mathbb{F}_p$, instantiated with the permutation $P_{\varphi,n}$. In the ideal-permutation model (modeling $P_{\varphi,n}$ as a uniformly random permutation, the modeling whose soundness for the recommended parameters is asserted by Design Claim 5.1), H_{Mambo} provides collision resistance with advantage*

$$\mathrm{Adv}^{\mathrm{coll}}_{H_{\mathrm{Mambo}}}(q) \;\leq\; \frac{q(q+1)}{2\,p^c}\,,$$

where q is the number of adversary queries to the underlying permutation. Equivalently, finding a collision requires approximately $q \geq \sqrt{p^c}$ queries.

Proof sketch. Immediate from Theorem 6.1 (BDPV08) with capacity space size $N_c = p^c$. The birthday-bound collision threshold is $q \approx \sqrt{2N_c} = \sqrt{2p^c}$. □

Concrete security for Mambo parameters. For $r = 1, c = 1$ (standard Mambo sponge):

- BN254 ($\log_2 p \approx 254$): 127-bit collision resistance.
- BLS12-381 ($\log_2 p \approx 255$): 127-bit collision resistance.
- Goldilocks ($\log_2 p = 64$): 32-bit collision resistance (internal use only).

For 128-bit collision resistance on public outputs, use a field with $\log_2 p \geq 256$ (capacity space $\geq 2^{256}$).

Collision resistance is independent of n. The collision bound depends on the capacity space size p^c, not the number of rounds n. Once $P_{\varphi,n}$ satisfies the indifferentiability hypothesis (requiring sufficient algebraic degree and diffusion), additional rounds do not improve collision resistance. The round count n is therefore set by the permutation-level CICO floor $D_{\mathrm{eff}} = 2^{t(n-1)-\max(t-2,0)} \geq 2^{\lambda}$ (Section 5.2, Table 6), not by the sponge collision bound.

7 Instantiations

7.1 Permutation Mode

The Mambo permutation $P_{\varphi,82}\colon \mathbb{F}_p^2 \to \mathbb{F}_p^2$ (the recommended 128-bit cell; the analysis minimum is $n = 65$) outputs the full state (θ_{82}, ψ_{82}).

- R1CS cost: $2 \times 82 = 164$ NL constraints (130 at the minimum cell).
- Security: CICO hardness (Conjecture 5.2); cf. Design Claim 5.1.
- No known attacks on the permutation.

Table 9: Sponge security levels by field at $c = 1$ (top), and the resulting minimum field sizes per construction (bottom).

Field	$\log_2 p$	Preimage (bits)	Collision (bits)	Status
BabyBear	31	31	15	Toy only
Mersenne-61	61	61	30	Insufficient
Goldilocks	64	64	32	Insufficient
BN254 (scalar)	254	254	127	Near-standard
BLS12-381	255	255	127	Near-standard
256-bit prime	256	256	128	Full security

Construction	Min $\log_2 p$	Reason
Permutation (ZK witness)	31	Internal use only
PRNG (private output)	31	State is secret
Hash (public, $\geq$64-bit)	254	Collision $\geq 2^{127}$
Hash (public, $\geq$128-bit)	256	Collision $\geq 2^{128}$

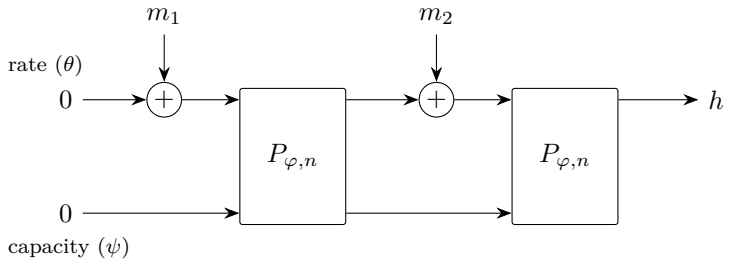


Figure 2: The Mambo 2-to-1 sponge hash at $t = 2$ (rate 1, capacity 1; Definition 7.1). The capacity lane is never exposed: it is zero-initialised, passes only through the permutation, and is discarded at squeeze time. At width $t \geq 3$ the rate lane carries $t - 1$ cells, both message elements are absorbed at once, and a single permutation call suffices.

7.2 Sponge Hash

The 2-to-1 sponge mode is shown in Figure 2.

Definition 7.1 (Mambo sponge hash)**.** Given messages $m_1, m_2 \in \mathbb{F}_p$:

1. Initialize: $(\theta, \psi) \leftarrow (0, 0)$.
2. Absorb m_1: $\theta \leftarrow \theta + m_1$; $(\theta, \psi) \leftarrow P_{\varphi,n}(\theta, \psi)$.
3. Absorb m_2: $\theta \leftarrow \theta + m_2$; $(\theta, \psi) \leftarrow P_{\varphi,n}(\theta, \psi)$.
4. Squeeze: output θ.

R1CS cost depends on state width. At $t=2$ (rate 1, two permutation calls per 2-to-1 hash), the theoretical count is $4n$; at the recommended $n=82$ this is 328 NL. At the $n=65$ minimum cell, Circom 2.2.3 measures 259 nonlinear constraints (Table 13), one below the theoretical $4n = 260$ because the capacity is zero-initialised, making the first permutation call's round-1 squaring $\psi_0^2 = 0 \cdot 0$ a constant that the compiler folds; the same structural fold applies at every n. At wider state $t \geq 3$ (rate $t-1$, single permutation call): tn. Recommended 128-bit 2-to-1 hash costs: 165 NL at $t=3, n=55$; 172 NL at $t=4, n=43$ (Table 1).

7.3 PRNG Mode

Definition 7.2 (Mambo PRNG)**.** Given a secret seed (θ_0, ψ_0) and step parameter k:

1. Apply k Mambo steps.

2. Extract: output $\theta_k + \psi_k \pmod p$.
3. Repeat.

R1CS cost: $2k$ NL per output. At $k = 1$: 2 NL per output element.

Security caveat. At $k = 1$ with public output, additive extraction is broken (Proposition 5.19). The PRNG at $k = 1$ is secure only when output is a private ZK witness. For public output the analysis minimum is $k \geq 65$ for 128-bit security; the recommended configuration is $k = 82$ (164 NL per output), applying the margin rule of Section 5.13.

Conjecture 7.3 (Hidden Coupling Property)**.** *For the Mambo PRNG with $k \geq 65$, the output sequence is computationally indistinguishable from random over $\mathbb{F}_p$.*

Lemma 7.4 (HCP in the ideal-permutation model)**.** *Let $P_{\varphi,k}$ be the k-round Mambo permutation on $\mathbb{F}_p^2$ and let PRNG_k be the Mambo PRNG of Definition 7.2 with step parameter k. For any distinguisher $\mathcal{D}$ making q output queries to PRNG_k, there is a permutation distinguisher $\mathcal{B}$ of comparable running time such that*

$$\mathrm{Adv}^{\mathrm{HCP}}_{\mathcal{D}}(\lambda) \;\leq\; \mathrm{Adv}^{\mathrm{perm}}_{\mathcal{B}}(\lambda) \;+\; \frac{q(q+1)}{2\,p^2},$$

where $\mathrm{Adv}^{\mathrm{perm}}_{\mathcal{B}}$ denotes $\mathcal{B}$'s advantage in distinguishing oracle access to $P_{\varphi,k}$ from a uniformly random permutation on $\mathbb{F}_p^2$. In the ideal-permutation model the first term vanishes and the Mambo PRNG output sequence is $\frac{q(q+1)}{2p^2}$-close to uniform on $\mathbb{F}_p$; for the concrete permutation, Design Claim 5.1 asserts that the first term stays below the target threshold at the recommended parameters with $k \geq 65$ (recommended $k = 82$; Section 7.3).

The proof is a standard three-hybrid argument (permutation switch, birthday-plus-cycle-closure switch, exact uniformity of the sum extraction) and is given in Appendix F; the second hybrid contributes $\binom{q}{2}/p^2$ for collisions plus q/p^2 for the event that the random-permutation orbit of the seed closes within q steps.

Why the sum extraction. The reduction uses only that the extraction map $f : \mathbb{F}_p^2 \to \mathbb{F}_p$, $f(\theta, \psi) = \theta + \psi$ is balanced (the pre-image of every output has size p). Any balanced linear extraction enjoys the same bound; the choice $f(\theta, \psi) = \theta + \psi$ in Definition 7.2 is for circuit economy (the extraction is a linear combination and costs 0 nonlinear constraints; the $2k$ NL per output is the step cost alone), not security.

The reduction in Lemma 7.4 extends verbatim to Mambo-t (Section 7.4): in the hybrid argument, $\mathbb{F}_p^2$ is replaced by $\mathbb{F}_p^t$, the birthday term becomes $q(q+1)/(2p^t)$, and the sum-extraction balance argument generalises to any balanced linear extraction $\mathbb{F}_p^t \to \mathbb{F}_p^r$ for rate $r \leq t - 1$. The lemma reduces HCP to a permutation-distinguishing advantage without algebraic hypotheses on the underlying permutation; the independent empirical evidence for the structural plausibility of Design Claim 5.1 and Conjecture 5.2—the 34/34 and 62/62 eliminant-degree batteries—is collected in Section 5.3.

Corollary 7.5 (HCP redundancy)**.** *Conjecture 7.3 follows from Lemma 7.4 in the ideal-permutation model and need not be assumed independently. The Mambo security argument therefore rests on a single assumption (Design Claim 5.1, with falsifiable algebraic form Conjecture 5.2), occupying the same epistemic position as the ideal-permutation assumptions underlying Keccak, Poseidon, and Anemoi.*

7.4 Wider-State Variants ($t \geq 3$)

The triangular coupling generalizes to any state width $t \geq 2$. Let $(x_1, \ldots, x_t) \in \mathbb{F}_p^t$. The Mambo-$t$ step applies t sequential squarings, each feeding into the next:

$$x_1' = \varphi \cdot x_1 + x_t^2 + c_{r,1},$$
$$x_i' = \varphi \cdot x_i + (x_{i-1}')^2 + c_{r,i} \quad \text{for } i = 2, \ldots, t.$$

Proposition 7.6 (ALFOM preservation under wider state)**.** *Mambo-t achieves $\eta = \log_2(2^t)/t = 1.0$ for every $t \geq 2$. Each step costs t nonlinear constraints (one squaring per element) while multiplying the algebraic degree by 2^t.*

The non-telescoping property (Proposition 4.6) extends directly to wider state.

Corollary 7.7 (Non-telescoping at wider state)**.** *For Mambo-t, $t \geq 2$, in the rate-1 sponge convention (capacity zeroed at the input, first output coordinate fixed to τ) and with $\varphi \neq 0$: the elimination ideal of the round system in the rate input is principal, generated by an eliminant of degree exactly $2^{t(n-1)} = D_I$ with leading coefficient $\varphi^{2^{t(n-1)}}$, and parts (i)–(iii) of Proposition 4.6 hold verbatim with these values.*

Proof. The round system is triangular in the same variable order, so the evaluation argument of Proposition 4.6(i) applies unchanged. For the degree and leading coefficient: in round 1 the chain of t squarings from the zeroed capacity gives degrees $1, 2, \ldots, 2^{t-1}$ and leading coefficients $\varphi, \varphi^2, \ldots, \varphi^{2^{t-1}}$, and in every later round each update has its top term from the single squaring ($2 \deg x_t^{(r-1)} = 2^{t(r-1)} > 2^{t(r-2)} = \deg x_1^{(r-1)}$ and $2 \deg x_{i-1}^{(r)} > \deg x_i^{(r-1)}$ at every step), giving $\deg x_1^{(n)} = 2^{t(n-1)}$ and $\operatorname{lc} x_1^{(n)} = \varphi^{2^{t(n-1)}}$ by induction. The discriminant argument of Proposition 4.6(iii) applies since $p \nmid 2^{t(n-1)}$ for odd p. □

Sponge CICO degree for wider state. For the sponge CICO problem with rate $r = 1$, capacity $c = t - 1$, fixing $x_2(0) = \cdots = x_t(0) = 0$ and $x_1(n) = \tau$, the univariate polynomial $x_1(n)$ in the unknown $x_1(0)$ has formal degree $D_I = 2^{t(n-1)}$ over the algebraic closure (the first round consumes one factor of $2^{(t-1)}$ because the initial capacity is zero; subsequent rounds multiply by 2^t). We verified $D_I = 2^{t(n-1)}$ computationally through $n = 5$ for $t = 3, 4$ over $\mathbb{F}_{65521}$ and $\mathbb{F}_{1048573}$, with square-free polynomial in all tested cases. Over $\mathbb{F}_p$ the effective bound is $\min(2^{t(n-1)}, p^t)$; the field-saturation condition $t \log_2 p \geq \lambda$ determines which base fields admit a given tier, as tabulated in Table 8. The deployed capacity-1 hash surface at $t \geq 3$ additionally admits the rate-specialization attack of Section 5.6, which lowers the binding eliminant degree to $D_{\text{eff}} = 2^{t(n-1)-(t-2)}$; tier sizing uses D_{eff}, not the rate-1 figure above.

Rate–throughput summary. Mambo-t at the recommended 128-bit cells (Table 6) achieves the following 2-to-1 sponge hash costs:

- $t = 2, n = 82$: rate 1, two permutation calls, 328 NL (at the minimum cell $n = 65$: 260 NL theoretical, 259 measured; Table 13).
- $t = 3, n = 55$: rate 2, one permutation call, 165 NL.
- $t = 4, n = 43$: rate 3, one permutation call, 172 NL.

Each entry follows from t nonlinear constraints per round times n rounds (Section 3.2), multiplied by the number of permutation calls required to absorb the two message elements at the given rate.

7.5 Comparison Table

The constraint counts and figure of merit appear in Table 1 (Section 1.2).

8 Implementation and Verification

8.1 Implementations

Reference implementations are provided across the following back-ends. All six produce bit-identical outputs against a common Python reference on the test vectors of Appendix B.

Table 10: Reference implementations. All back-ends are tested, cross-verified reference code; none has undergone an external security audit, and "tested" should not be read as production-hardened.

Back-end	Field	Cost	Status
Python (ref)	multi	—	Specification
Circom (R1CS) [46]	BN254	$2n$	Constraint counting
Halo2 (Plonk/KZG) [33, 44]	BN254	$2n$ custom gates	Tested
Plonky3 (STARK/FRI) [45]	BB, KB, GL	$2n$ AIR rows	Tested
Noir / ACIR	BN254	$2n$ (custom gate: less)	Tested
Cairo [37]	Stark field	$2n$	Tested
Stwo (Circle STARK)	Mersenne-31	one perm / row	Tested

BB = BabyBear, KB = KoalaBear, GL = Goldilocks. Plonky3 also supports a one-permutation-per-row unrolled AIR variant.

8.2 Performance

The reference implementation produces the predicted constraint count of Proposition 3.4 and runs at single-permutation cost quoted below on commodity hardware. Comparison against existing primitives at the design level (constraint count, ALFOM, R1CS cost) is provided by Table 1 in Section 1.4. The single-anchor number in Table 11 fixes the predicted-cost match; Appendix E reports the cross-system wall-clock measurements at production proof depths, the UltraHonk custom-gate arithmetisation, the Plonky3 recursive-verifier prove time across round counts, and the Stwo Circle-STARK wider-state tier sweep.

Table 11: Reference single-permutation prove and verify time for Mambo $t = 2$ $n = 65$ on Plonky3/FRI over BabyBear, release mode, reference host. Configuration status: BabyBear fails the saturation condition $t \log_2 p \geq \lambda$ at $t = 2$ (Table 8), so this cell is not a recommended tier; it is retained as the cross-system scaling anchor because relative ratios across proof systems at matched (t, n) are the meaningful quantity, and absolute costs scale linearly in the round count.

Metric	Value	Unit
Prove	12	ms
Verify	1	ms
NL constraints	130	—

Median of 11 runs on the reference Linux cloud VM (8-vCPU AMD, 16 GB RAM); see Appendix E for the full cross-system table and provenance. The constraint-count column matches Proposition 3.4.

8.3 Test Vectors and Cross-Verification

Test vectors below are reproduced bit-identically by every back-end of Section 8.1.

Full 65-step test vectors for BN254, Goldilocks, and Mersenne-61 are provided in Appendix B.

Table 12: Step trace for $(\theta_0, \psi_0) = (1, 1)$, $\varphi = 7$, BN254, with round constants $c_i = d_i = 0$ (illustrative).

Step	θ_n	ψ_n
0	1	1
1	8	71
2	5 097	25 979 906
3	674 955 515 804 515	455 564 948 314 938 897 947 876 244 567

Production Mambo uses SHA-256-derived round constants (Section 3.4); test vectors with the production constants are in Appendix B.

Table 13: Machine-measured R1CS constraint counts (Circom 2.2.3), at the $t = 2$ L1 minimum cell $n = 65$ and $k = 1$ PRNG; counts scale linearly in the round count (e.g. 164/328 at the recommended $n = 82$).

Circuit	Theoretical	Measured	Status
Permutation ($n = 65$)	130	130	Exact match
Sponge hash (2-to-1)	260	259	One $0{\cdot}0$ squaring folded[f]
PRNG (100 outputs)	200	200	Exact match

[f]The deficit is exactly one constraint and is structural: the sponge initialises the capacity to zero, so the first permutation call's round-1 squaring $\psi_0 \cdot \psi_0$ is a constant and Circom folds it. The count is identical at the default and `--O2` optimization levels. An earlier draft of this table reported 255, which we traced to a 2-to-1 sponge compiled at $n = 64$ ($4 \cdot 64 - 1 = 255$); the figure above is the $n = 65$ measurement.

8.4 Measured Constraint Counts

9 Conclusion

We have introduced the Mambo map, a two-variable coupled quadratic permutation over prime fields, and proved that its algebraic degree grows as 4^n at R1CS cost $2n$, matching the theoretical maximum $\eta = 1.0$. Alongside Clotho ($\beta = 2$) from [1], which was introduced as a theoretical instance at this bound, Mambo is offered as a complementary family accompanied by concrete instantiations and a security study.

The core contribution is the exact algebraic characterization and its proofs. The degree growth of 4^n is a theorem (Theorem 4.2) with a constructive proof showing that no cancellations occur at any step. This theorem and the $t = 2$ closed-form regularity result $d_{\text{reg}} = 2n$ are machine-checked in Lean 4; the tier-floor Macaulay arithmetic is kernel-verified, conditional on the measured d_{reg} at $t \in \{3, 4\}$; the development is `sorry`-free, uses only the three standard axioms (none for the floor computations), and is provided as supplementary material. The formalisation verifies the mathematics, not the attack-cost model. We have provided concrete instantiations sized under the conservative cost model with a 25% recommended round margin (sponge hash at 165 NL at 128-bit with wider state $t = 3$, $n = 55$, or 172 NL at $t = 4$, $n = 43$; permutation at 164 NL for the 128-bit narrow state; PRNG at 2 NL per private-witness output) with cross-verified implementations in five proof systems.

The security of Mambo as a cryptographic primitive rests on a single explicitly stated assumption: that the permutation admits no structural distinguisher below the target cost at the recommended parameters (Design Claim 5.1), with the CICO-hardness conjecture (Conjecture 5.2) as its falsifiable algebraic form. The conjecture is supported by the explicit Gröbner-basis and eliminant analysis of Section 5, differential analysis, and statistical testing. We have self-disclosed the capacity-1 rate-specialization attack, which costs $t - 2$ bits at every wide state width and is priced into every parameter row (Section 5.6), self-disclosed a polynomial-time

attack on additive extraction at $k = 1$ (Proposition 5.19), disclosed all small-field cryptanalysis results, and framed hash security via the BDPV08 sponge indifferentiability theorem in the ideal-permutation model.

Several directions remain open. The relation $d_{\text{reg}} = 2n$ is now structurally established for all $n \geq 2$ (Theorem G.5 in Appendix G) via an explicit Gröbner basis with pairwise coprime leading monomials and Buchberger's first criterion; the value is one below the strict semi-regularity prediction $2n + 1$, reflecting a single non-generic syzygy in the CICO ideal that the proof identifies as the eliminated s_{n-1} relation. Reducing Conjecture 5.2 to a standard computational assumption remains the most impactful open problem.

The question this paper raises is whether the algebraic efficiency captured by $\eta = 1.0$ can coexist with cryptographic security. We invite the community to investigate whether: (i) Conjecture 5.2 reduces to a standard computational assumption; (ii) the closed-form prediction of the higher Hilbert coefficients beyond the first two—now *resolved at* $t = 4$, where the Hilbert series is $(1+z)^{4n-9}Q_4(z)$ with residue $Q_4 = z^5+11z^4+25z^3+19z^2+7z+1$ and $d_{\text{reg}} = 4n-3$ for all $n \geq 4$ (Theorem 5.8), the genericity of the underlying leading-monomial set being certified by random specialization—can be pushed to a fully deterministic symbolic-constant confirmation, and whether the analogous (n-stable, septic) residue at widths $t \geq 5$ admits a closed form; (iii) further cryptanalysis of the coupled-quadratic structure reveals any non-generic algebraic shortcut across the parameter ranges summarised in Table 1; and (iv) the $n \leq t$ analysis gap can be closed—regularity and specialization analysis at widths $t \geq 5$—which is the named open problem whose resolution would create the first recommended 128-bit configurations on 31-bit fields (Section 5.13).

We close by restating the paper's framing as an explicit cryptanalysis challenge, with the exact objects an attacker needs. The falsification targets are the *minimum* cells of Table 6 ($n = 65/44/34$ at $t = 2/3/4$ for $\lambda = 128$), sized at exact analysis equality $D_{\text{eff}} = 2^\lambda$ with zero margin: any algebraic solve of the capacity-1 CICO instance below $\widetilde{O}(\min(D_{\text{eff}}, p))$ at a minimum cell falsifies Conjecture 5.2 outright, while the recommended cells ($n = 82/55/43$) retain a $\geq$ 32-bit buffer so that such a result would not immediately translate to deployed parameters. The attacker's toolkit is supplied in full: the explicit degrevlex Gröbner basis of the CICO ideal (Appendix G), the closed-form Hilbert series and eliminant degrees $D_I = 2^{t(n-1)}$ and $D_{\text{eff}} = 2^{t(n-1)-(t-2)}$, the SHA-256-derived round constants (Section 3.4), the reproducer scripts of the supplementary material, and the self-disclosed attack surface (rate specialization, $k = 1$ extraction, small-field artifacts). Sharper targets than a full break are also stated: falsify the functional-indecomposability hypothesis behind the monodromy certificates (Section 5.3), beat the measured D_{eff} specialization optimum at any cell, exhibit an invariant structure escaping the classification of Section 5.12, or close the multi-block chaining-value question raised there in the attacker's favour. The permutation, in short, is offered to the community with its mathematics fully exposed and its weakest deliberate point marked.

Acknowledgments

The author thanks Aurélien Boeuf for feedback on fixed-point structure, which motivated the introduction of per-round constants (Section 3.4); Pierre Briaud for the correction of our initial D_I estimate, which refined our security analysis and clarified the applicability of the FreeLunch framework (Section 5.4); and Håvard Raddum for the observation that the algebraic-degree argument must account for finite-field saturation (Section 5.4, Table 8).

Intellectual Property Statement

A provisional patent application covering the Mambo construction was filed with the USPTO in March 2026 (patent pending). The reference implementation, formal development, and re-

production scripts accompanying this paper are released for research, academic verification, and audit use; production and commercial use are governed by the Business Source License 1.1 and the pending patent. This statement is provided for transparency; nothing in it restricts the reproduction or verification of the results reported here.

A Reference Implementation

The Mambo map in Python (reference):

```
import hashlib, struct

def derive_round_constants(n, p,
        seed=b"Mambo_round_constants"):
    vals = []
    for j in range(2 * n):
        h = hashlib.sha256(
            seed + j.to_bytes(4, "big")).digest()
        v = int.from_bytes(h, "big") % p
        vals.append(v if v != 0 else 1)
    return [(vals[i], vals[n + i])
            for i in range(n)]

def mambo_step(theta, psi, phi, c, d, p):
    theta_new = (phi * theta + psi * psi + c) % p
    psi_new   = (phi * psi
                 + theta_new * theta_new + d) % p
    return theta_new, psi_new

def mambo_permutation(theta, psi, phi, p, n=82):
    rc = derive_round_constants(n, p)
    for i in range(n):
        c_i, d_i = rc[i]
        theta, psi = mambo_step(
            theta, psi, phi, c_i, d_i, p)
    return theta, psi
```

The sequential coupling is on line 17: `psi_new` is a function of `theta_new`—the freshly computed θ—rather than of the input `theta`. This is what forces the non-telescoping property (Proposition 4.6): no round can be bypassed without solving both coupled constraints. Round constants are derived deterministically from SHA-256 in counter mode.

B Extended Test Vectors

All permutation test vectors below are pinned at $n = 65$ rounds (the $t = 2$ L1 minimum cell, the analysis anchor); pass `n=65` explicitly when reproducing them. The recommended deployment round count is $n = 82$ (Table 6).

BN254 scalar field. $\varphi = 7$. The BN254 scalar field prime is:

```
p = 21888242871839275222246405745257275088548364400
    41603434369820418657580849561 7
```

Permutation ($n = 65$, seed $(1, 1)$):

```
theta_65 = 173947325353572303607972234493069428551 6
           189422867445817798347307623504401900 7
psi_65   = 727645115688839322486340676067575104992 3
           791015985763477840281348092940222991
```

Goldilocks field. $p = 2^{64} - 2^{32} + 1 = 18446744069414584321$, $\varphi = 7$.
Permutation ($n = 65$, seed $(1, 1)$):

$$\theta_{65} = 14260991711422744827$$
$$\psi_{65} = 4124660317069805710$$

C R1CS Circuit for One Mambo Step

A single Mambo step $(\theta_i, \psi_i) \to (\theta_{i+1}, \psi_{i+1})$ at round i with constants (c_i, d_i):

Witness variables: $\theta_i, \psi_i, u_i, \theta_{i+1}, v_i, \psi_{i+1}$.

Constraints:

1. (Nonlinear) $\psi_i \cdot \psi_i = u_i$
2. (Linear) $\varphi \cdot \theta_i + u_i + c_i = \theta_{i+1}$
3. (Nonlinear) $\theta_{i+1} \cdot \theta_{i+1} = v_i$
4. (Linear) $\varphi \cdot \psi_i + v_i + d_i = \psi_{i+1}$

Total: 2 nonlinear constraints, 2 linear constraints. Only the nonlinear constraints contribute to the cost C. The round constants c_i, d_i appear only in the linear constraints and add no nonlinear cost.

D Security Analysis Summary

Table 14 summarizes all attack vectors analyzed for the Mambo permutation and their current status.

Table 14: Security analysis checklist for Mambo at the L1 minimum cell ($n = 65$, $t = 2$, BN254)—the falsification target of the cost model. Every entry is monotone in the round count, so the recommended cell ($n = 82$) inherits each status with margin.

Attack Vector	Result	Section	Status
Interpolation	$D = 4^{65} \approx 2^{130}$	5.4	Secure
Gröbner basis (GB)	basis explicit (free); Macaulay vs blind F4/F5 $> 2^{300}$	5.5	See D_I row
Ideal degree (D_I)	$D_I = 4^{n-1} = 2^{128}$; univariate solve $\widetilde{O}(D_I)$	5.2	Secure
Higher-order diff.	$\deg(\theta_{65}) = 2^{128}$ in sponge	5.4	Secure at $n = 65$
Differential	MDP $= 1/p$ per round (exact); trail $\leq p^{-65}$	5.7	Secure
Boomerang / BCT	Max BCT $\leq 6\times$ random; infeasible	5.7	Secure
Linear	Quadratic S-box, no high-bias trails	5.7	Secure
Slide attacks	Round constants eliminate	5.8	Secure
Invariant subspaces	No invariant/trail lines at $t = 2$ (complete); $t \geq 3$ partial	5.12	See section
Fixed points	≤ 4 per round; no common round fixed point	5.12	Analyzed
Multi-block CICO	Chaining-value specialization unanalyzed	5.12	Open
$k = 1$ additive attack	Polynomial-time; recommended public-output $k = 82$	5.10	Disclosed
$c = 1$ specialization ($t \geq 3$)	$D_{\text{eff}} = 2^{t(n-1)-(t-2)}$; priced into tiers	5.6	Disclosed
Statistical tests	PractRand 1 TB, BigCrush 160/160	5.3	Passed

E Extended Performance Measurements

This appendix reports the proof-system-specific wall-clock and proof-size data complementary to the single-anchor measurement of Section 8.2 (Table 11). All numbers are from the same reference Linux cloud VM (8-vCPU AMD, 16 GB RAM, Ubuntu 24.04) and the same bench harness; provenance, commit hashes, and full p10/p50/p90 distributions are listed in the supplementary materials.

E.1 Single-permutation cost across proof systems

Table 15: Single-permutation Mambo performance at reference parameters, release mode. Halo2 and Plonky3 figures are for the narrow $t = 2$ $n = 65$ permutation (the L1 minimum cell). UltraHonk gate counts are for $t = 4$ $n = 18$ (the dense-FGLM-model L1 cell of Table 7; the conservative L1 minimum cell at $t = 4$ is $n = 34$, recommended $n = 43$). KZG proof size is constant; FRI grows with trace.

Metric	Halo2/KZG	Plonky3/FRI	UltraHonk	Unit
Prove	174	12	—	ms
Verify	13	1	—	ms
Proof size	960	var	—	B
Gates/perm	—	—	133 (std.) / **61** (custom)	gates
Trusted setup	Yes	No	No	—
Post-quantum	No	Yes	No	—

UltraHonk custom gate. A custom-gate extension to the UltraHonk arithmetisation reduces Mambo $t = 4$ $n = 18$ to 61 UltraHonk gates per permutation, compared to 133 in the default non-custom arithmetisation. The reduction is obtained by consolidating the coupled-quadratic round into a single degree-2 gate instance rather than separate squaring constraints.

E.2 Halo2/KZG Merkle verifier prove time at production depths

Table 16: Halo2/KZG Merkle verifier prove time on BN254 at production depths, Mambo $t = 4$ $n = 18$ (the dense-FGLM-model L1 cell of Table 7). The local symbol κ denotes the KZG circuit-size parameter (trace domain size 2^κ); this is unrelated to the PRNG step parameter k of Section 7.3.

Depth	κ	Prove (ms)
10	9	367
16	10	529
20	10	596
24	10	573
64	12	908
128	13	1313
256	14	2411
512	15	4300

E.3 Plonky3/FRI production Merkle proof-size scaling

Table 17: Plonky3/FRI production Merkle verifier proof size for Mambo $t = 2$ $n = 18$ (speed proxy). KoalaBear results within 2–5% of BabyBear at matched configuration. $n = 18$ at $t = 2$ is a speed-proxy configuration for cross-depth scaling; the conservative 128-bit $t = 2$ minimum cell is $n = 65$, recommended $n = 82$ (cf. Section 5.13). A tier-secure wider-state Plonky3 Merkle verifier AIR is under construction and will supersede this table when shipped.

Depth	Proof size (B)	Field
2	27 519	BabyBear
4	30 311	BabyBear
8	39 327	BabyBear
16	51 485	BabyBear
64	85 401	BabyBear
128	107 159	BabyBear
512	160 275	BabyBear

E.4 Plonky3 recursive-verifier end-to-end prove time

End-to-end recursive prove-time measurement on a Plonky3 FRI-only recursive verifier at 32 k rows on BabyBear, median of 11 trials with p10 and p90 bands.

Table 18: Plonky3 recursive verifier total prove time, Mambo. BabyBear is sub-$L1$-secure at $t = 2$ for any n; round counts shown are scaling references. Empirical hash share of this FRI-only recursive verifier on the tested round counts: 72–84%, higher than typical arithmetic-verifier-heavy recursive stacks because a pure FRI recursive verifier is hash-dominated in its inner loop.

Configuration	Median (ms)	p10–p90 (ms)
Mambo $t = 4$ $n = 18$	744	715–858
Mambo $t = 2$ $n = 33$	715	687–785
Mambo $t = 2$ $n = 49$	905	870–985
Mambo $t = 2$ $n = 65$	1155	1120–1215

The recursive-verifier measurements are prove-cost measurements and field-independent; they apply at any $t = 2$-admissible field at the same round counts. Tier-level security labels are omitted here because the underlying field is BabyBear, for which the field-saturation condition of Section 5.13 caps accessible security below $L1$ at $t = 2$ regardless of n.

E.5 Stwo (Circle STARK, Mersenne-31)

Mambo runs natively in the Circle STARK framework underlying Stwo at all state widths tested ($t \in \{2, 16, 32\}$) because the permutation has constraint degree exactly 2 by construction. The framework enforces a degree-2 upper bound on constraint polynomials in its present form; primitives whose rounds have algebraic degree above 2 require auxiliary witness columns to be represented under this ceiling, while Mambo fits it natively.

Table 19: Stwo prove time at $\log_n = 14$ (16 384 parallel permutations), median of five runs, reference host. The $t \in \{16, 32\}$ widths are *research configurations — insecure-until-analyzed*: they lie in the $n \leq t$ regime for which no regularity or specialization analysis exists (Section 5.13), so no tier label or security claim attaches to any row. Round counts were fixed purely as reference points by the pre-specialization rule $D_I = 2^{t(n-1)} \geq 2^\lambda$ for $\lambda \in \{128, 192, 256\}$: $n_{128} = 9$ at $t = 16$ and $n_{128} = 5$ at $t = 32$; $n_{192} = 13$ at $t = 16$ and $n_{192} = 7$ at $t = 32$; $n_{256} = 17$ at $t = 16$ and $n_{256} = 9$ at $t = 32$. Under the rate-specialization correction $D_{\text{eff}} \geq 2^\lambda$ (Section 5.6) each reference round count rises by exactly one ($t = 16$: $n = 10/14/18$; $t = 32$: $n = 6/8/10$); the measured times below are per-round-count references only, and cross-configuration relative ratios are the meaningful quantity.

Round count (reference only)	$t = 16$ (ms)	$t = 32$ (ms)
n_{128}	453	806
n_{192}	533	981
n_{256}	794	882

Two scoping notes apply to these wide-width cells. First, sponge collision resistance is bounded by $p^{c/2}$, so on a 31-bit field the 128-bit collision level requires capacity $c \geq \lceil 2 \cdot 128/31 \rceil = 9$ cells; the usable rate on M31 is therefore $t-9$, not $t-1$ (rate 7 at $t = 16$, rate 23 at $t = 32$). At the corrected reference round counts this gives $160/7 \approx 22.9$ nonlinear constraints per absorbed element at $t = 16$, $n = 10$, and $192/23 \approx 8.3$ at $t = 32$, $n = 6$; a 2-to-1 node over two 9-cell digests (18 cells) fits in a single $t = 32$ permutation call (192 nonlinear constraints), while $t = 16$ requires three calls. Second—the reason for the research-configuration label—the wide-width cells lie in the $n \leq t$ regime, entirely outside the analyzed range of the regularity and specialization measurements (the D_{eff} closed form was measured at $t \in \{3, 4, 5\}$, $n \in \{2, \ldots, 5\}$; Section 5.6), and no recommended 128-bit configuration currently exists on 31-bit fields (Section 5.13). We flag the $t \in \{16, 32\}$ configurations as the priority target for further cryptanalysis; closing the $n \leq t$ analysis gap is the named open problem of Section 9.

F Structural Observations and Deferred Proofs

Remark F.1 (First syzygy module is exactly Koszul)**.** Computation of the minimal free resolution of the CICO ideal (via the same `syzygy_hunt_sage` scripts) shows that the first syzygy module coincides with the Koszul complex at every tested n: the non-genericity reflected by Corollary 5.6 is not visible at the level of pairwise relations among the generators. Homogenizing the ideal by introducing an auxiliary degree-1 variable h restores the semi-regular numerator $(1 + z)^{2n}$ exactly, confirming that the observed density gap is a purely affine phenomenon arising from the interaction of the two degree-1 CICO endpoint constraints with the $2n$ degree-2 propagation equations.

Remark F.2 (Joint origin of the half-density and $1/p$ signals)**.** Two near-tight non-genericity observations on the Mambo round map admit a common structural explanation:

1. Corollary 5.6: $\text{hilb}(G, 2) = (1 + z)^{2n-1}$, one factor of $(1 + z)$ shorter than the semi-regular baseline $(1 + z)^{2n}$.
2. Proposition 5.17: the maximum differential probability per round equals $1/p$, half the $2/p$ generic upper bound for a degree-2 permutation on $\mathbb{F}_p^2$.

Both observations are consistent with a single structural property of the round map, which we term its *bilinear-quadratic asymmetry*: the output coordinate θ' depends on θ affinely and on ψ quadratically, while ψ' depends on ψ affinely and on θ' quadratically. Each output coordinate depends on its "own" input variable affinely and on the cross variable quadratically.

Algebraic side (signal 1). Under homogenization, the leading forms at infinity factor through one variable per equation: $\theta_{i+1} = \psi_i^2$, $\psi_{i+1} = \theta_{i+1}^2$. The bilinear-quadratic asymmetry is

therefore present in the ideal of leading forms; the closed-form factorization $(1+z)^{2n-1}$ of $\mathrm{hilb}(G,2)$ is empirically observed for $n \in \{2,\dots,10\}$ (Corollary 5.6) and is consistent with this leading-form bilinearity. A formal derivation of the factorization from the leading-form ideal is open work; we record the structural connection here as the most parsimonious explanation compatible with the data and the Koszul syzygy observation (Remark F.1).

Differential side (signal 2). The case analysis in the proof of Proposition 5.17 shows that every DDT row is either concentrated in a single output column with p preimages per cell (input differences $(a,0)$), or assigns p preimages to exactly one cell and one preimage to the rest (input differences with $\Delta\psi \neq 0$). In both witness families the affine dependence on the "own" variable is what forces the concentration, and the quadratic dependence on the cross variable is what spreads the remainder uniformly—the same asymmetry as on the algebraic side.

The bilinear-quadratic asymmetry is intentional: it minimizes nonlinear constraint count (one squaring per output coordinate per round) and is the source of the half-pack and quarter-pack circuit savings reported in Section 8.2.

Proof of Lemma 7.4

Proof. Let $(\theta_0,\psi_0) \sim \mathcal{U}(\mathbb{F}_p^2)$ denote a uniformly random seed and write the PRNG output $\sigma_i = \theta_{ki} + \psi_{ki} \pmod p$ where $(\theta_{ki},\psi_{ki}) = P_{\varphi,k}^i(\theta_0,\psi_0)$ is the i-th iterate of $P_{\varphi,k}$ on the seed.

Hybrids. Define three games, each producing a transcript $(\sigma_1,\dots,\sigma_q) \in \mathbb{F}_p^q$:

- $\mathbf{H}_0$: real PRNG. Permutation $= P_{\varphi,k}$; $\sigma_i = (\theta_{ki}+\psi_{ki})$ as above.
- $\mathbf{H}_1$: $P_{\varphi,k}$ replaced by a uniform random permutation $\pi : \mathbb{F}_p^2 \to \mathbb{F}_p^2$. Output $\sigma_i = a_i + b_i$ where $(a_i,b_i) = \pi^i(\theta_0,\psi_0)$.
- $\mathbf{H}_2$: π-orbit replaced by q i.i.d. uniform samples $(u_i,v_i) \sim \mathcal{U}(\mathbb{F}_p^2)$. Output $\sigma_i = u_i + v_i$.

Step 1: $\mathbf{H}_0 \to \mathbf{H}_1$ (permutation switch). Given $\mathcal{D}$ distinguishing $\mathbf{H}_0$ from $\mathbf{H}_1$ with advantage ε_{01}, build a permutation distinguisher $\mathcal{B}$ as follows: $\mathcal{B}$ samples $(\theta_0,\psi_0) \sim \mathcal{U}(\mathbb{F}_p^2)$ and maintains a state variable $s_0 = (\theta_0,\psi_0)$. For $i = 1,\dots,q$, $\mathcal{B}$ queries its permutation oracle on s_{i-1} to obtain s_i, records σ_i as the sum of coordinates of s_i, and feeds σ_i to $\mathcal{D}$. When $\mathcal{D}$ halts with a guess, $\mathcal{B}$ outputs the same guess. If the oracle is $P_{\varphi,k}$ (resp. a random permutation), $\mathcal{B}$ simulates $\mathbf{H}_0$ (resp. $\mathbf{H}_1$) perfectly. Hence $\varepsilon_{01} \le \mathrm{Adv}_{\mathcal{B}}^{\mathrm{perm}}(\lambda)$. $\mathcal{B}$ runs in time $\mathrm{time}(\mathcal{D}) + q \cdot \mathrm{cost}(\text{1 oracle query})$ and uses q forward queries.

Step 2: $\mathbf{H}_1 \to \mathbf{H}_2$ (birthday switch). Let E be the event that the π-orbit of the seed closes within the first q steps, i.e. $\pi^i(\theta_0,\psi_0) = (\theta_0,\psi_0)$ for some $1 \le i \le q$. Since the cycle length of a uniformly random point under a uniformly random permutation of a domain of size p^2 is uniform on $\{1,\dots,p^2\}$, $\Pr[E] \le q/p^2$. Conditioned on $\neg E$, the orbit $(a_1,b_1),\dots,(a_q,b_q)$ is distributed as q uniformly random *distinct* points of $\mathbb{F}_p^2$. The total variation distance between this distribution and $\mathcal{U}(\mathbb{F}_p^2)^q$ (i.i.d. uniform samples) is bounded by the probability of collision among q i.i.d. samples on a domain of size p^2. Adding the cycle-closure term,

$$\Delta(\mathbf{H}_1^{(a,b)}, \mathbf{H}_2^{(a,b)}) \;\le\; \binom{q}{2} \Big/ p^2 \;+\; \frac{q}{p^2} \;=\; \frac{q(q+1)}{2\,p^2}.$$

The transcript $(\sigma_i)_i$ is a deterministic post-processing of $(a_i,b_i)_i$; by the data-processing inequality this bound transfers to the distinguisher's view.

Step 3: $\mathbf{H}_2$ is uniform. For independent $u,v \sim \mathcal{U}(\mathbb{F}_p)$, the sum $u+v \bmod p$ is distributed as $\mathcal{U}(\mathbb{F}_p)$. The samples in $\mathbf{H}_2$ are mutually independent, so $\mathbf{H}_2$ produces q i.i.d. uniform elements of $\mathbb{F}_p$. Hence $\mathcal{D}$'s advantage in distinguishing $\mathbf{H}_2$ from a uniform reference equals zero.

Combining the three steps via the triangle inequality yields the claimed bound. □

G Structural Proof of $d_{\mathrm{reg}} = 2n$ via an Explicit Gröbner Basis

This appendix upgrades the empirical observation $\mathrm{hilb}(G, 2) = (1+z)^{2n-1}$ (Corollary 5.6, verified bit-exact through $n = 10$) to a theorem by exhibiting an explicit Gröbner basis of the Mambo CICO ideal whose leading monomials are pairwise coprime. The proof has two ingredients: (i) a square-reduction lemma collapsing every free variable's square to a linear polynomial in free variables; (ii) an explicit list of $2n + 2$ ideal elements whose leading monomials are powers of distinct variables, so Buchberger's first criterion gives the Gröbner basis property without any explicit S-pair reduction. The standard monomials read directly off the leading-monomial structure as the squarefree monomials in a $(2n-1)$-element set, giving Hilbert series $(1+z)^{2n-1}$ and $d_{\mathrm{reg}} = 2n$ for all $n \geq 2$.

Throughout this appendix we use the shorthand $t_i := \theta_i$ and $s_i := \psi_i$ to match the variable-ordering convention of the Singular reproducer `analysis/mambo_proof_verify.sage`, which independently verifies every algebraic identity below at $n \in \{2, 3, 4, 5, 6\}$ over $\mathbb{F}_{257}$.

G.1 Setup

Let $R = \mathbb{F}_p[t_0, t_1, \ldots, t_n, s_0, s_1, \ldots, s_n]$ with $N = 2(n + 1)$ variables. Let $I \subseteq R$ be the Mambo CICO ideal at state width $t = 2$ and round count n, generated by the $2n + 2$ polynomials

$$\begin{aligned} g_0 &:= t_0, & g_{2n+1} &:= t_n - \tau, \\ g_{2i+1} &:= t_{i+1} - \varphi t_i - s_i^2 - c_i, & g_{2i+2} &:= s_{i+1} - \varphi s_i - t_{i+1}^2 - d_i, \end{aligned}$$

for $i = 0, 1, \ldots, n - 1$, where $\tau \in \mathbb{F}_p$ is the rate output target, $\varphi \in \mathbb{F}_p^*$ is the multiplier, and $(c_i, d_i)_{i=0}^{n-1}$ are the round constants.

G.2 Free variables and the candidate basis

Define the *free variable set*

$$\mathcal{F}_n := \{t_1, t_2, \ldots, t_{n-1}\} \cup \{s_0, s_1, \ldots, s_{n-2}\} \cup \{s_n\}$$

and the *fixed variable set* $\mathcal{X}_n := \{t_0, t_n, s_{n-1}\}$. Note $|\mathcal{F}_n| = 2n-1$ and $|\mathcal{X}_n| = 3$, so $|\mathcal{F}_n| + |\mathcal{X}_n| = 2n + 2 = N$.

Let $\mathcal{B}_n$ denote the set of squarefree monomials in $\mathcal{F}_n$:

$$\mathcal{B}_n = \left\{ \prod_{x \in S} x \ : \ S \subseteq \mathcal{F}_n \right\}, \qquad |\mathcal{B}_n| = 2^{|\mathcal{F}_n|} = 2^{2n-1}.$$

G.3 Reduction relations

Each fixed variable in $\mathcal{X}_n$ is reducible modulo the generators of I:

$$t_0 \equiv 0 \pmod{I}, \tag{R1}$$

$$t_n \equiv \tau \pmod{I}, \tag{R2}$$

$$s_{n-1} \equiv \varphi^{-1}(s_n - \tau^2 - d_{n-1}) \pmod{I}, \tag{R3}$$

where (R3) is obtained from $g_{2n} = s_n - \varphi s_{n-1} - t_n^2 - d_{n-1}$ combined with (R2).

Lemma G.1 (Square reduction)**.** *For every $x \in \mathcal{F}_n$, there exist scalars $a_y \in \mathbb{F}_p$ ($y \in \mathcal{F}_n$) and $a_0 \in \mathbb{F}_p$ such that*

$$x^2 \equiv a_0 + \sum_{y \in \mathcal{F}_n} a_y \, y \pmod{I}.$$

Proof. We treat the four cases.

Case $x = s_i$, $i \in \{0, 1, \ldots, n-2\}$. Generator $g_{2i+1} = t_{i+1} - \varphi t_i - s_i^2 - c_i \in I$ gives $s_i^2 \equiv t_{i+1} - \varphi t_i - c_i \pmod I$. Both t_i and t_{i+1} lie in $\mathcal{F}_n \cup \{t_0\}$; using (R1) when $i = 0$, the right-hand side reduces to a linear polynomial in $\mathcal{F}_n$.

Case $x = t_i$, $i \in \{1, 2, \ldots, n-2\}$. The generator $g_{2i} = s_i - \varphi s_{i-1} - t_i^2 - d_{i-1} \in I$ gives $t_i^2 \equiv s_i - \varphi s_{i-1} - d_{i-1} \pmod I$. Both s_{i-1} and s_i lie in $\mathcal{F}_n$ since $i - 1 \geq 0$ and $i \leq n - 2$.

Case $x = t_{n-1}$. The same generator $g_{2(n-1)} = s_{n-1} - \varphi s_{n-2} - t_{n-1}^2 - d_{n-2}$ gives $t_{n-1}^2 \equiv s_{n-1} - \varphi s_{n-2} - d_{n-2} \pmod I$. Now $s_{n-1} \in \mathcal{X}_n$, but (R3) reduces s_{n-1} to $\varphi^{-1}(s_n - \tau^2 - d_{n-1})$, a linear polynomial in $\mathcal{F}_n$.

Case $x = s_n$. Combine g_{2n} with (R2) to obtain $s_{n-1} \equiv \varphi^{-1}(s_n - \tau^2 - d_{n-1}) \pmod I$. Squaring, $s_{n-1}^2 \equiv \varphi^{-2}(s_n - \tau^2 - d_{n-1})^2 \pmod I$. Independently, $g_{2n-1} = t_n - \varphi t_{n-1} - s_{n-1}^2 - c_{n-1}$ combined with (R2) gives $s_{n-1}^2 \equiv \tau - \varphi t_{n-1} - c_{n-1} \pmod I$. Equating the two expressions for s_{n-1}^2 and solving for s_n^2:

$$s_n^2 \equiv 2(\tau^2 + d_{n-1})s_n - \varphi^3 t_{n-1} + \big(\varphi^2(\tau - c_{n-1}) - (\tau^2 + d_{n-1})^2\big) \pmod I,$$

which is linear in $\{s_n, t_{n-1}\} \subseteq \mathcal{F}_n$. □

G.4 Spanning and linear independence

Proposition G.2 ($\mathcal{B}_n$ spans R/I)**.** *The classes $\{\bar{m} : m \in \mathcal{B}_n\}$ span R/I as an $\mathbb{F}_p$-vector space.*

Proof. Every monomial $m \in R$ is, by Lemma G.1 and relations (R1)–(R3), congruent modulo I to a polynomial in the variables of $\mathcal{F}_n$ in which each variable appears to power at most 1. Such a polynomial is an $\mathbb{F}_p$-linear combination of monomials in $\mathcal{B}_n$. Hence $\mathcal{B}_n$ spans R/I. □

Proposition G.3 ($\mathcal{B}_n$ is linearly independent)**.** *The classes $\{\bar{m} : m \in \mathcal{B}_n\}$ are $\mathbb{F}_p$-linearly independent in R/I.*

Proof. We exhibit an explicit Gröbner basis $\mathcal{G}$ of I under the degrevlex order with variable ranking $t_0 > t_1 > \cdots > t_n > s_0 > s_1 > \cdots > s_n$, and read off the standard monomials.

The candidate basis $\mathcal{G}$. Define the following $2n + 2$ polynomials.

(G1) $\gamma_0 := t_0$. Leading monomial t_0.

(G2) $\gamma_{2n+1} := t_n - \tau$. Leading monomial t_n.

(G3) $\rho := s_{n-1} - \varphi^{-1}(s_n - \tau^2 - d_{n-1})$. Leading monomial s_{n-1}.

(G4) For $i \in \{0, 1, \ldots, n-2\}$, $\eta_i := s_i^2 - (t_{i+1} - \varphi t_i - c_i)$. Leading monomial s_i^2.

(G5) For $i \in \{1, 2, \ldots, n-1\}$, $\kappa_i := t_i^2 - (s_i - \varphi s_{i-1} - d_{i-1})$. Leading monomial t_i^2.

(G6) $\sigma := s_n^2 - 2(\tau^2 + d_{n-1})s_n + \varphi^3 t_{n-1} - \big(\varphi^2(\tau - c_{n-1}) - (\tau^2 + d_{n-1})^2\big)$. Leading monomial s_n^2.

Set $\mathcal{G} := \{\gamma_0, \gamma_{2n+1}, \rho\} \cup \{\eta_i\}_{i=0}^{n-2} \cup \{\kappa_i\}_{i=1}^{n-1} \cup \{\sigma\}$. Then $|\mathcal{G}| = 3 + (n-1) + (n-1) + 1 = 2n + 2$.

Step 1: $\mathcal{G} \subseteq I$. $\gamma_0 = g_0 \in I$ and $\gamma_{2n+1} = g_{2n+1} \in I$ directly. $\eta_i = -g_{2i+1} \in I$ for $i \leq n - 2$. $\kappa_i = -g_{2i} \in I$ for $i = 1, \ldots, n-1$. For ρ: from $g_{2n} = s_n - \varphi s_{n-1} - t_n^2 - d_{n-1}$ and $\gamma_{2n+1} = t_n - \tau$, we have $\varphi\rho = -g_{2n} - (t_n + \tau)\gamma_{2n+1}$, so $\varphi\rho \in I$, hence $\rho \in I$. For σ: by Lemma G.1 (case $x = s_n$), s_n^2 reduces to the displayed linear polynomial in $\{s_n, t_{n-1}\}$ modulo I, so $\sigma \in I$.

Step 2: $\mathcal{G}$ generates I. The original generators $\{g_0, g_{2n+1}\} \cup \{g_{2i+1}\}_{i=0}^{n-1} \cup \{g_{2i+2}\}_{i=0}^{n-1}$ all lie in $\langle\mathcal{G}\rangle$: $g_0 = \gamma_0$, $g_{2n+1} = \gamma_{2n+1}$, $g_{2i+1} = -\eta_i$ for $i \leq n - 2$, $g_{2i+2} = -\kappa_{i+1}$ for $i \leq n - 2$. The remaining two original generators, g_{2n-1} and g_{2n}, follow: $g_{2n} = -\varphi\rho - (t_n + \tau)\gamma_{2n+1}$ as above;

and g_{2n-1} is recovered from σ, γ_{2n+1}, and ρ by the same closure derivation as in Lemma G.1. Hence $\langle \mathcal{G} \rangle = I$.

Step 3: leading monomials of $\mathcal{G}$ are pairwise coprime. The set of leading monomials is

$$\mathrm{LM}(\mathcal{G}) = \{t_0,\ t_n,\ s_{n-1}\} \cup \{s_i^2\}_{i=0}^{n-2} \cup \{t_i^2\}_{i=1}^{n-1} \cup \{s_n^2\}.$$

Each entry is supported on a single variable, and the variables are distinct. Therefore $\gcd(m_1, m_2) = 1$ for every pair $m_1 \neq m_2$ in $\mathrm{LM}(\mathcal{G})$.

Step 4: $\mathcal{G}$ is a Gröbner basis of I. By Buchberger's first criterion (coprime leading monomials imply trivial S-pair reduction; cf. Cox–Little–O'Shea *Ideals, Varieties, and Algorithms*, Thm. 3.6.4), pairwise coprime leading monomials of generators that span I imply that the generators form a Gröbner basis of I. By Steps 1–3, $\mathcal{G}$ satisfies the hypotheses, hence $\mathcal{G}$ is a Gröbner basis of I under the chosen monomial order.

Step 5: standard monomials $= \mathcal{B}_n$. Macaulay's theorem identifies R/I with the $\mathbb{F}_p$-vector space spanned by monomials not divisible by any element of $\mathrm{LM}(\mathcal{G})$. A monomial $m \in R$ avoids $\mathrm{LM}(\mathcal{G})$ iff (a) m is not divisible by t_0, t_n, or s_{n-1}, and (b) m is not divisible by any of $s_0^2, \ldots, s_{n-2}^2, t_1^2, \ldots, t_{n-1}^2, s_n^2$. The first condition restricts m to monomials supported on $\mathcal{F}_n = \{t_1, \ldots, t_{n-1}, s_0, \ldots, s_{n-2}, s_n\}$. The second restricts those to squarefree monomials. The intersection is exactly $\mathcal{B}_n$. The standard monomials are an $\mathbb{F}_p$-basis of R/I, so the classes $\{\bar{m} : m \in \mathcal{B}_n\}$ are linearly independent. □

The $t = 4$ residue: leading-monomial classification

The $t = 4$ closed form of Theorem 5.8 rests on an explicit classification of the initial ideal. Unlike the $t = 2$ case, the quotient is not a Boolean ring: the residue factor $Q_4(z) = z^5 + 11z^4 + 25z^3 + 19z^2 + 7z + 1$ records a fixed finite set of non-Boolean leading monomials that survive the Gröbner reduction.

Lemma G.4 ($t = 4$ leading-monomial classification, $n \geq 4$)**.** *For the rate-fixed CICO ideal I at width $t = 4$ and every $n \geq 4$, the degree-reverse-lex (round-major) initial ideal is exactly the finite monomial certificate* $\mathrm{in}_<(I) = C$*, where C is generated by:*

(a) six linear *monomials: the three input pins and the anti-diagonal linear chain $x_{n,0}, x_{n-1,1}, x_{n-2,2}$;*
(b) the Boolean square *x^2 of every other free variable;*
(c) a fixed finite non-Boolean certificate *of* 57 *monomials: the single tower w^6 with $w = x_{n,3}$, and* 56 *mixed monomials of the form $w^a \cdot u$ with u a squarefree monomial on the nine-variable window of rounds $n-3, \ldots, n$.*

Proof. Window-local membership. Define the four-round window subideal $J_n \subseteq I$ generated by the round-map relations on rounds $n-4 \to \cdots \to n$ together with the output pin; J_n involves only the state variables of those rounds. A direct (finite, n-independent) computation verifies, at $n \in \{4, 5, 6, 7\}$, that every monomial of the certificate (c) and every linear monomial of (a) is divisible by a leading monomial of a Gröbner basis of J_n; hence for each certificate monomial μ there is $h \in J_n \subseteq I$ with $\mathrm{LM}(h) \mid \mu$, so $(\mu / \mathrm{LM}(h))\, h \in I$ has leading monomial exactly μ and $\mu \in \mathrm{in}_<(I)$. The Boolean squares of (b) are leading monomials of the original round-map generators directly. The window bound 4 is minimal: a three-round window fails to capture the certificate monomials engaging round $n-3$.

n-uniformity. In window coordinates the generators of J_n are, for every n, the same polynomials up to the values of the round constants, which the round-index shift merely re-indexes; the monomial order restricts to the window with the relative order preserved, so a window element with a given leading monomial maps under the shift to one of $J_{n'}$ with the shifted leading monomial, *provided membership is independent of the constants.* We establish constant-independence by upper-semicontinuity of the initial ideal: regarding J as a family over the parameter space of

round constants, $\mathrm{in}_<(J_v) \supseteq \mathrm{in}_{\mathrm{gen}}$ for every parameter value v, the generic initial ideal $\mathrm{in}_{\mathrm{gen}}$ being attained on a dense open stratum. Random specialization identifies $\mathrm{in}_{\mathrm{gen}}$: 80 parameter points across two primes all return the bit-identical leading-monomial set, which covers the certificate; by semicontinuity the certificate is then covered at *every* constant schedule, hence for all $n \geq 4$. The certificate is moreover exactly round-shift invariant (the window monomial set read off at $n = 5$ and $n = 6$ is bit-identical over both primes). *The fully deterministic symbolic-constant block-order Gröbner computation—which would replace the sampling-based identification of* $\mathrm{in}_{\mathrm{gen}}$ *by a symbolic certificate—is beyond current compute; by upper-semicontinuity it is a strengthening, not a logical necessity, and constant-independence is therefore certified by sampling, not by a deterministic symbolic Gröbner basis.*

Completeness. The standard monomials of C number 2^{4n-3} exactly (inclusion–exclusion on (a)–(c); $Q_4(1) = 64 = 2^6$). The membership above gives $\mathrm{in}_<(I) \supseteq C$, hence $\mathrm{colength}(I) \leq 2^{4n-3}$; the established wider-width colength formula $\mathrm{colength}(I) = 2^{tn-(t-1)} = 2^{4n-3}$ (following Proposition 5.10) forces equality, whence $\mathrm{in}_<(I) = C$. □

Theorem G.5 (Closed-form Hilbert series, structural)**.** *The Hilbert series of the Mambo CICO quotient ring satisfies*

$$\mathrm{HS}_{R/I}(z) = (1+z)^{2n-1}$$

for every $n \geq 2$, and consequently $\dim_{\mathbb{F}_p}(R/I) = 2^{2n-1}$ and $d_{\mathrm{reg}} = 2n$.

Proof. By Propositions G.2 and G.3, $\{\bar{m} : m \in \mathcal{B}_n\}$ is an $\mathbb{F}_p$-basis of R/I. Each $m \in \mathcal{B}_n$ has total degree $|S|$ where $S \subseteq \mathcal{F}_n$ is the support of m, and there are exactly $\binom{2n-1}{|S|}$ subsets of each size. Hence

$$\mathrm{HS}_{R/I}(z) = \sum_{S \subseteq \mathcal{F}_n} z^{|S|} = \sum_{k=0}^{2n-1} \binom{2n-1}{k} z^k = (1+z)^{2n-1}.$$

The dimension is $\mathrm{HS}_{R/I}(1) = 2^{2n-1}$. The degree of regularity equals $\deg \mathrm{HS}_{R/I} + 1 = (2n-1) + 1 = 2n$. □

G.5 Computational support

The unified verification script `analysis/mambo_proof_verify.sage` confirms every step of the above argument over the small prime field $\mathbb{F}_{257}$ at $n \in \{2, 3, 4, 5, 6\}$, exhausting the explicit-basis identities of Lemma G.1 and Propositions G.2–G.3: $\dim_{\mathbb{F}_p}(R/I) = 2^{2n-1}$ exactly; Hilbert function $H(d) = \binom{2n-1}{d}$ at every degree d; Singular's `kbase` returns exactly 2^{2n-1} monomials; every monomial in `kbase` is squarefree in $\mathcal{F}_n$; and $x^2 \bmod I$ reduces to a linear polynomial in $\mathcal{F}_n$ for every $x \in \mathcal{F}_n$.

The scope of this computational verification is intentionally narrower than Table 3 of Section 5.5, which reports the empirical regularity-degree observation $d_{\mathrm{reg}} = 2n$ over the rationals at $n \in \{2, \ldots, 10\}$. The two scopes are complementary: Table 3 samples a wider range over a different ring (rationals) to probe the characteristic-independence of the relation, while the verification here tightens the explicit-basis identities of the structural proof over a fixed small prime where Singular's exact-arithmetic stack runs fastest. Theorem G.5 is independent of both verification ranges.

Structural source of the half-density gap

The structural source of the $(1+z)^{2n-1}$ closed form is twofold. First, the round-map structure forces an explicit quadratic relation $x^2 \equiv \ell(x) \pmod{I}$ for every free variable, where ℓ is linear (Lemma G.1); this collapses the algebra to a Boolean ring on $|\mathcal{F}_n| = 2n - 1$ generators. Second, the fixings $t_0 = 0$ and $t_n = \tau$, combined with the last ψ-recurrence, place exactly three variables $(t_0,\ t_n,\ s_{n-1})$ into the fixed set $\mathcal{X}_n$, leaving the remaining $2n - 1$ variables in the free set $\mathcal{F}_n$. Those $2n - 1$ free variables span a Boolean algebra on 2^{2n-1} elements, identifying R/I with

the squarefree monomial algebra on $\mathcal{F}_n$ as graded vector spaces. The factor of 2 between $\mathrm{HS}_{R/I}(1) = 2^{2n-1}$ and the naive Bezout bound 2^{2n} is precisely the contribution of the eliminated s_{n-1}: were s_{n-1} free, the basis would be $2 \cdot 2^{2n-1} = 2^{2n}$. The half-density gap is structural, not coincidental.

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